



Metal-Organic Frameworks (MOFs)

- from basic research to potential applications

Christoph Janiak

University of Düsseldorf, Germany

Email: janiak@uni-duesseldorf.de

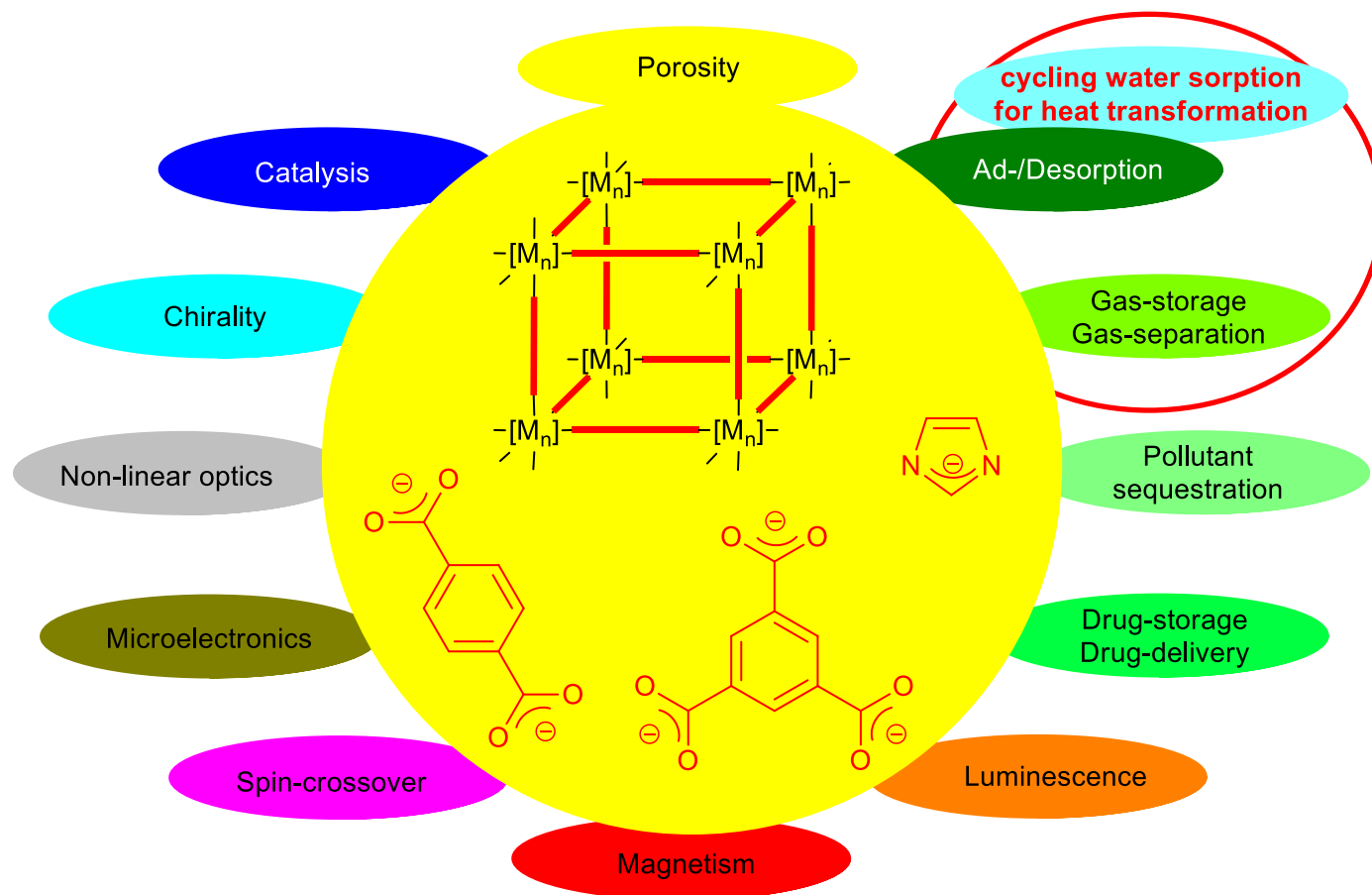


Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis
6. Selected potential applications
 - gas storage
 - gas separation
 - cycling water sorption
 - drug delivery
 - luminescence
 - spin crossover
 - (chiral) catalysis

Metal-Organic Frameworks (MOFs):

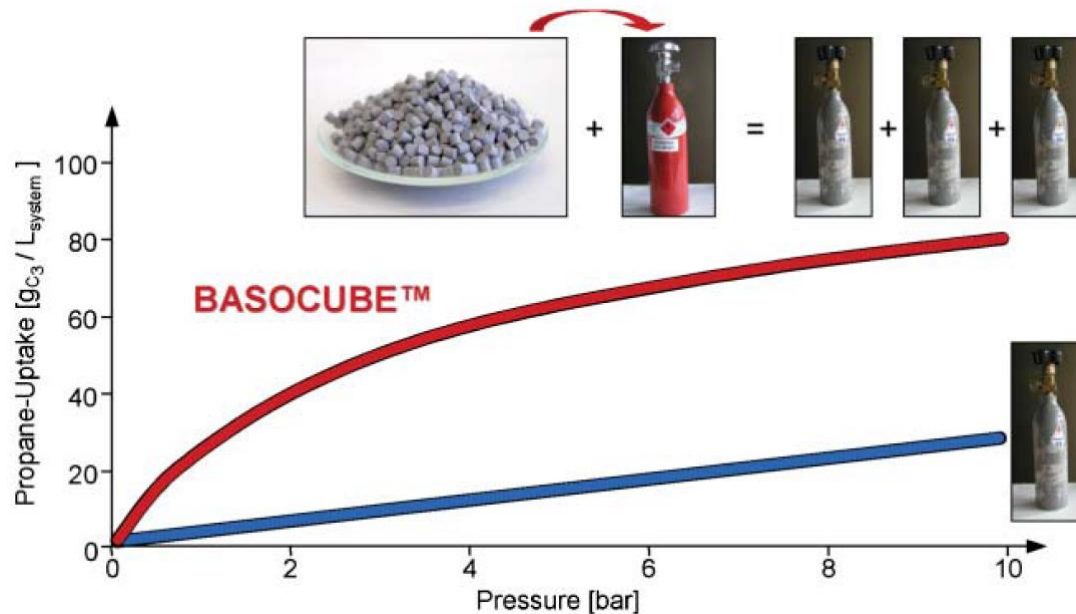
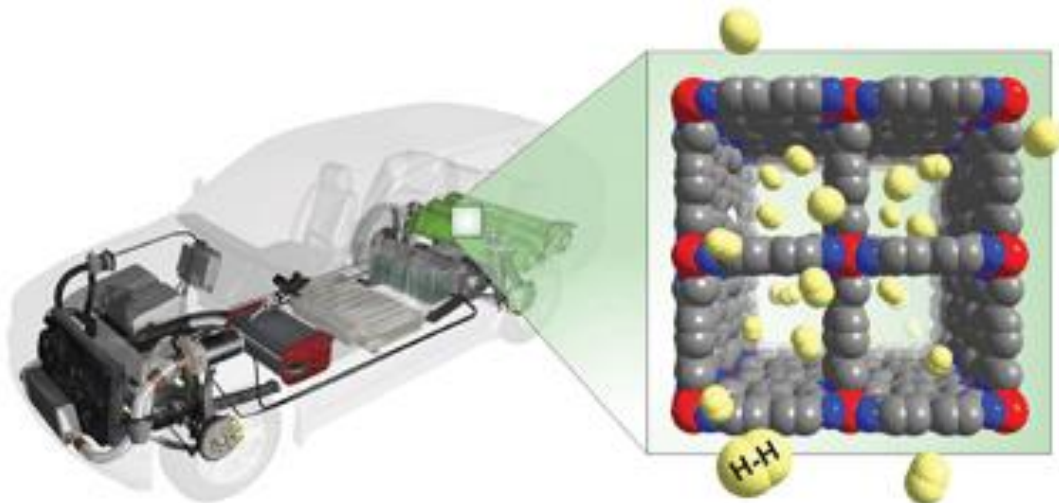
Application-oriented properties



Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E. (>400 citations)
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b. (>2500 citations)

Metal-Organic Frameworks (MOFs): H₂-storage for fuel cells in mobile applications

Gas-storage
-separation

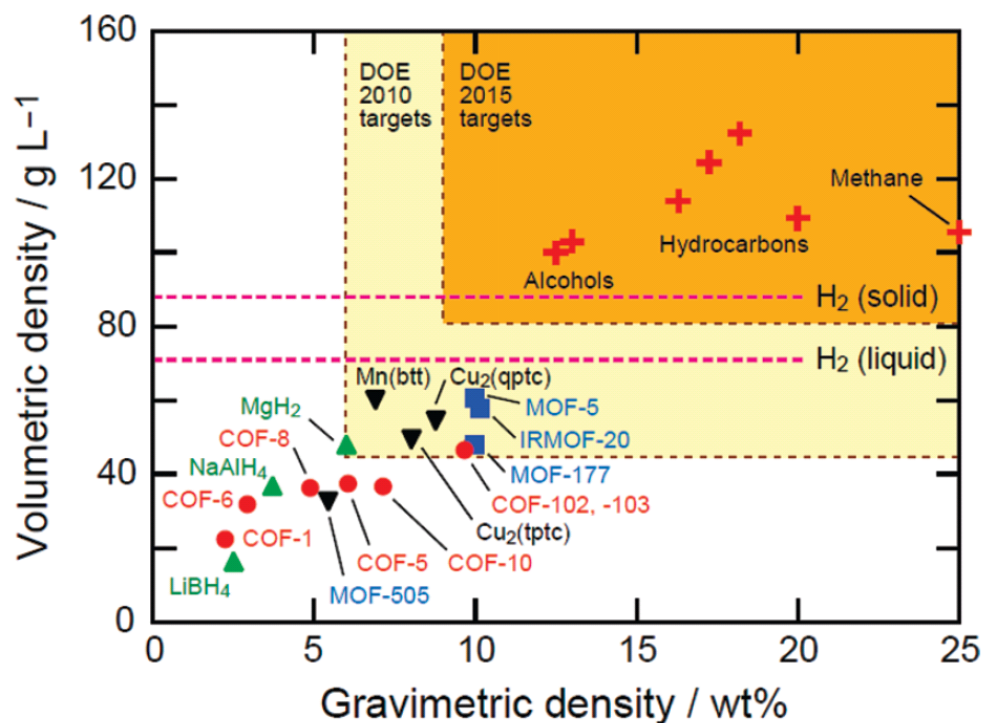


Murray et al., *Chem. Soc. Rev.* **2009**, 38, 1294.

Mueller et al., *J. Mater. Chem.* **2006**, 16, 626.

Metal-Organic Frameworks (MOFs): H_2 -storage

A crucial aspect for H_2/O_2 fuel cells in mobile applications is the dihydrogen storage. The storage of gaseous hydrogen at high pressure (up to 700 bar) or of liquid hydrogen at very low temperatures ($-250\text{ }^{\circ}\text{C}$) in mass-scale, for example in automotive applications, raises issues of safety, handling, insulation, leakage, volumetric and gravimetric storage density. Metal-organic frameworks (MOFs) and also covalent organic frameworks (COFs) have raised some hopes as potential hydrogen storage media because of their high porosity and large specific surface area.

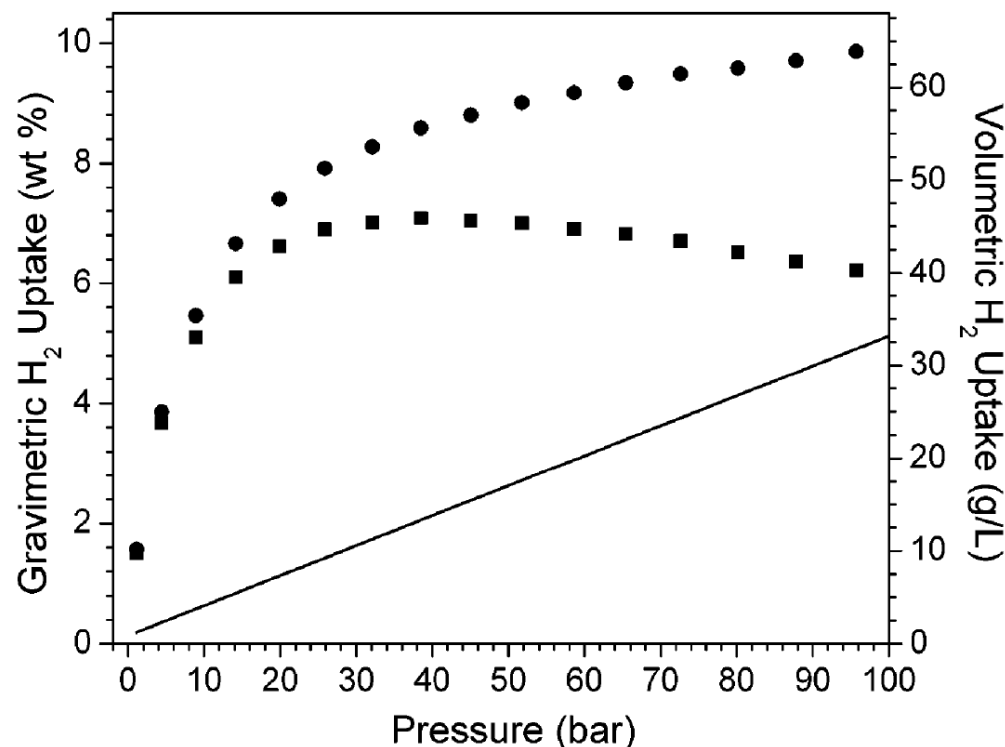


Storage density of dihydrogen, H_2 per mass and per volume. Figure from H. Furukawa and O. M. Yaghi, *J. Am. Chem. Soc.*, 2009, **131**, 8875-8883.

Metal-Organic Frameworks (MOFs): H_2 -storage

**Gas-storage
-separation**

H_2 uptake of 7.1 wt% at 77 K and 40 bar



(Most articles dealing with hydrogen storage in MOFs report the H_2 uptake capacity at a pressure of ca. 1 bar, where excess and total adsorption values are nearly identical.)

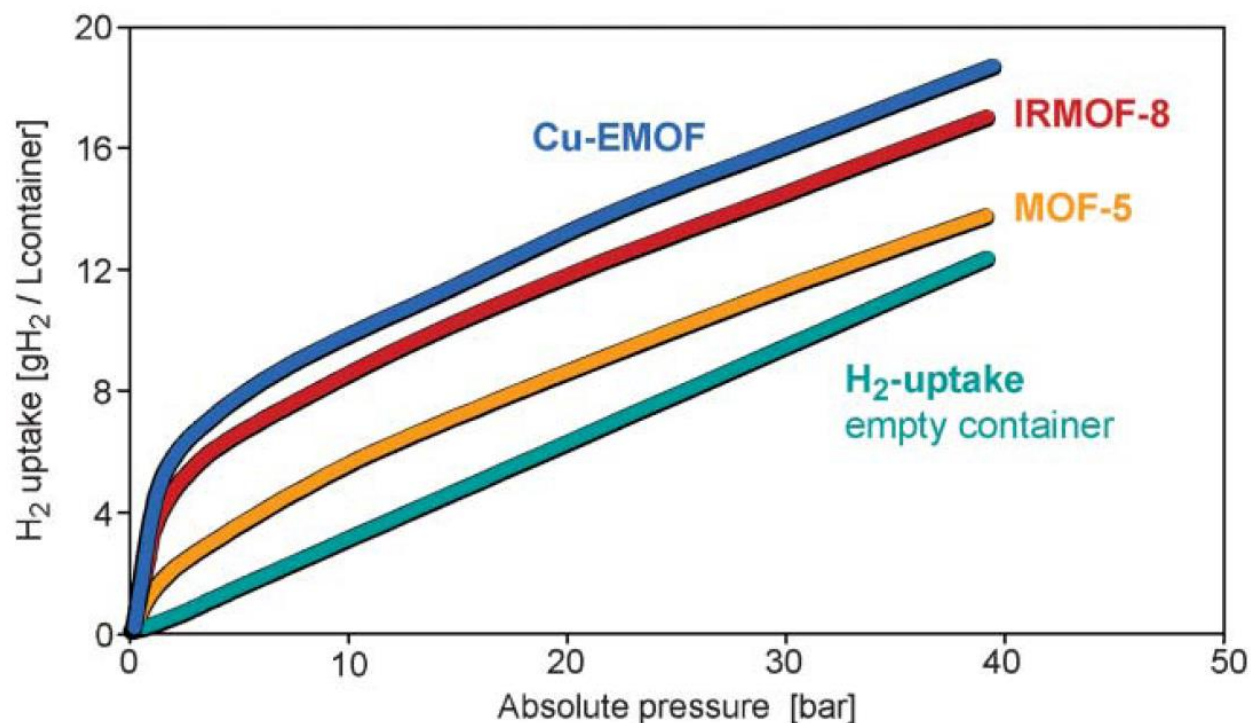
Excess (squares) and total (circles) hydrogen uptake for MOF-5 at 77 K. The solid line represents the density of compressed hydrogen over the given pressure range at this temperature.

Metal-Organic Frameworks (MOFs): H_2 -storage

much interest is attributed to the storage of hydrogen for mobile and portable fuel-cell application by using hydrogen as a carrier for electric power supply.

**Gas-storage
-separation**

Comparing to the empty container MOF-5, IRMOF-8 and Cu-BTC-MOF take up higher amounts of hydrogen.



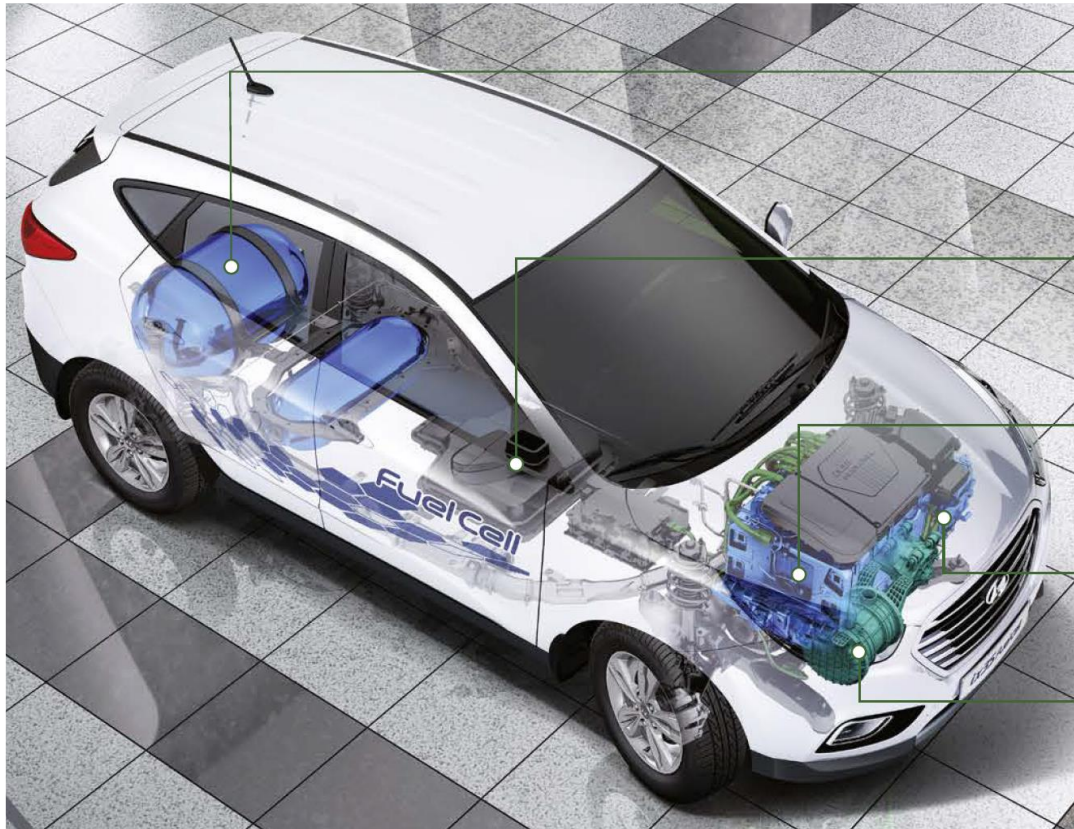
Volume-specific storage curves of hydrogen on different MOF-materials in comparison to compression curve of hydrogen into empty gas containers (77 K)

Presently (2006), a high amount of storage at room temperature has not been attained.

New –H₂-storage in mobile applications

Hyundai ix35 Fuel Cell

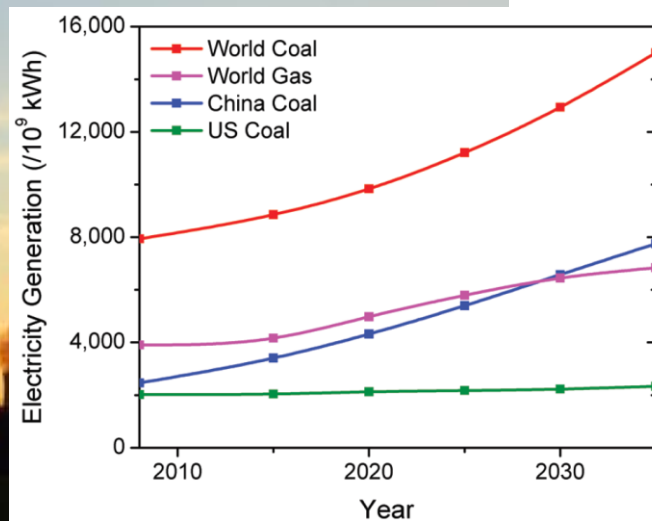
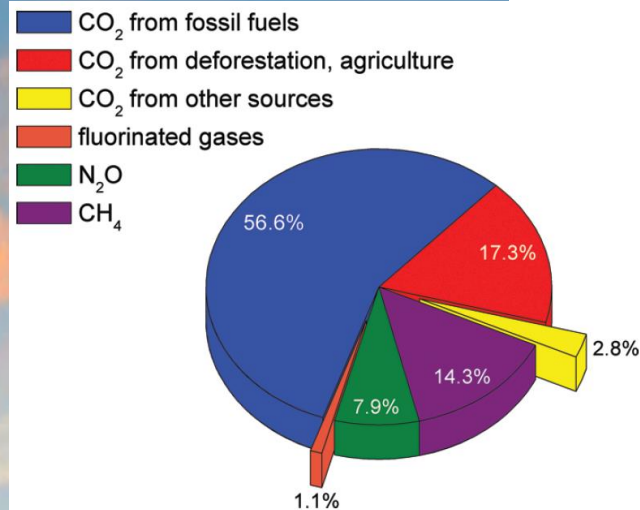
high-pressure hydrogen storage with 700 bar gas tank,
driving range up to 594 km per tank filling



(list price in Germany:
starting 65 450 EUR)

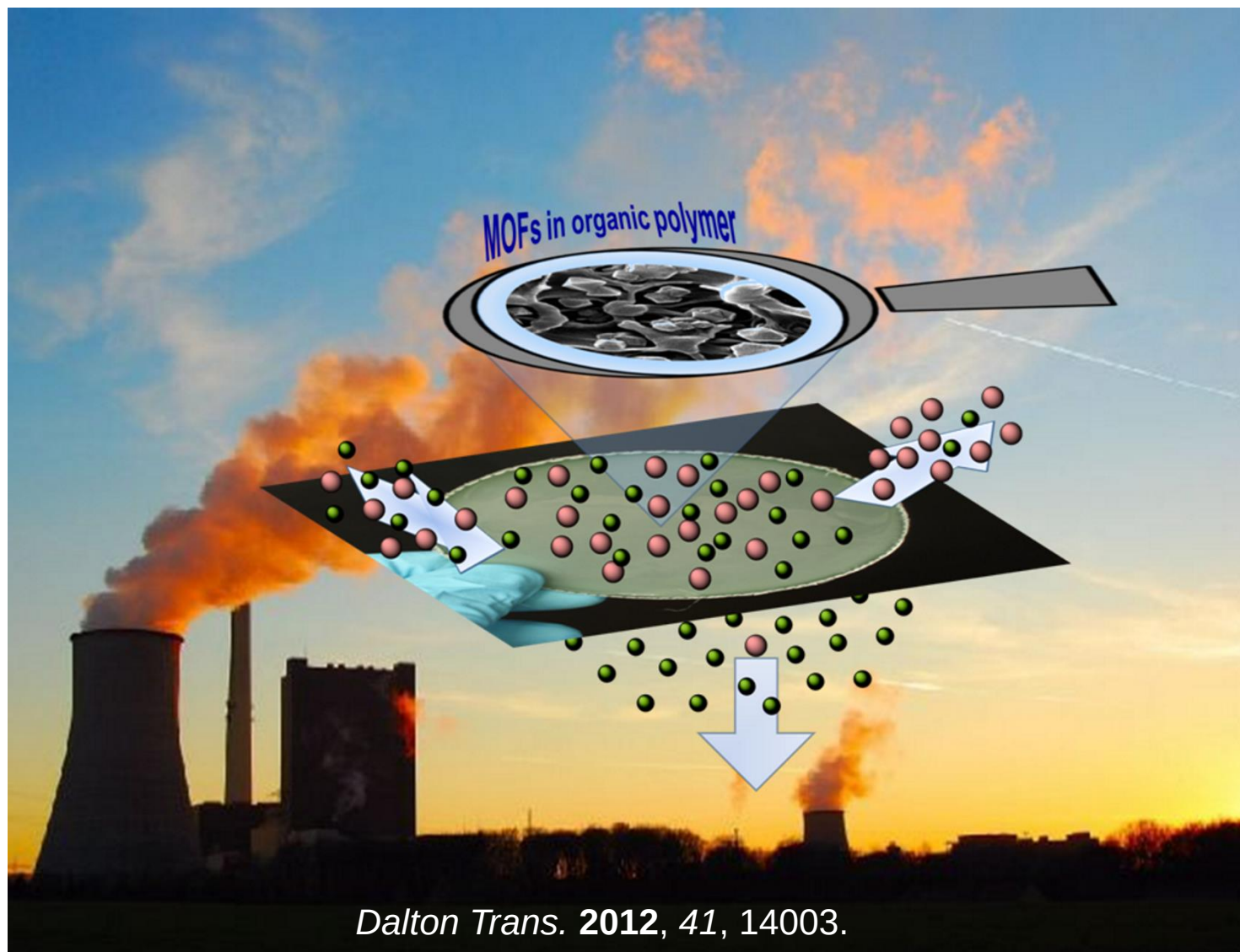
MOFs for gas separation

– e.g. CO₂ from flue gas ("CO₂/N₂")

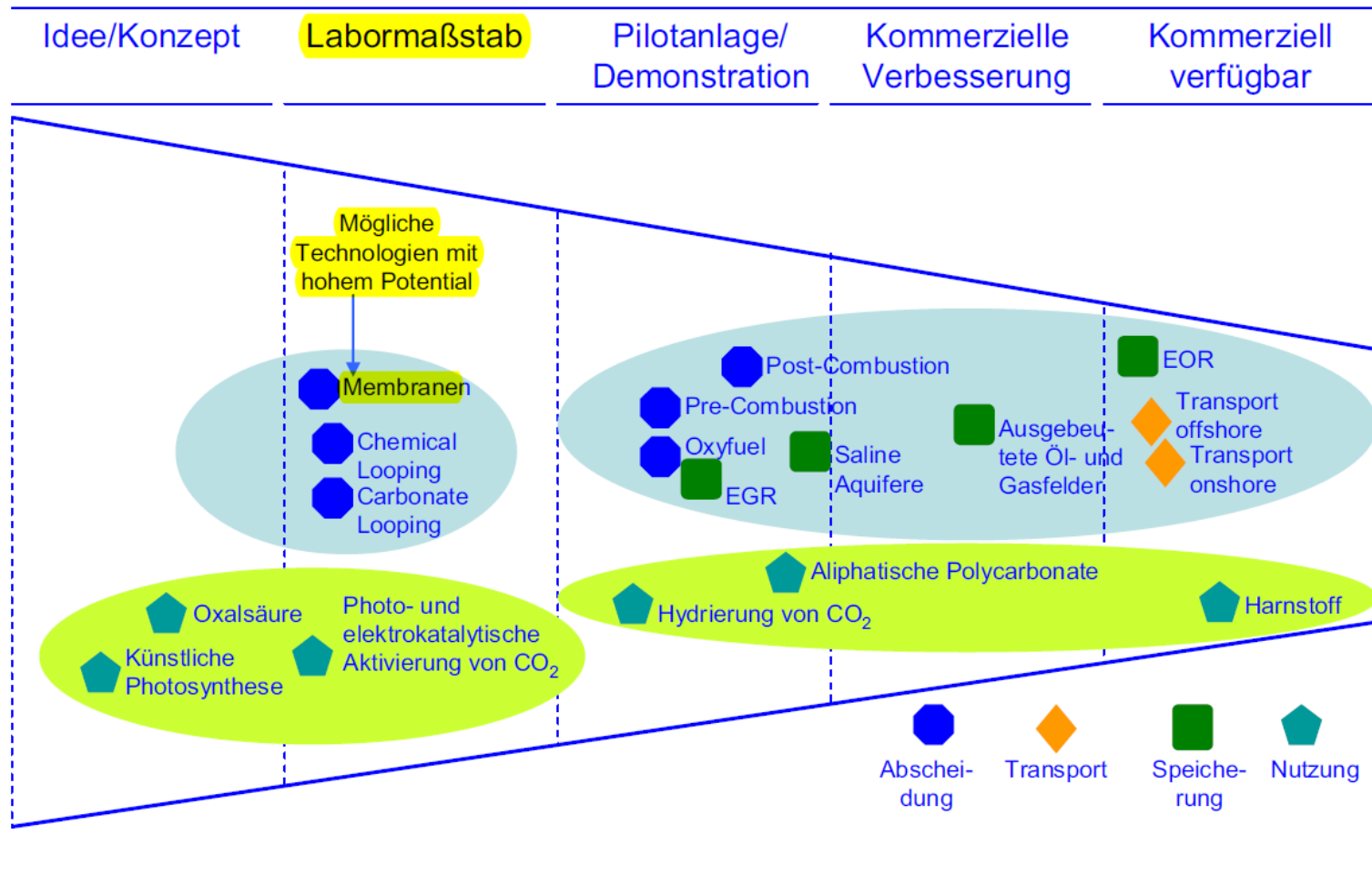


inserts from K. Sumida, D. L. Rogow, J. A. Mason, T. M. McDonald, E. D. Bloch, Z. R. Herm, T.-H. Bae, and Jeffrey R. Long, *Chem. Rev.* **2012**, *112*, 724–781.

Mixed-matrix membranes with MOFs for gas separation – the goal



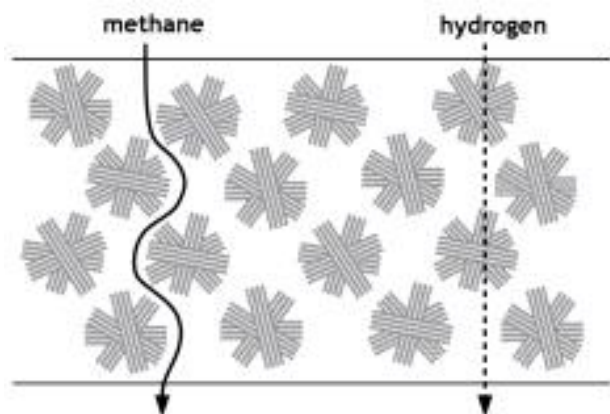
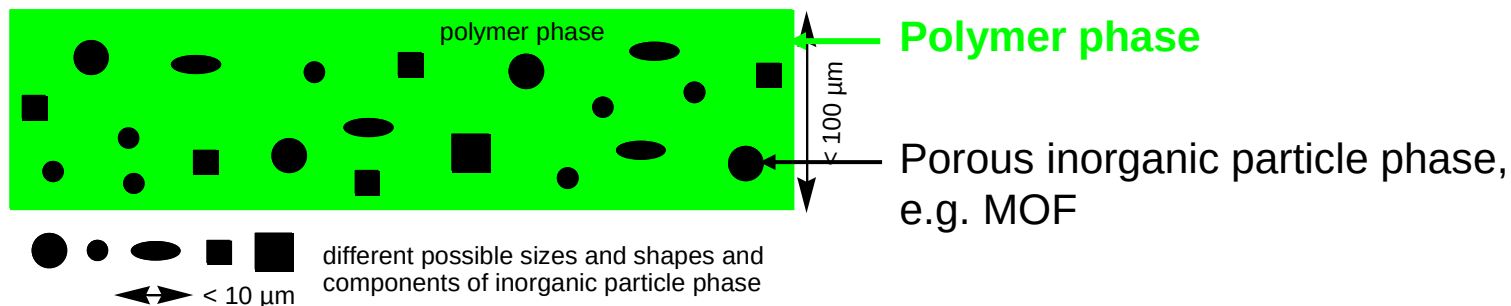
Technologies for separation, transport, storage and uses of CO₂



aus: Weltweite Innovationen bei der Entwicklung von CCS-Technologien und Möglichkeiten der Nutzung und des Recyclings von CO₂
 Studie im Auftrag des Bundesministeriums für Wirtschaft und Technologie (BMWi)
 Projektnummer 25/08, Endbericht März 2010

Mixed-matrix membranes (MMMs) for gas separation:

Polymer-MOF membranes



	H ₂	O ₂	N ₂	CO ₂	CH ₄
kinetic diameter (Å)	2.9	3.46	3.64	3.3	3.8

Chung et al., *Prog. Polym. Sci.*, **2007**, 32, 483.

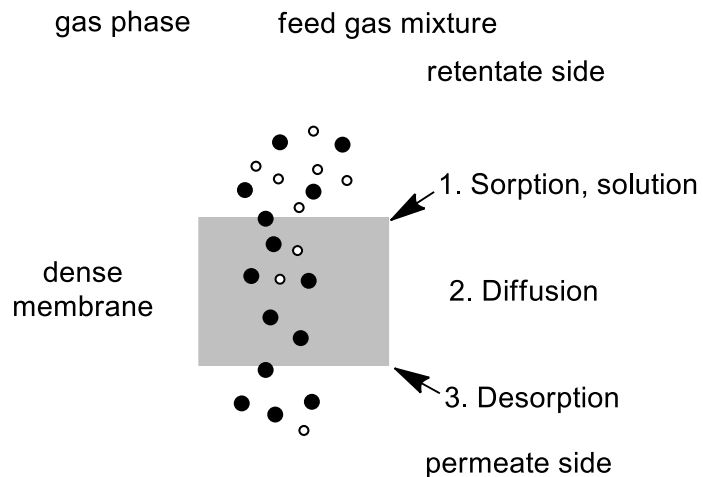
Li et al., *Coord. Chem. Rev.*, **2011**, 255, 1791.

Review: *Dalton Trans.* **2012**, 41, 14003-14027, DOI: 10.1039/C2DT31550E.

B. Seoane, J. Coronas, I. Gascon, M. E. Benavides, O. Karvan, J. Caro, F. Kapteijn, J. Gascon, *Chem. Soc. Rev.* **2015**, 44, 2421–2454.

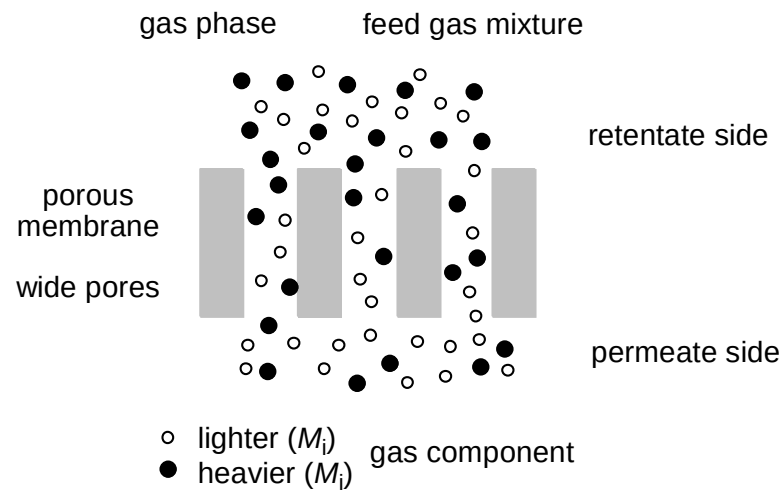
Separation mechanism

solution-diffusion in dense membranes:

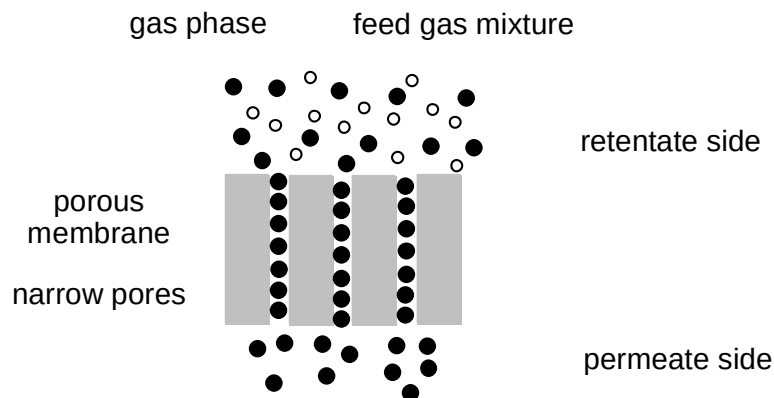


porous membranes:

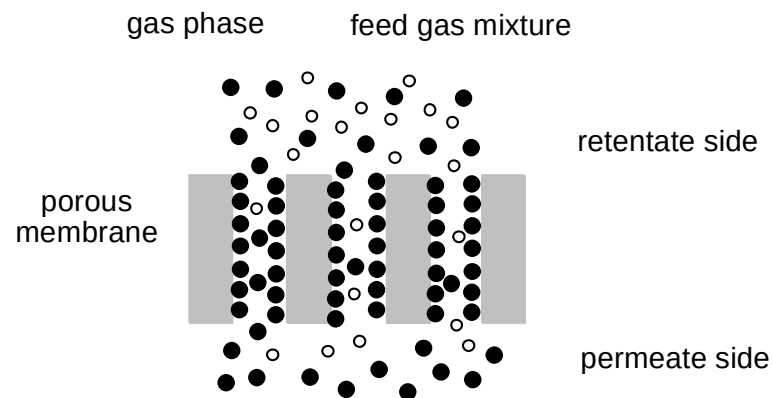
- Knudsen diffusion



- molecular sieving

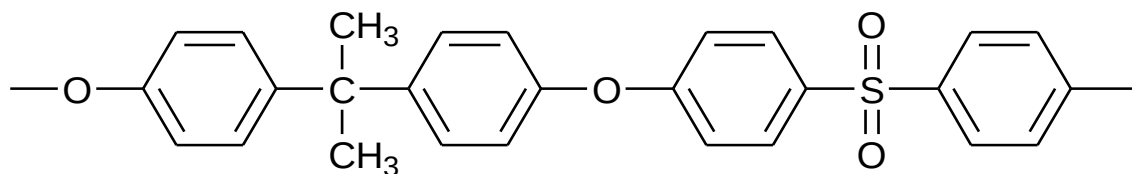


- surface diffusion

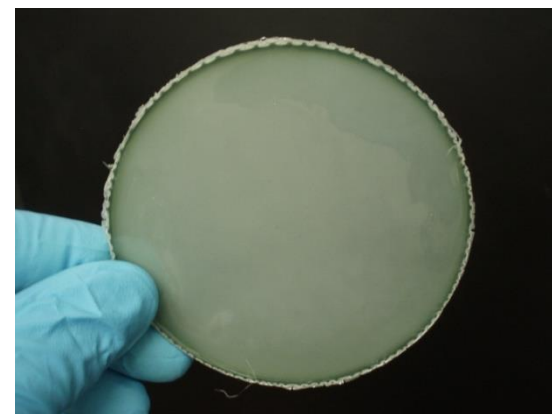
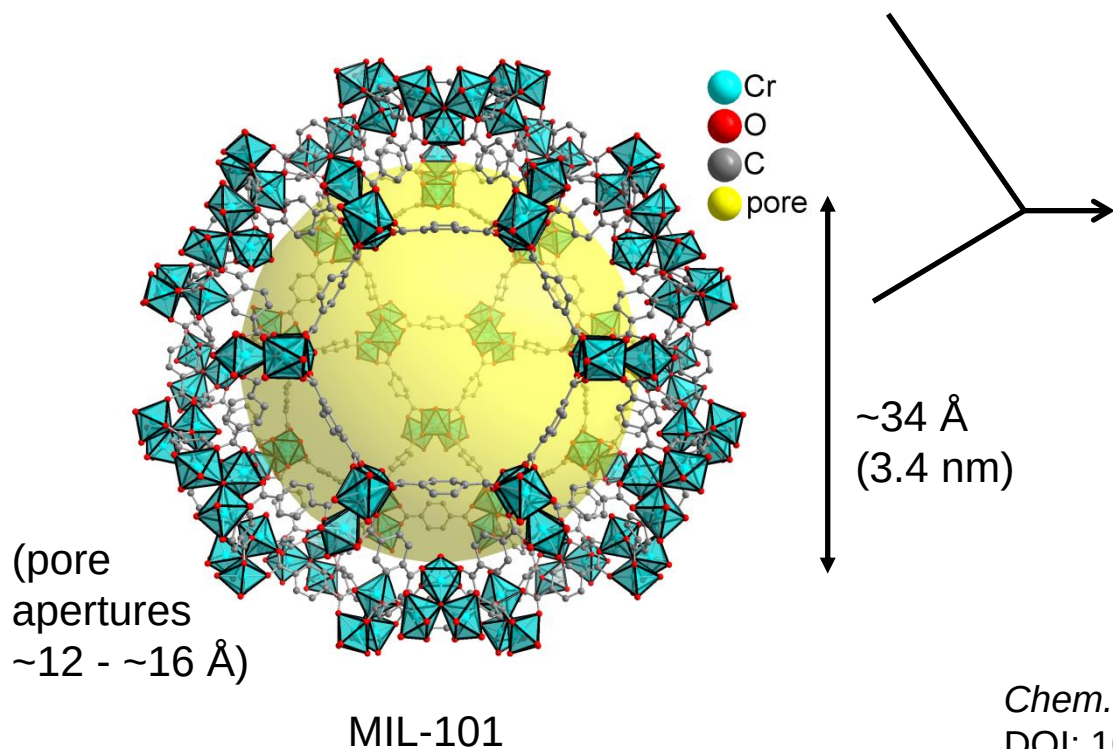


Mixed-matrix membranes with MOFs for gas separation

– PSF-MIL-101 membrane



Polysulfone (PSF) repeating unit



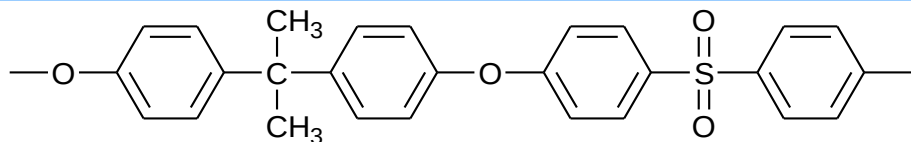
PSF-MIL-101

MMM,

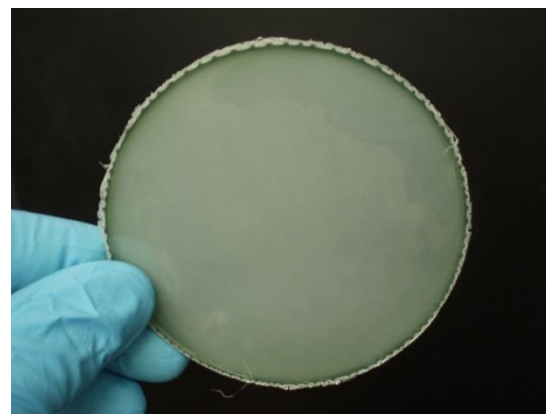
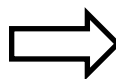
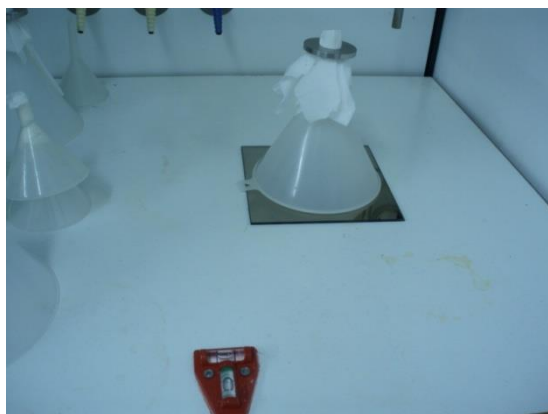
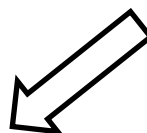
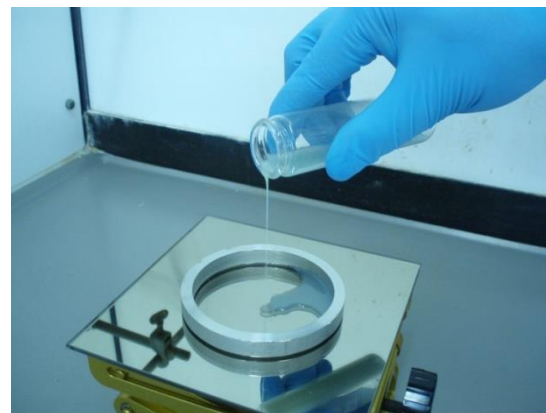
↓
20-60 μm
↑

Chem. Commun. **2012**, 48, 2140-2142,
DOI: 10.1039/C2CC16628C.

Mixed-matrix membranes with MOFs for gas separation – membrane preparation



Polysulfone (PSF) repeating unit

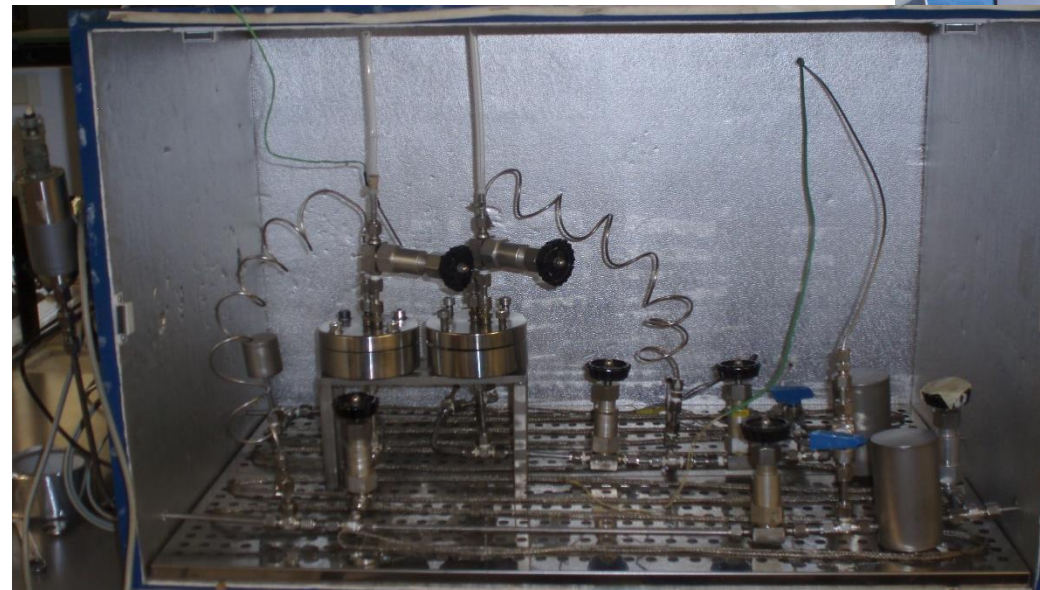


PSF-MIL-101

Mixed-matrix membranes with MOFs for gas separation – membrane preparation

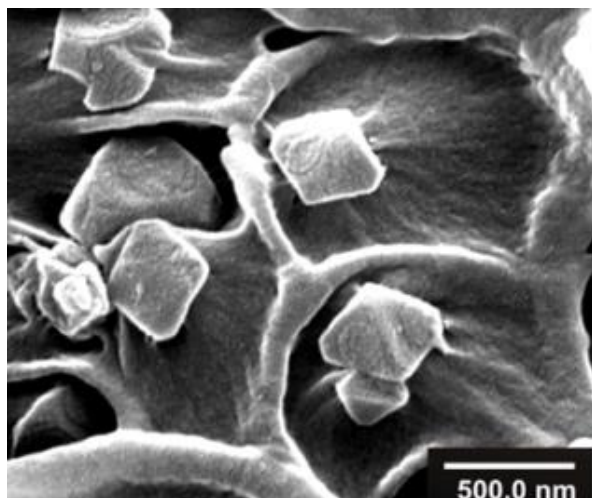
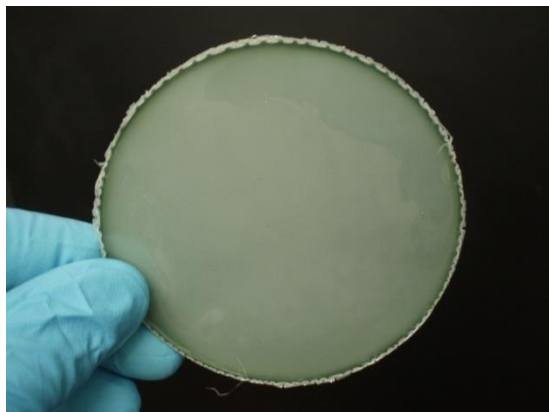


measuring gas permeability
- gas selectivity



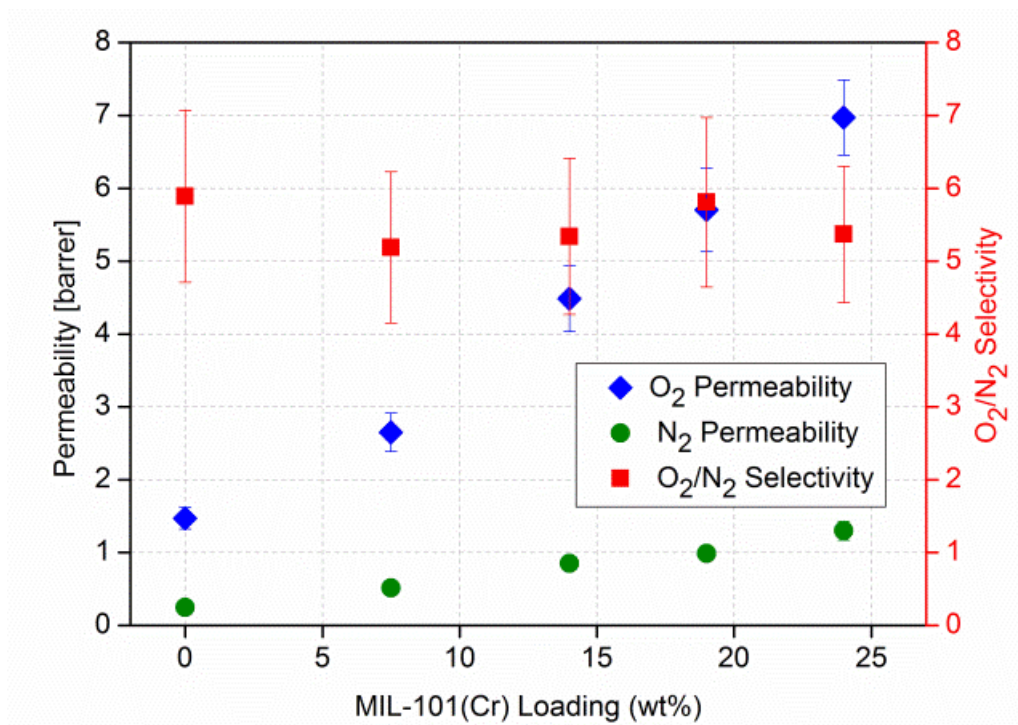
Mixed-matrix membranes with MOFs for gas separation

– O₂/N₂ separation with PSF-MIL-101(Cr) membrane



7.5 wt% MIL-101(Cr)@PSF

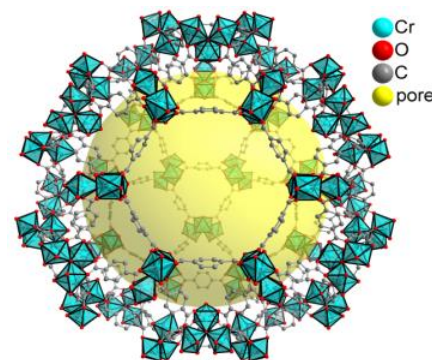
O₂/N₂ separation



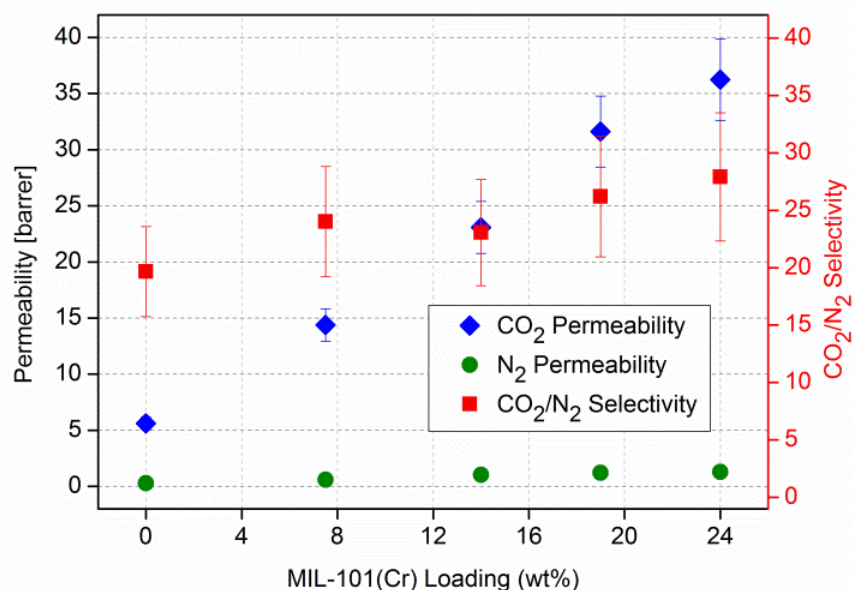
Chem. Commun. **2012**, 48, 2140-2142,
DOI: 10.1039/C2CC16628C.

PSF-MIL-101(Cr) membrane

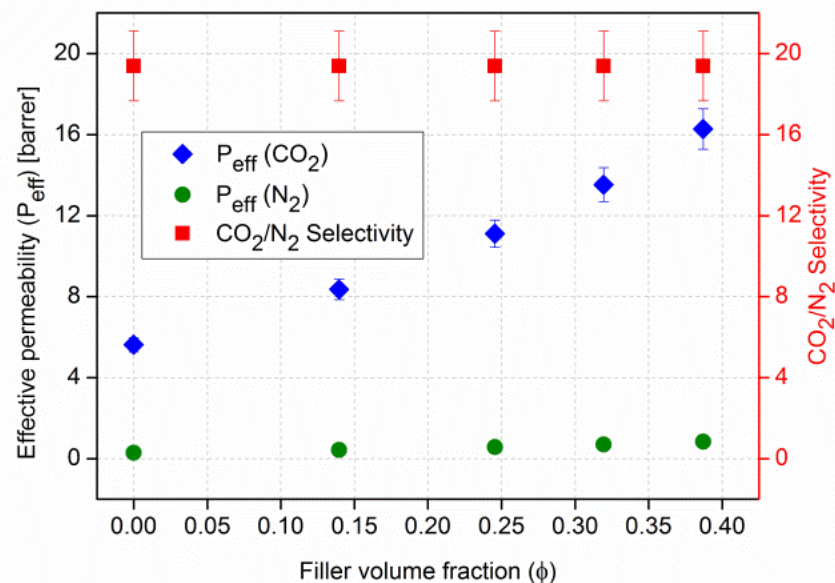
– CO₂ separation from N₂ or CH₄



ideal single gas: CO₂ / N₂



Maxwell model calculation:

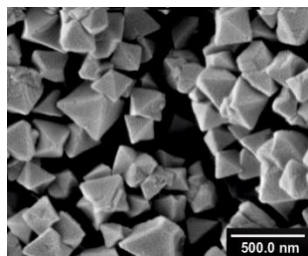


Membranes **2012**, 2, 727-763, DOI: 10.3390/membranes2040727.

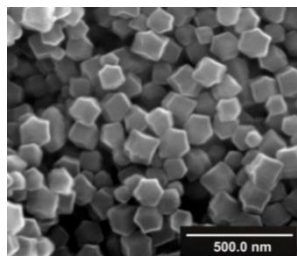
Membranes **2013**, 3, 331-353, DOI: 10.3390/membranes3040331.

mixed-MOF MMM: MIL-101(Cr)+ZIF-8@PSF

– CO₂ separation from CH₄

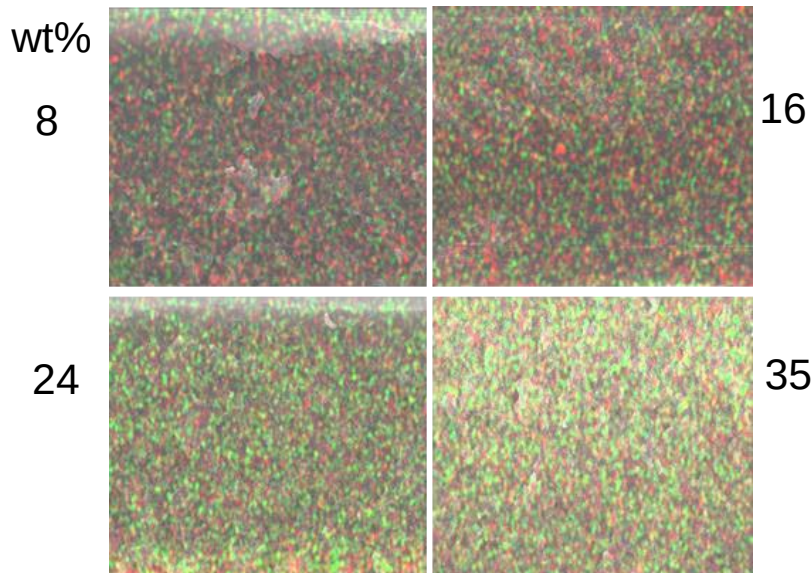


MIL-101(Cr)



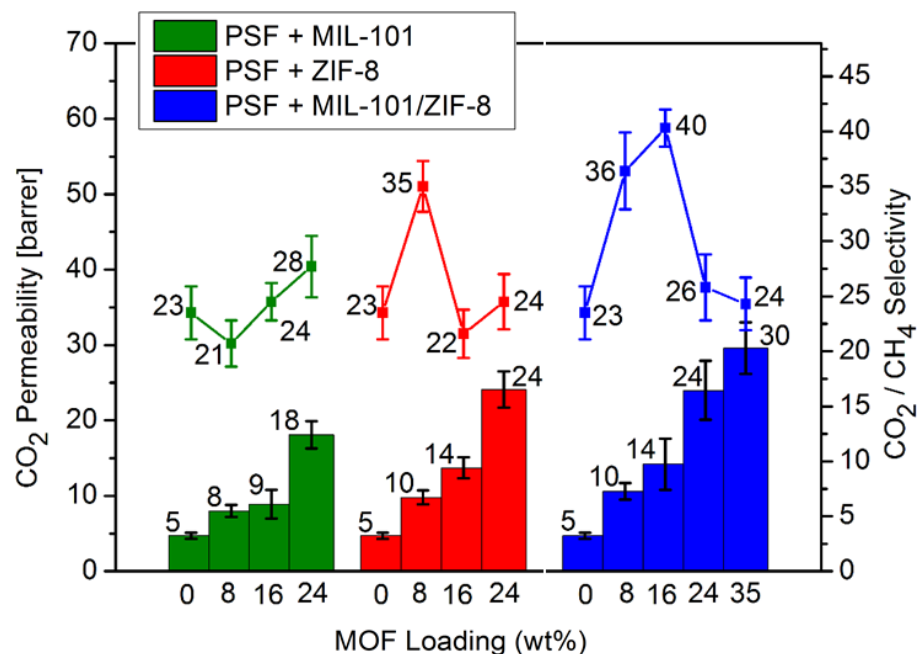
ZIF-8

SEM-EDX element mapping:



Cr-red, Zn-green (1:1)

"two MOFs are better than one"

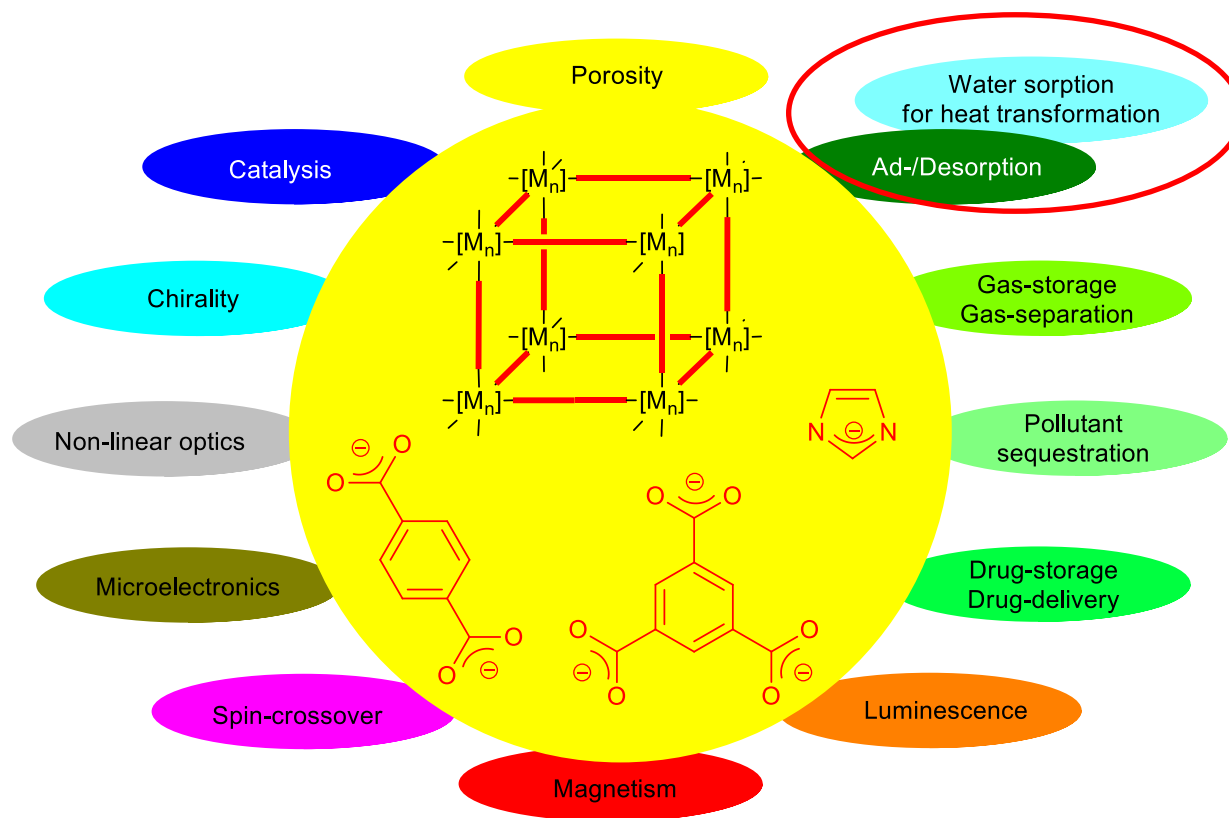


(mixed-gas measurements
of CO₂:CH₄ 1:1 (v:v)
mixture)

Eur. J. Inorg. Chem. **2016**, 4363–4367.
DOI: 10.1002/ejic.201600190.

Metal-Organic Frameworks (MOFs):

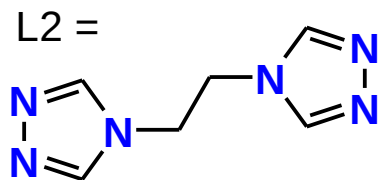
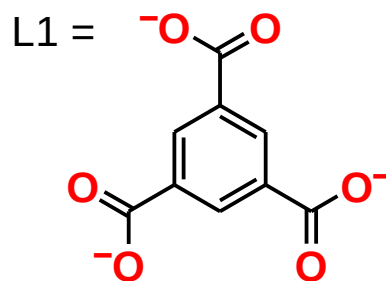
Application-oriented properties



Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E. (>400 citations)
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b. (>2500 citations)

MOF example with neutral ligand - additional bridging anionic ligands

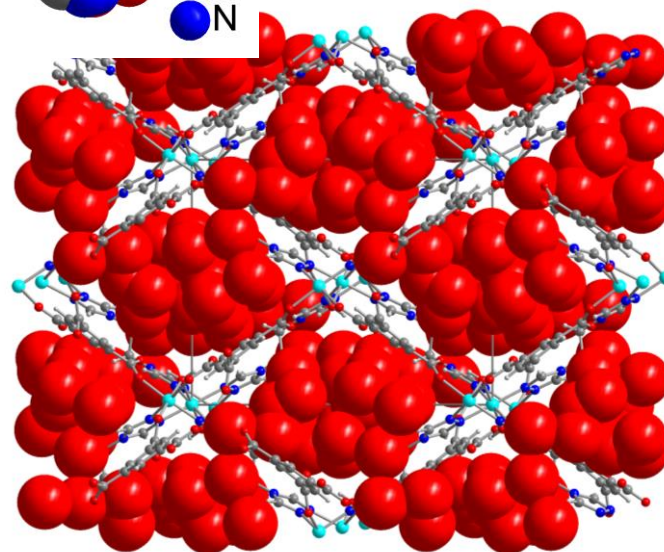
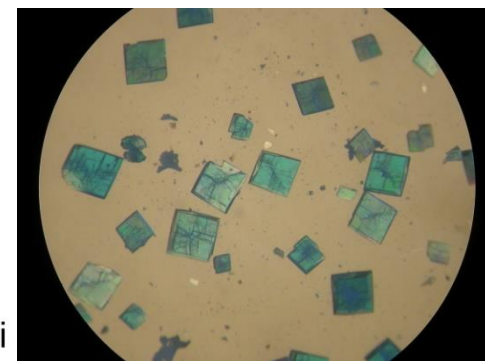
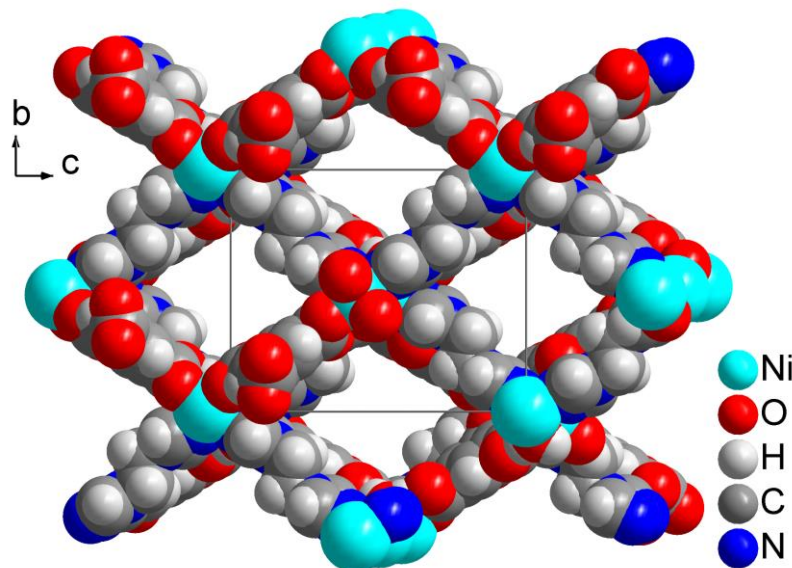
- **mixed-ligand MOF** $3D-\{[Ni_3(\mu_3-L1)_2(\mu_4-L2)_2(\mu-H_2O)_2]\cdot\sim 22H_2O\}$



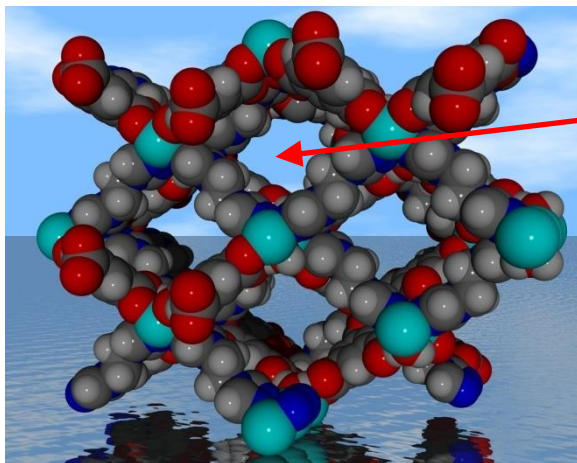
potential solvent volume
1621 Å³/unit cell
= 52%
(H₂O ~40 Å³)

MOF for reversible
water sorption

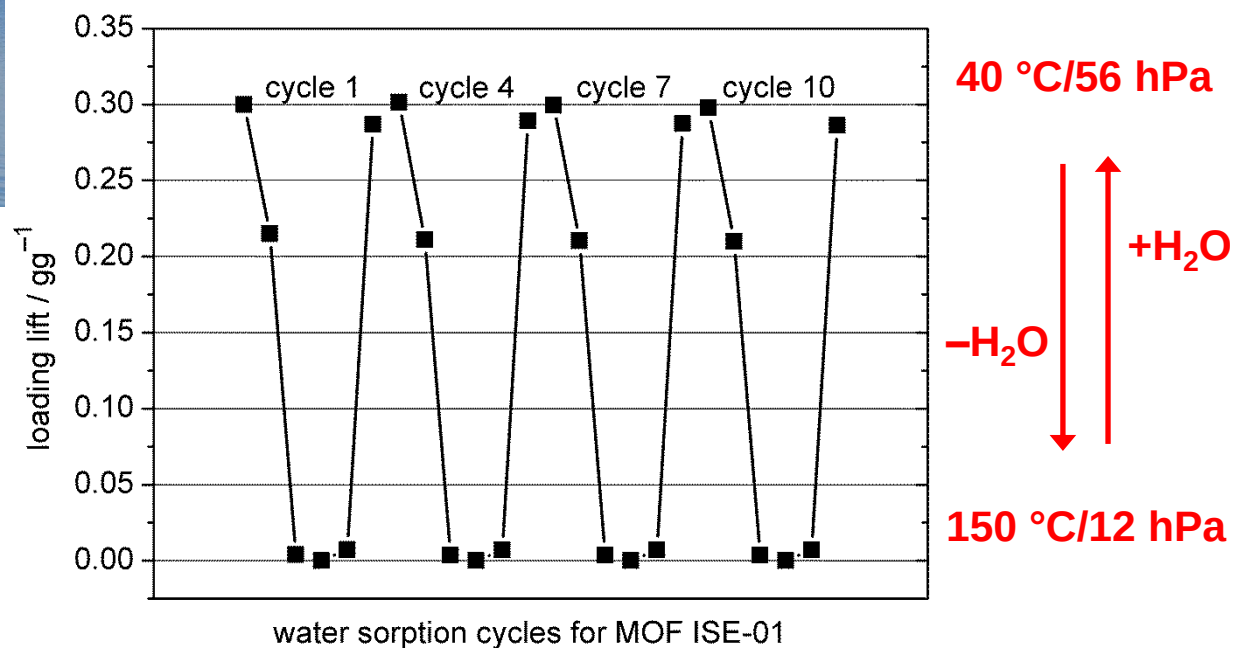
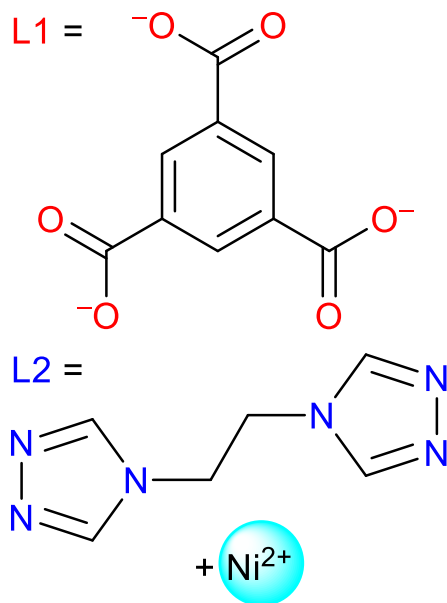
H. A. Habib, J. Sanchiz, C. Janiak,
Dalton Trans. **2008**, 1734-1744,
DOI: 10.1039/b715812b



Mixed-ligand MOF: Water ad-/desorption



water filled/emptied pores



Dalton Trans. **2008**, 1734-1744, DOI: 10.1039/b715812b.

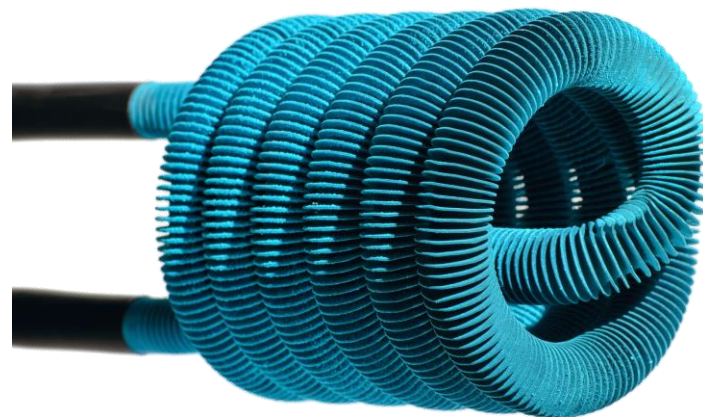
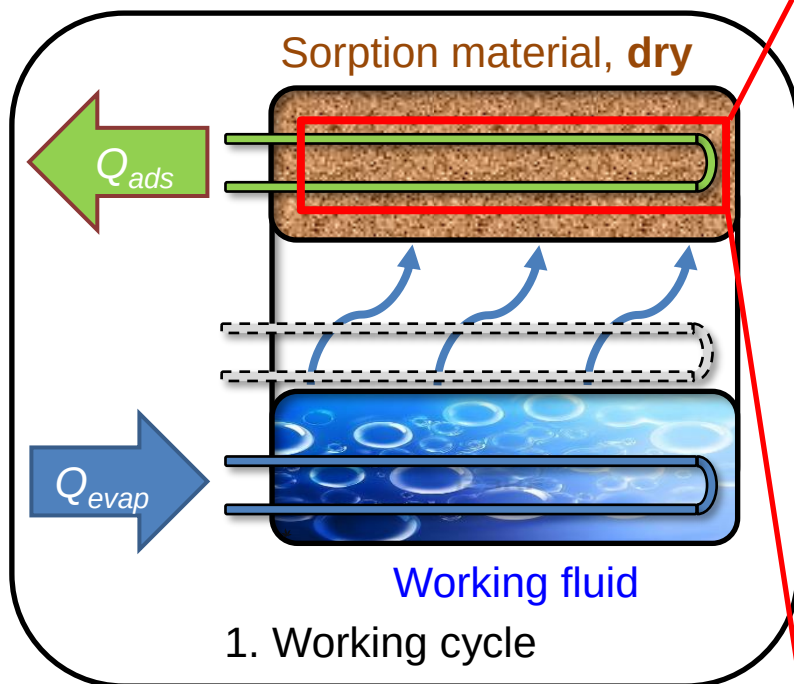
J. Am. Chem. Soc. **2009**, 131, 2776-2777, DOI: 10.1021/ja808444z.



MOFs for heat transformation through cycling (water) sorption in an adsorption chiller / heat pump

HEINRICH HEINE
UNIVERSITÄT DÜSSELDORF

MOF-coated heat exchangers



© Fraunhofer ISE



Chem. Commun. **2012**, 48, 9708-9710,
DOI: 10.1039/C2CC34782B..
RSC Advances **2014**, 4, 24073-24082,
DOI: 10.1039/c4ra03794d.
J. Mater. Chem. A **2016**, 4, 11859-11869,
DOI: 10.1039/C6TA01757F.

Comfortable living requires air-conditioning – in many parts of the world

More than **50% of the energy** consumption ... of modern buildings originates **from air conditioning** processes.

(*J. Mater. Chem.* **2012**, 22, 10148.)



conventional
air-conditioner
=
electricity
consumption
=
largely
fossil fuel
burning

Comfortable living requires air-conditioning – in many parts of the world

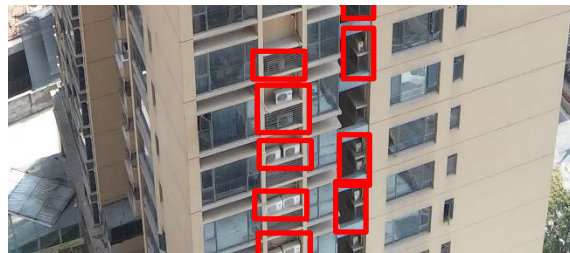
More than **50%** of the energy consumption ... of modern buildings originates **from air conditioning** processes.

(*J. Mater. Chem.* **2012**, 22, 10148.)



conventional
air-conditioner
=
electricity
consumption
=
largely
fossil fuel
burning

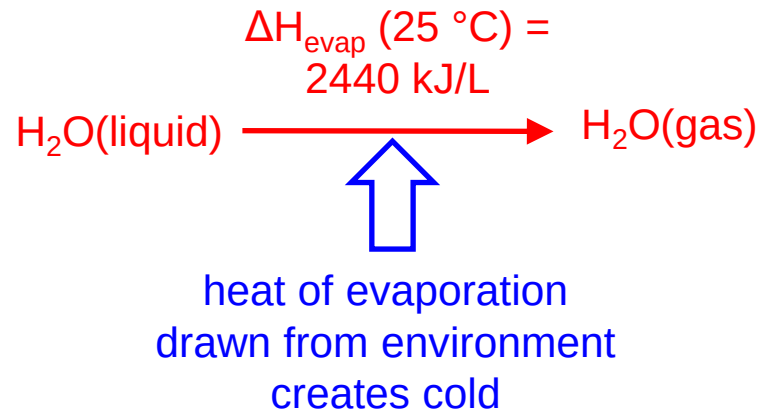
Alternatives?



Alternatives?



Alternatives: Water evaporation – evaporative cooling



sweating = evaporate cooling !

Cooling through water evaporation = evaporative cooling

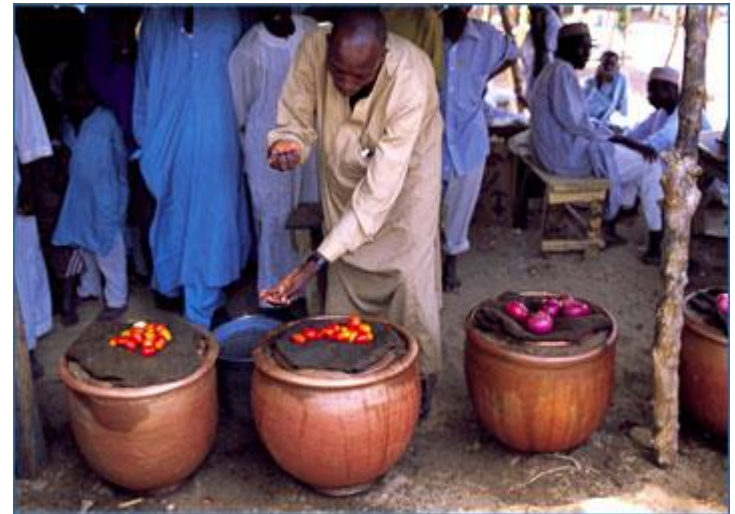
Pot-in-pot refrigerator

- with wet sand in-between

zeer (Arabic: زير)

- porous outer earthenware pot,
- + layer with wet sand
- + inner pot within which the food is placed
- the evaporation of the water draws heat from the inner pot.

dry air drives water evaporation



used already in

- old kingdom of Egypt, around 2500 BC
- Indus Valley Civilization around 3000 BC
- present-day ghara or matki used in India and Pakistan

from Wikipedia, the free encyclopedia

Chilling through **evaporative cooling** - common in Australia and other countries



Evaporative Cooling Sys

How Evaporative Cooling Works

dry air drives water evaporation



Outside ambient temperature °C	Percentage relative humidity						
	20%	30%	40%	50%	60%	70%	80%
20	12,0	13,0	14,5	15,5	16,5	17,5	18,5
25	16,0	17,0	18,5	20,0	21,0	22,0	23,0
30	19,5	21,0	22,5	24,0	25,0	26,5	28,0
35	23,0	25,0	26,5	28,5	30,0	31,5	32,5
40	26,5	29,0	31,0	32,5	34,5	36,0	-
45	29,0	32,5	35,0	-	-	-	-

Evaporative Cooling Effectiveness, by Region

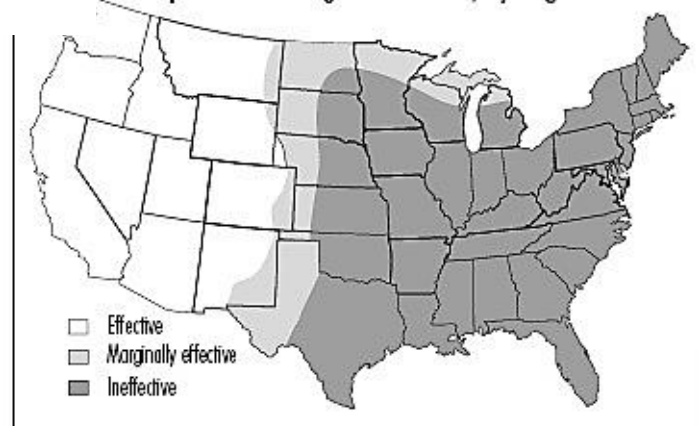
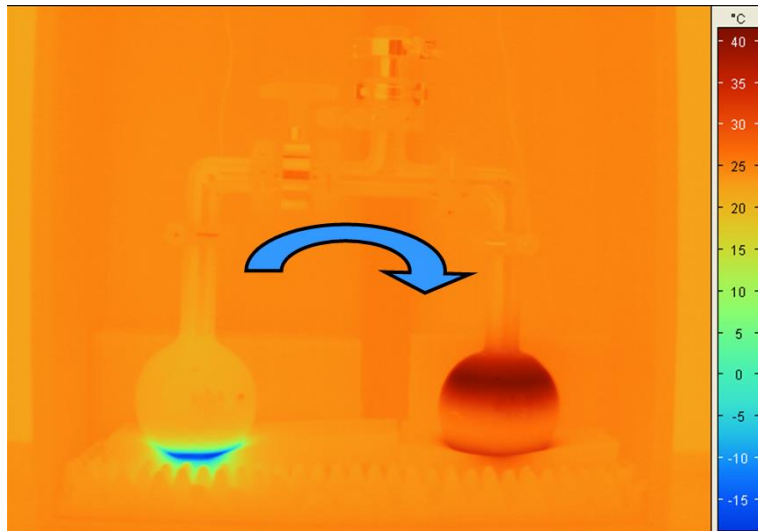
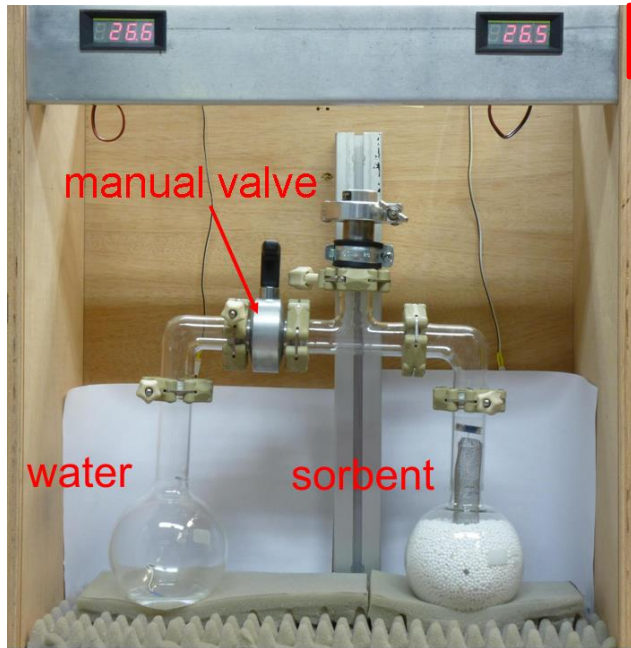


Figure 2. In areas of the United States where humidity tends to run high in the summertime, evaporative cooling is not the best way to stay cool; in the West, it's a

Chilling through evaporative cooling

- no dry air? – porous materials as driving force!

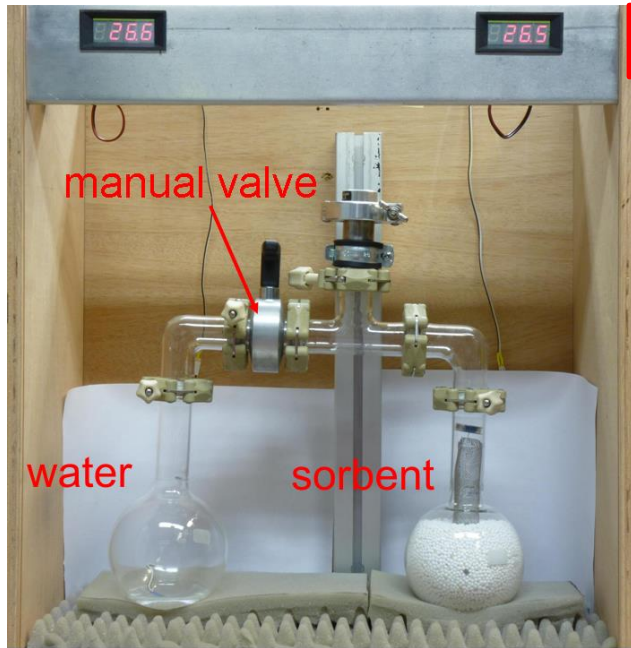
Adsorption chilling



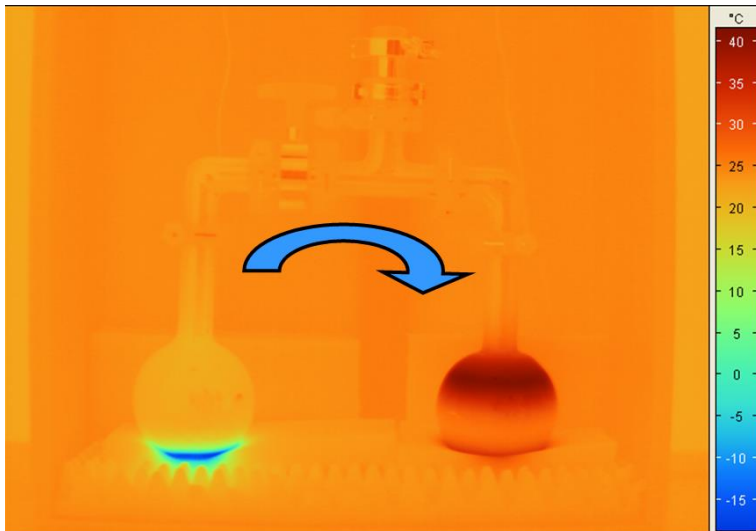
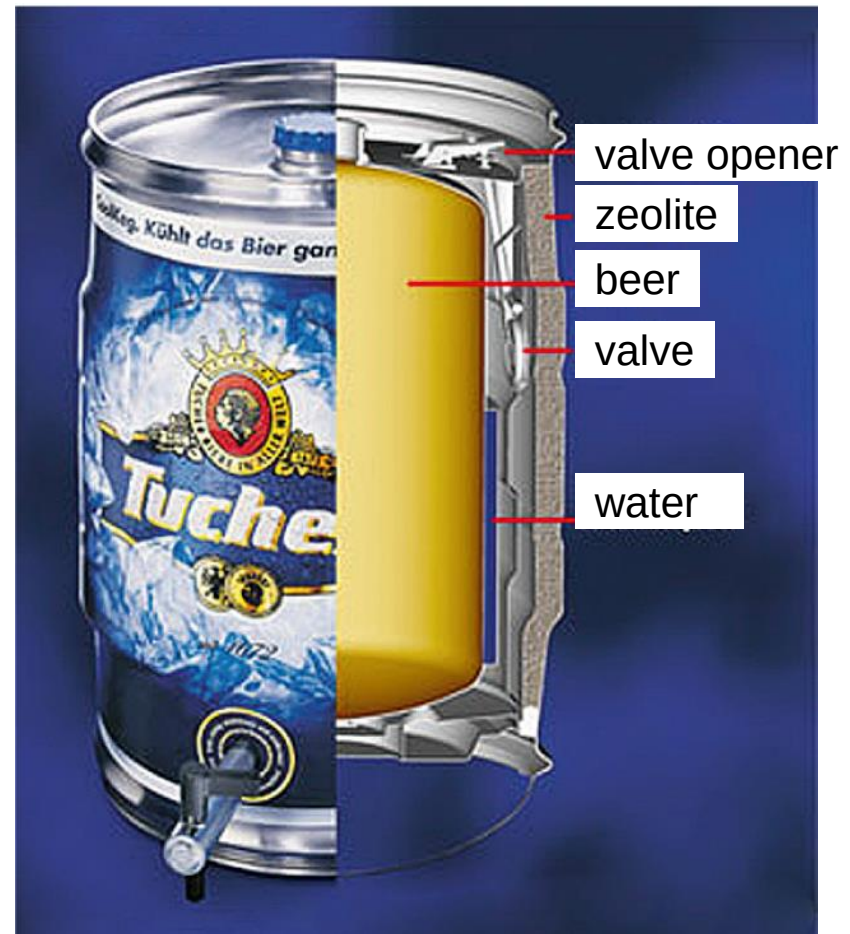
Chilling through evaporative cooling

- no dry air? – porous materials as driving force!

Adsorption chilling

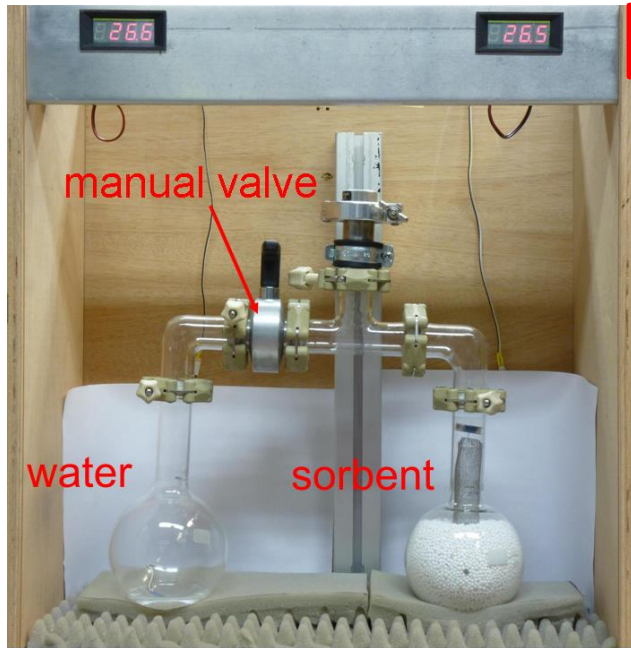


an established technology:
"Cool-Keg"



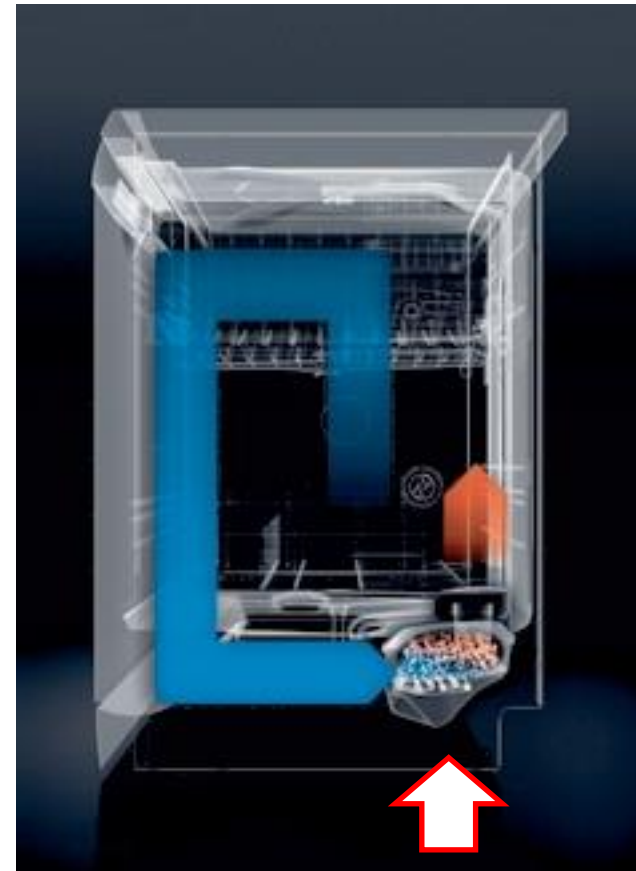
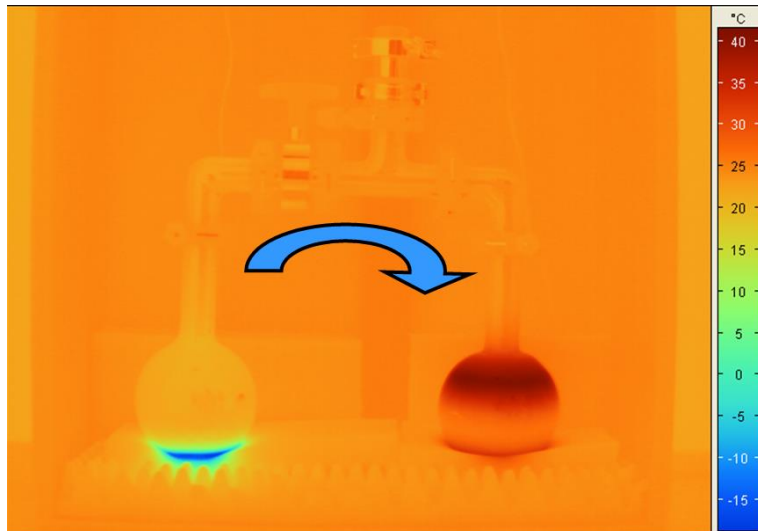
Chilling through evaporative cooling

- no dry air? – porous materials as driving force!



Adsorption (chilling) and de-humidification

an established technology:
"zeolite" dishwasher (from BSH)

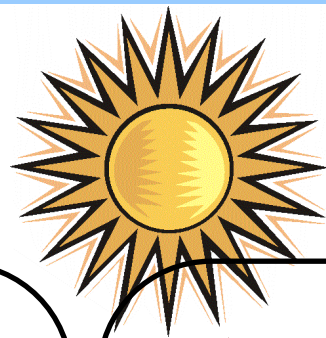


1 kg zeolite

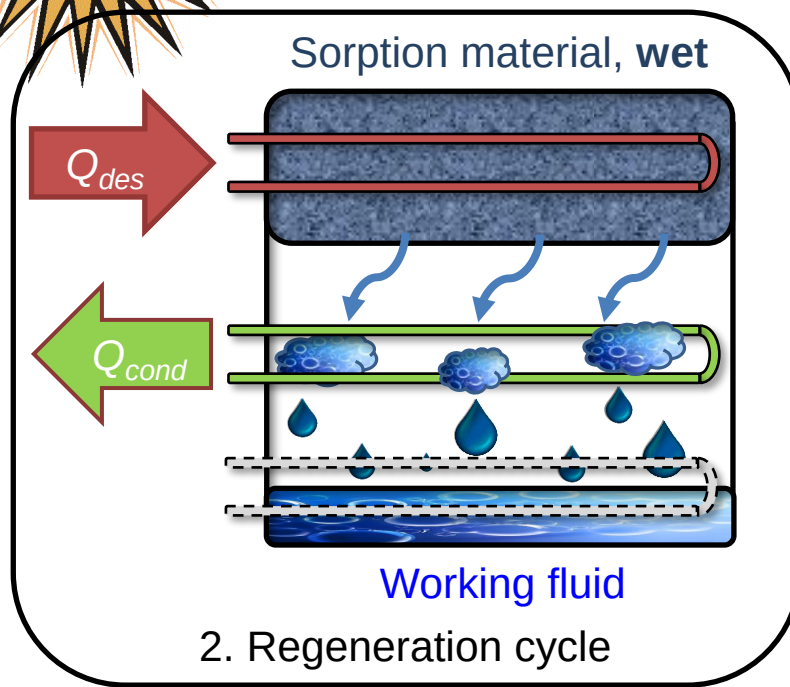
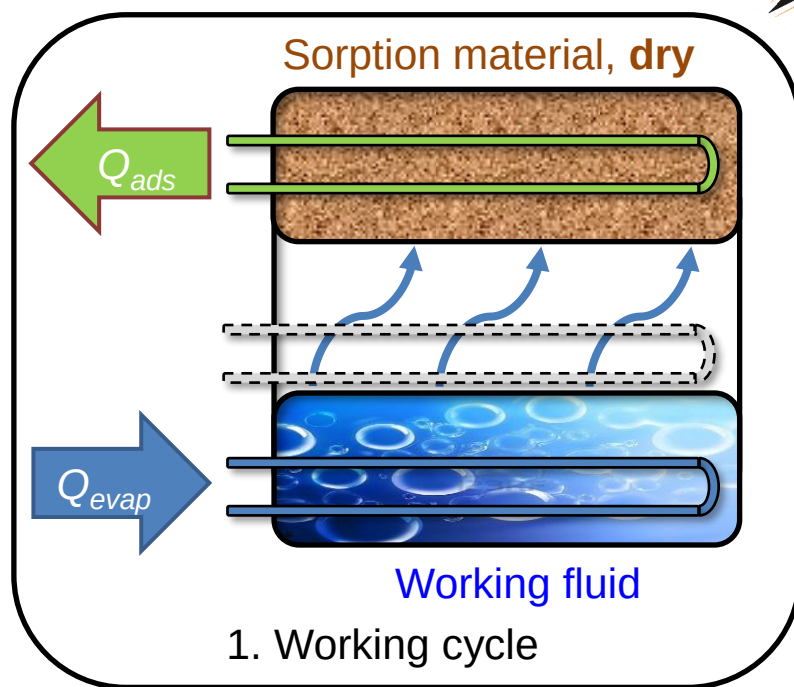
Principle process in an adsorption chiller / heat pump

-- a "new" type of air-conditioner?

only ~1/3 electrical energy of
conventional air-conditioner



solar thermal energy
for regeneration, driving heat



Eur. J. Inorg. Chem. **2011**, 471.
J. Mater. Chem. **2012**, 22, 10148.
Chem. Mater. **2013**, 25, 790.
RSC Advances **2014**, 4, 24073.

Reviews: *Eur. J. Inorg. Chem.* **2012**, 2625.
Chimia **2013**, 67, 419.
New J. Chem. **2014**, 38, 1846.

Adsorption chiller (through evaporative cooling) – commercially available



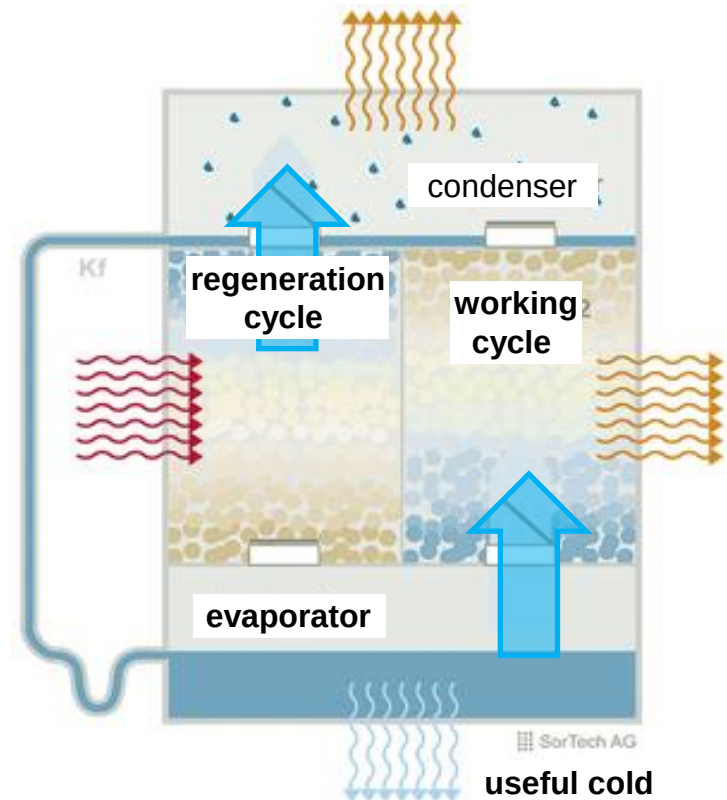
<https://fahrenheit.cool/>



www.inversor.com

sorbent: silica gel or zeolite

large scale adsorption chillers (50-1000 kW) produced by Power Partners (USA), Mayekawa (Japan), GBU mbH (Germany), Fahrenheit (Germany), HIJC (USA, ex-Nishiyodo),;
small scale (5-25 kW) by Mitsubishi Plastics (Japan), SolabCool (Netherlands), Jiangsu Huineng (China), Fahrenheit, Inversor (both Germany),



- water vapor
- liquid process water
- back flush valves
- Kf condenser re-use
- driving heat
- excess heat
- useful cold

Example of **adsorption chiller with solar thermal assistance** at University hospital in Freiburg



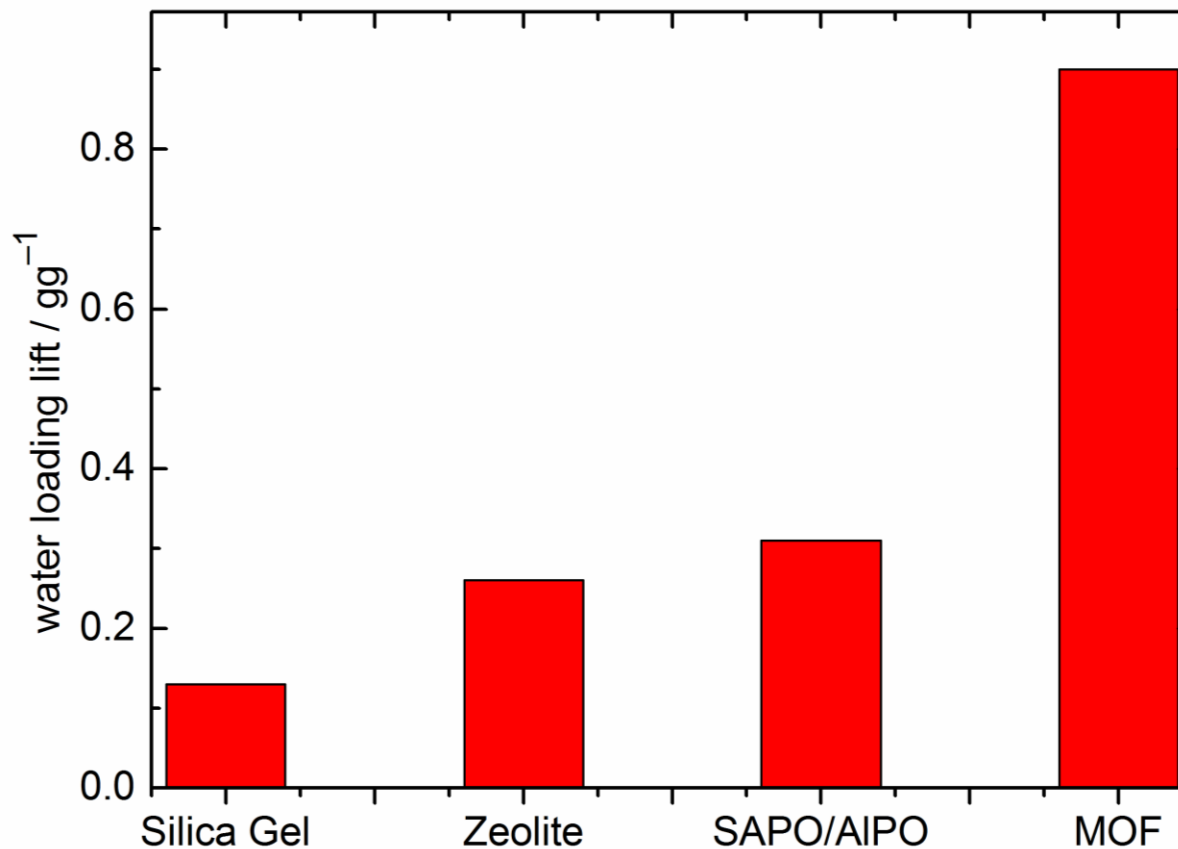
Capacity of 70 kW, adsorbent: silica gel

Henning, *Appl. Therm. Engin.* **2007**, 27, 1734.
Núñez et al., *Appl. Therm. Engin.* **2007**, 27, 2205.

Solar collector field



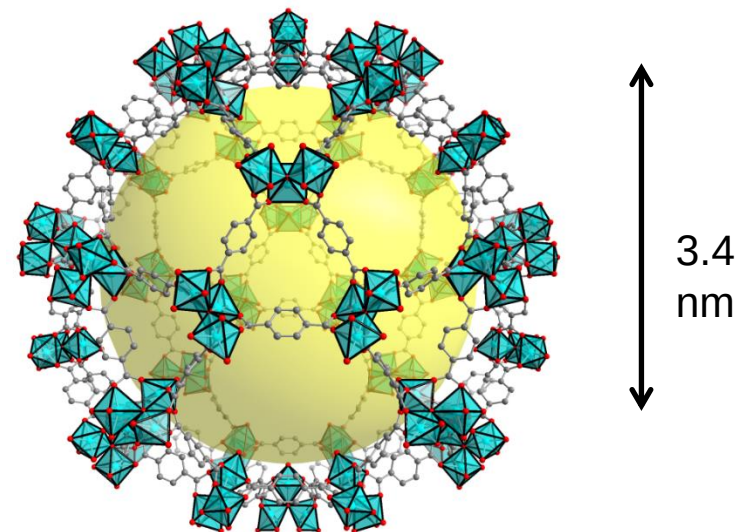
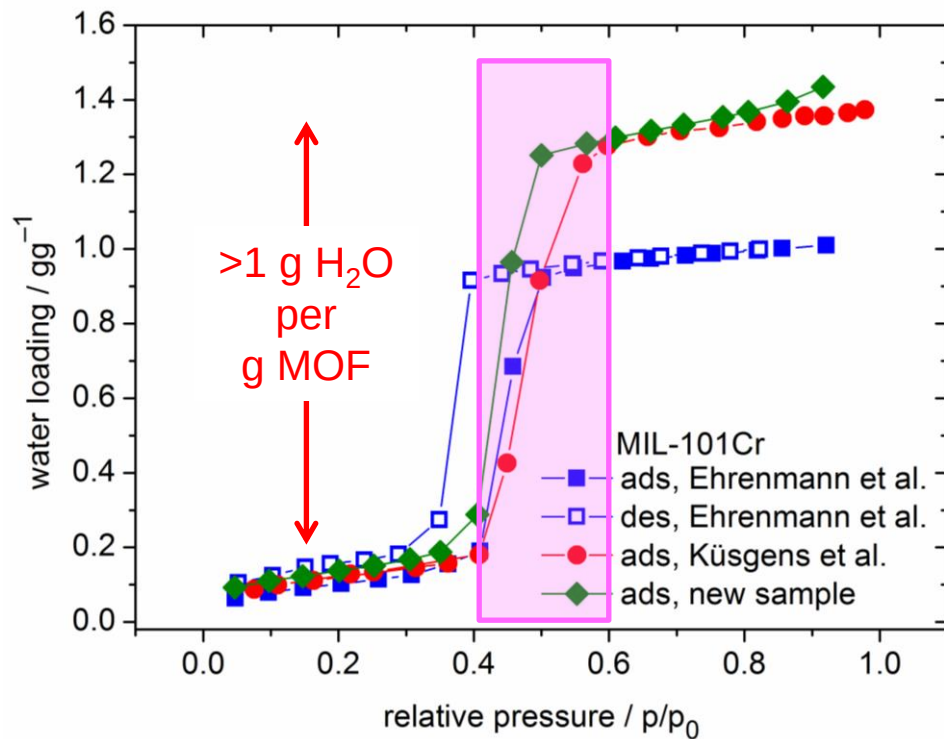
Adsorption chillers: Water loading lifts



MOF =
metal-organic
framework

Reviews: *Eur. J. Inorg. Chem.* **2012**, 2625-2634. DOI: 10.1002/ejic.201101056
Chimia **2013**, 67, 419-424. DOI: 10.2533/chimia.2013.419

Water loading lifts MIL-101 water vapor cycle stability



$[\text{Cr}_3\text{O}(\text{F},\text{OH})(\text{H}_2\text{O})_2(\text{bdc})_3] \cdot \sim 25\text{H}_2\text{O}$
inner surface up to $4000 \text{ m}^2/\text{g}$
(Férey et al., *Science* **2005**, 309, 2040.)

hydrothermal stress test over 40 cycles:

water uptake [g/g]:

after 1 cycle 0.97

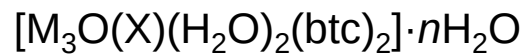
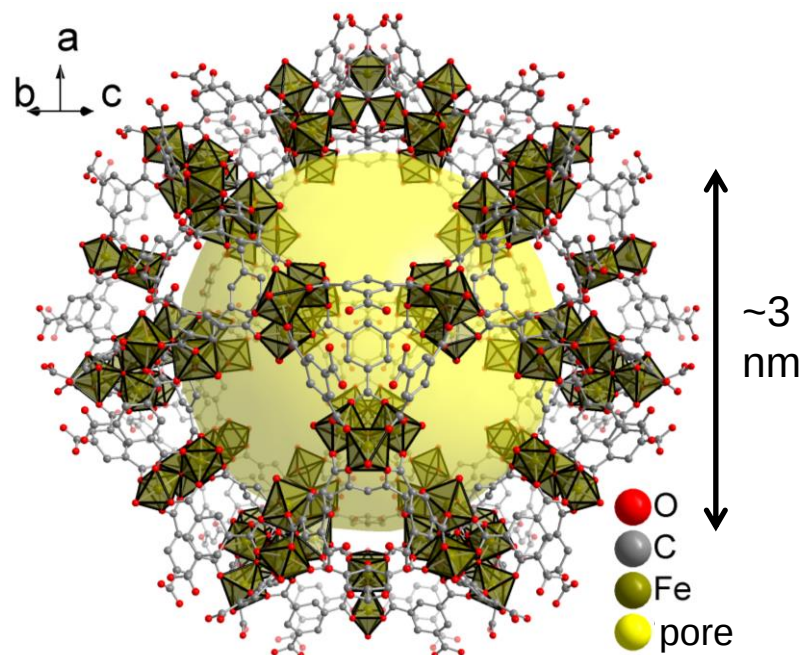
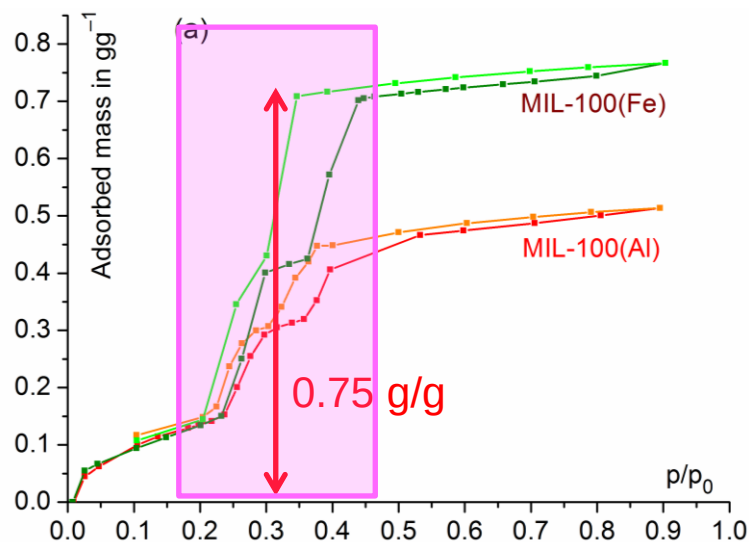
after 40 cycles 0.94

BET surface [m^2/g]:

untreated 2059

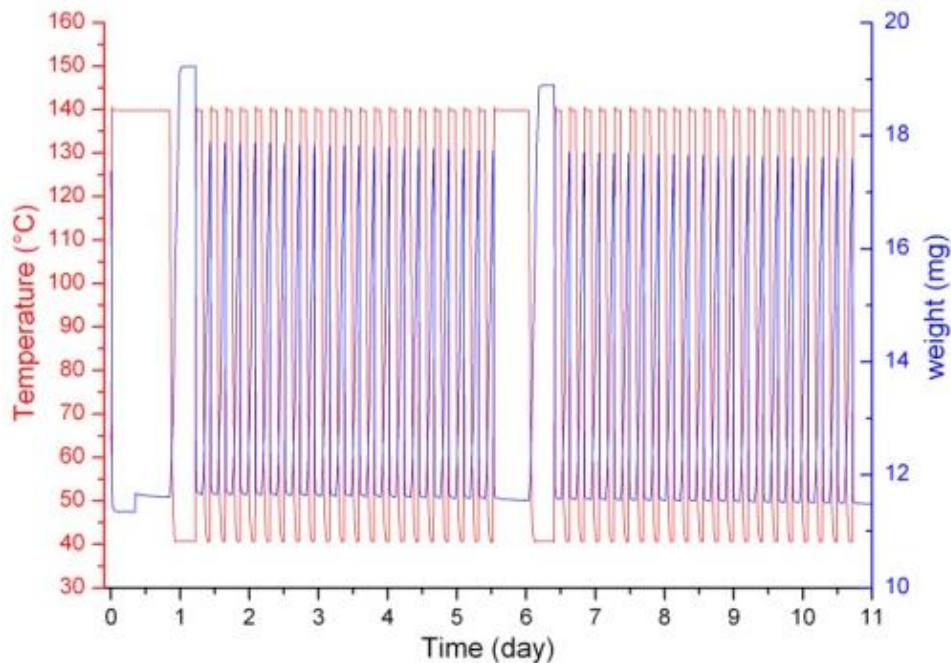
after 40 cycles 2047

Water stability, water loading lifts MIL-100(Fe and Al)



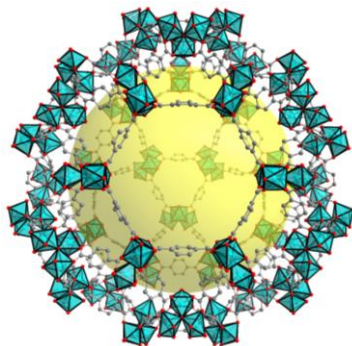
(Horcajada, Férey et al.
Chem. Commun. **2007**, 2820.)

S(BET): 1917 m² / g
after 40 cycles: 1891 m² / g

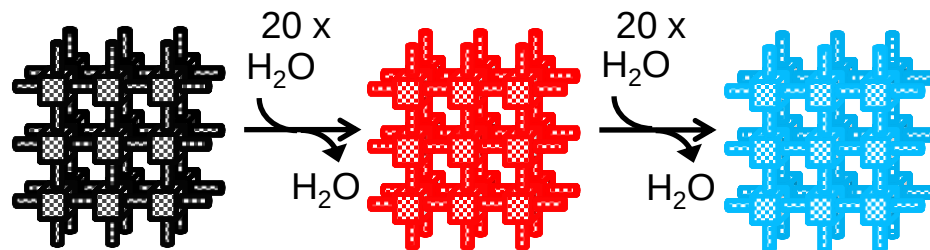


J. Mater. Chem. **2012**, 22, 10148.

MOFs: Water loading lifts and water vapor cycle stability = hydrothermal stress test



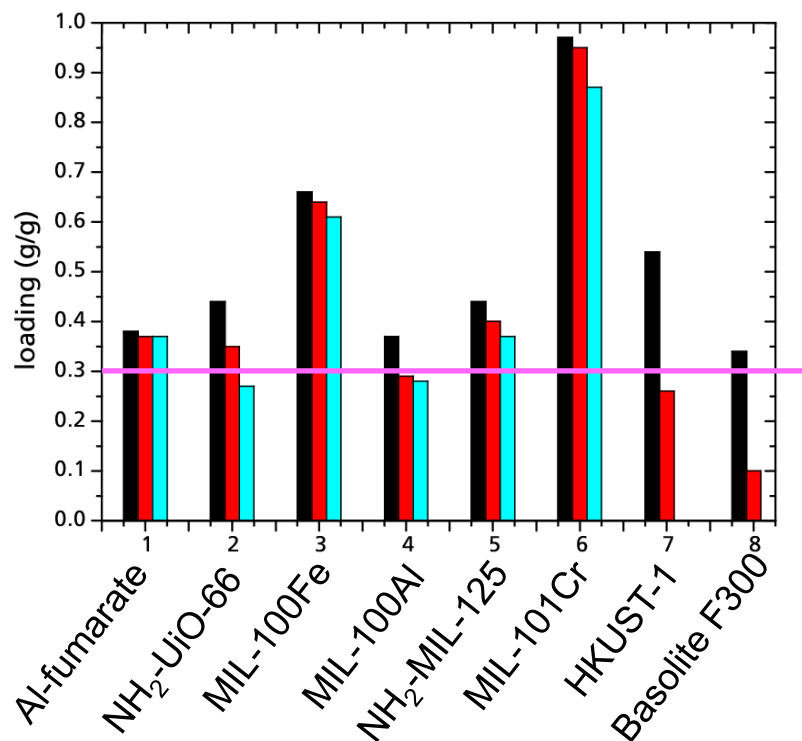
stability



Eur. J. Inorg. Chem. **2011**, 471.
J. Mater. Chem. **2012**, 22, 10148.
Chem. Mater. **2013**, 25, 790.
Dalton Trans. **2013**, 42, 15967.
RSC Advances **2014**, 4, 24073.
J. Mater. Chem. A **2016**, 4, 11859.

Reviews:

Eur. J. Inorg. Chem. **2012**, 2625.
Chimia **2013**, 67, 419.
New J. Chem. **2014**, 38, 1846.

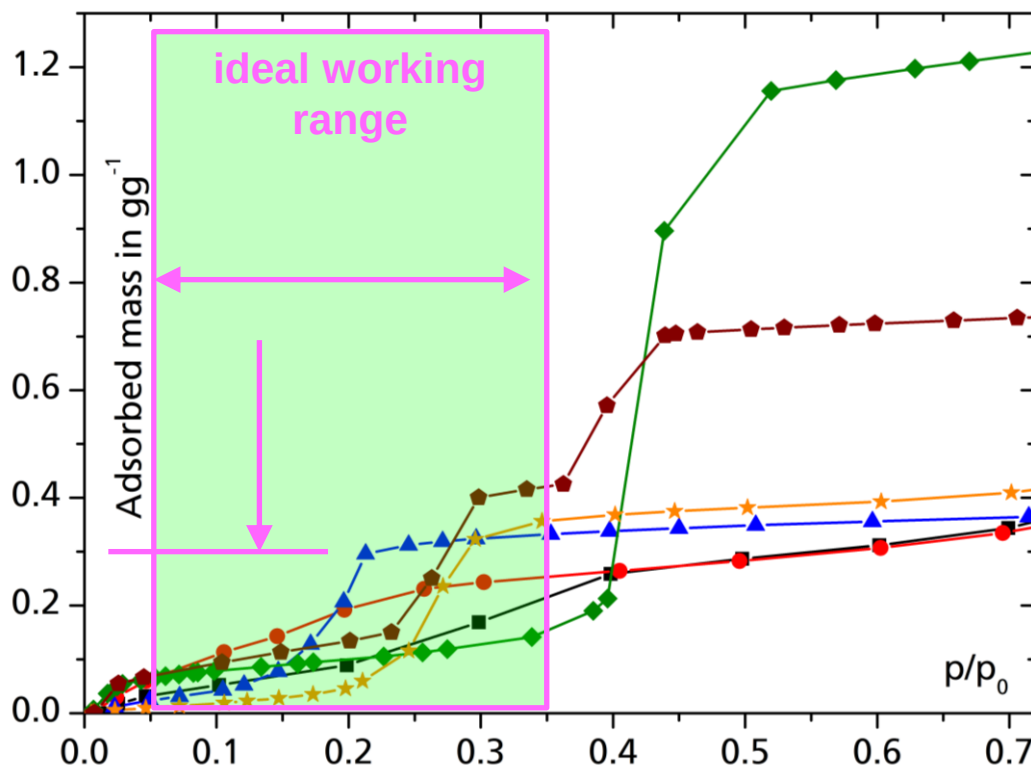


min.
lift
needed

Water loading

■ after activation, ■ after 20, ■ after 40
water vapor ad/desorption cycles.

Water loading increasing the hydrophilicity



water vapor adsorption isotherms
at 25 °C

increasing hydrophilicity

▲ NH₂-MIL-125 ● NH₂-UiO-66 ★ Al-fumarate ◆ MIL-100Fe ■ UiO-66 ◆ MIL-101Cr

Reviews:

Eur. J. Inorg. Chem. **2012**, 2625.

Chimia **2013**, 67, 419.

New J. Chem. **2014**, 38, 1846.

Eur. J. Inorg. Chem. **2011**, 471.

J. Mater. Chem. **2012**, 22, 10148.

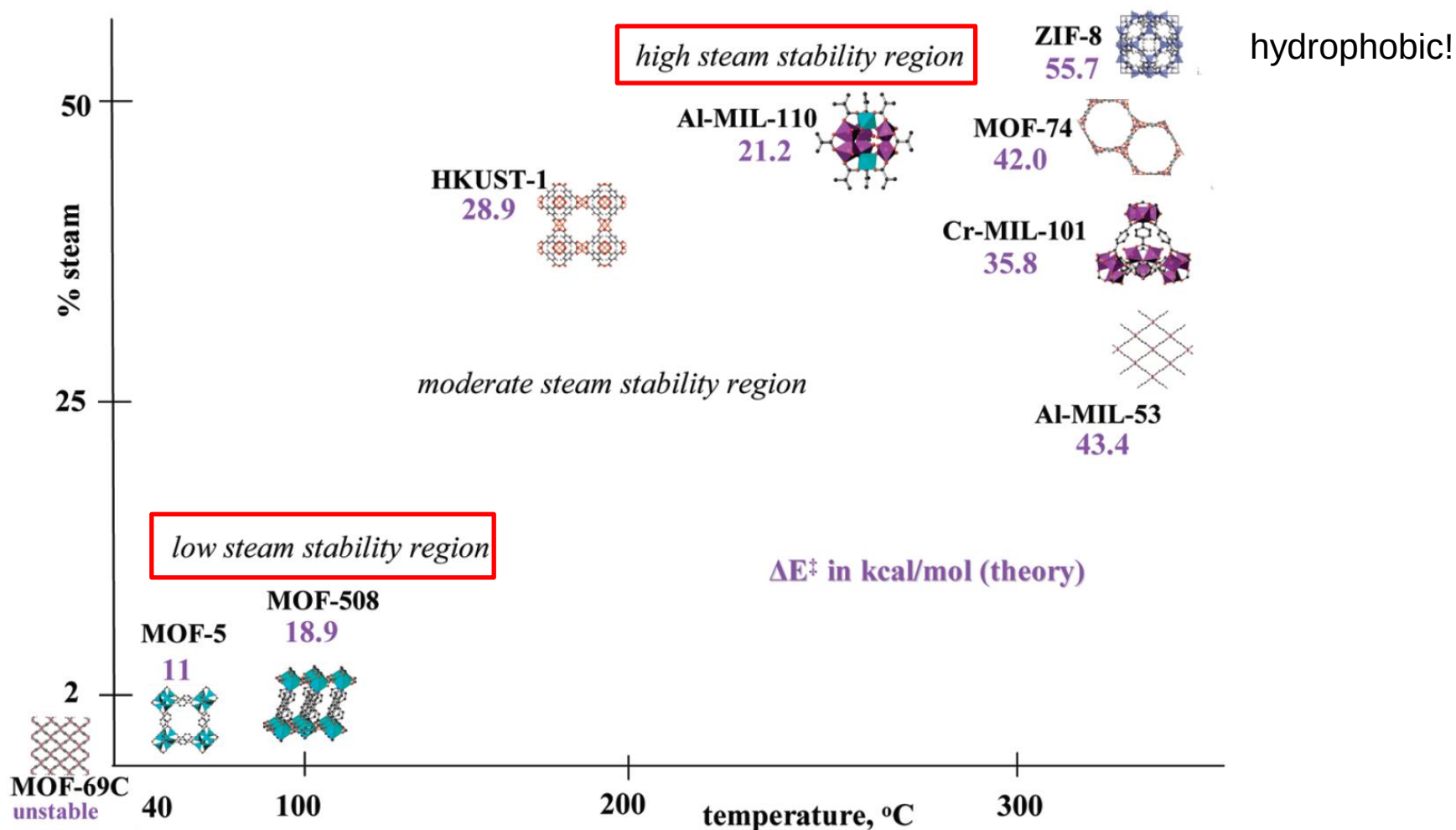
Dalton Trans. **2013**, 42, 15967.

RSC Advances **2014**, 4, 24073.

J. Mater. Chem. A **2016**, 4, 11859.

Water stability of MOFs

- an inherent problem (as water is almost everywhere)



Water stability of MOFs

- an inherent problem (as water is almost everywhere)

	3 days												2 months												Steam test		
	pH=0		pH=4		pH=12		H ₂ O ₂		H ₂ O		air		pH=0		pH=4		pH=12		H ₂ O ₂		H ₂ O		air				
	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	XRD	N ₂	
MIL-101 (Cr)																											
NH ₂ -MIL-101 (Al)																											
MIL-53 (Al)																											
NH ₂ -MIL-53		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP	
UiO-66																											
NH ₂ -UiO-66																											
UiO-67																											
CuBTC																											
ZIF-8																											

Powder X-ray diffractogram
and surface area, S_{BET}

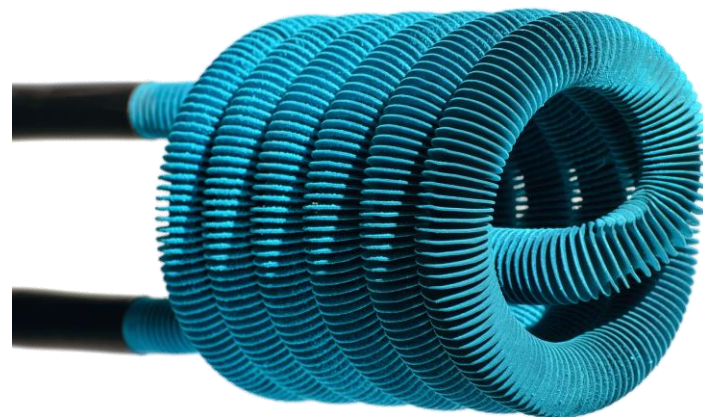
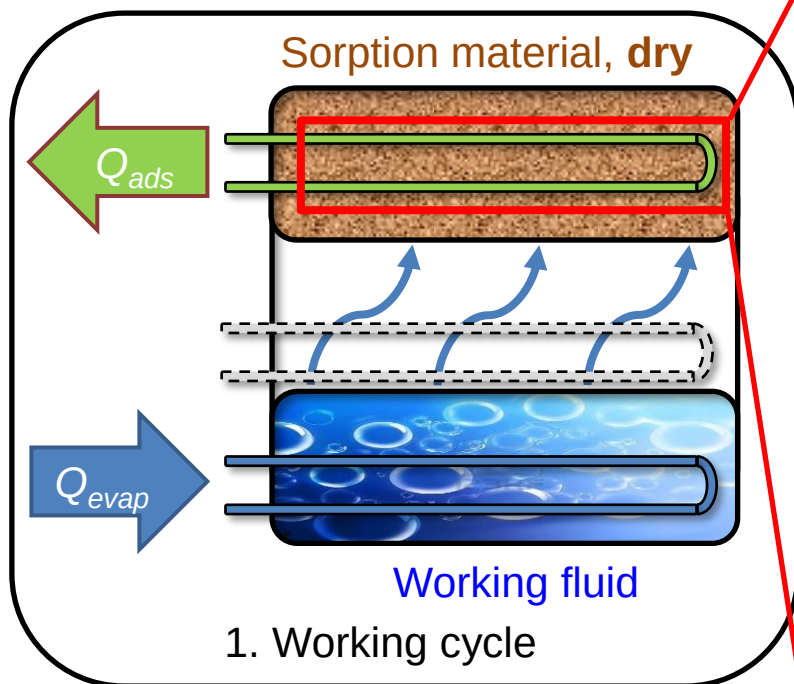
 completely
 partially
destroyed

K. Leus, T. Bogaerts, J. De Decker, H. Depauw,
K. Hendrickx, H. Vrielinck, V. Van Speybroeck,
P. Van Der Voort,
Micropor. Mesopor. Mater. **2016**, 226, 110-116

Principle process in an adsorption chiller / heat pump

Aim: MOF as coating on metal for heat transfer

MOF-coated heat exchangers



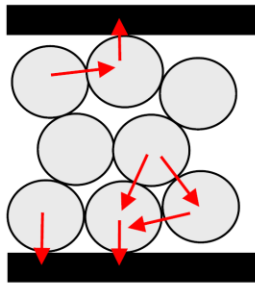
© Fraunhofer ISE



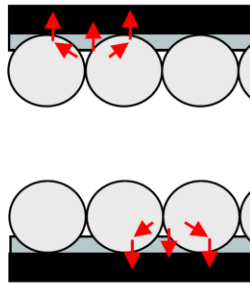
Chem. Commun. **2012**, 48, 9708-9710,
DOI: 10.1039/C2CC34782B..
RSC Advances **2014**, 4, 24073-24082,
DOI: 10.1039/c4ra03794d.
J. Mater. Chem. A **2016**, 4, 11859-11869,
DOI: 10.1039/C6TA01757F.

MOF shaping for handling and good heat transfer kinetics

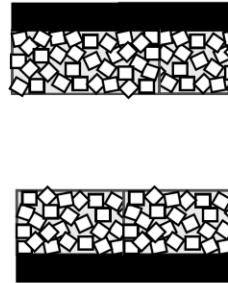
possible shapings for metal contact for heat transfer



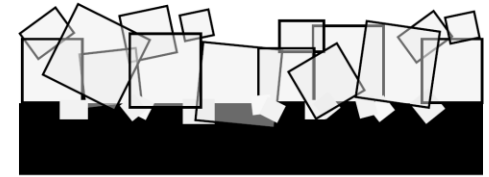
fixed bed



fixed monolayer



binder coating

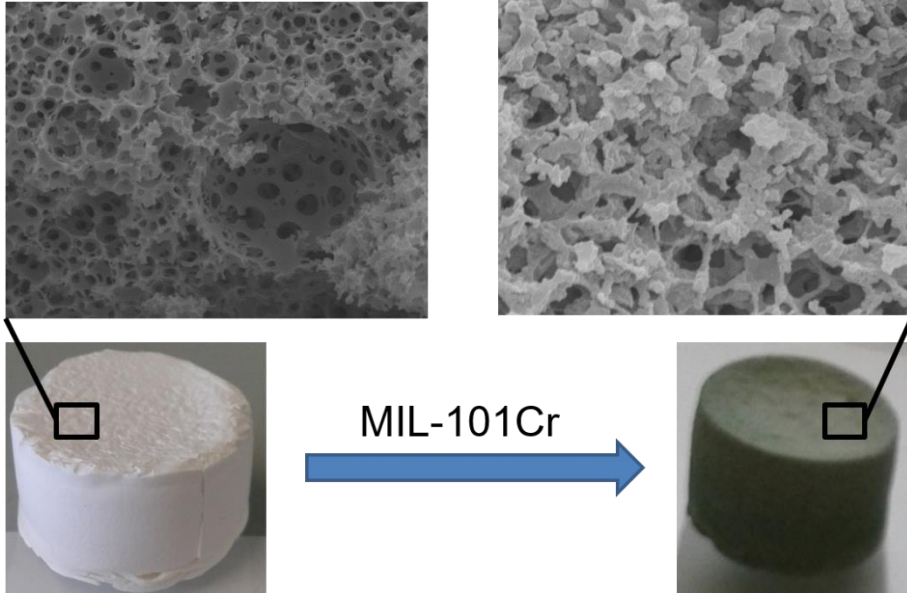


partial support transformation

increasing heat transfer
increasing adsorption rate
increasing cooling power

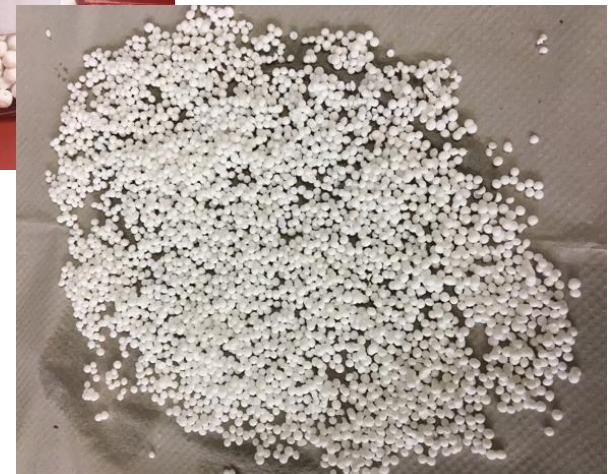
MOF formulation – formulating MOFs for devices

- as monoliths:



MIL-101(Cr)@poly(HEMA)HIPE

- as grains



Micropor. Mesopor. Mater. **2015**, 204, 242–250.

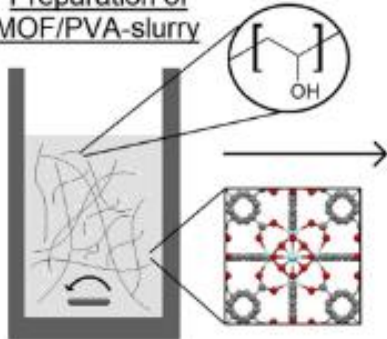
Micropor. Mesopor. Mater. **2015**, 215, 143–153.

Micropor. Mesopor. Mater. **2016**, 220, 258–269.

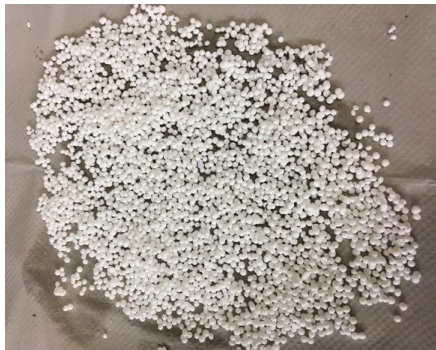
Micropor. Mesopor. Mater. **2019**, 280, 277–287.

MOF formulation – formulating MOFs in grains/pellets

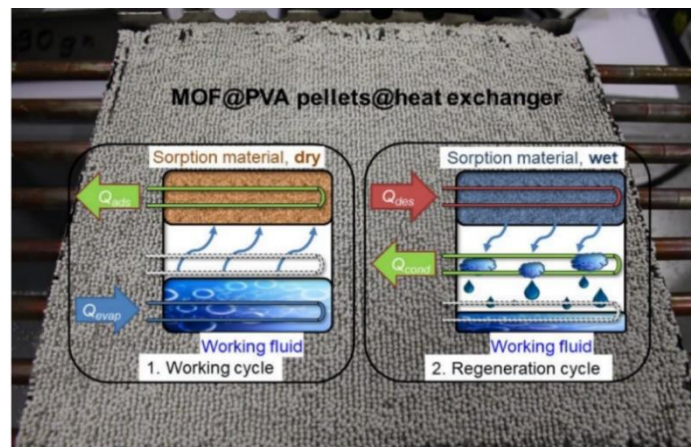
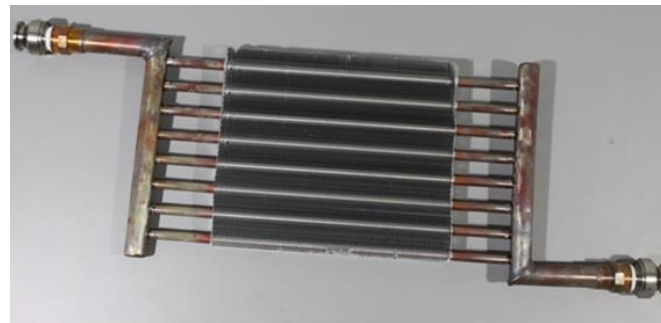
Preparation of
MOF/PVA-slurry



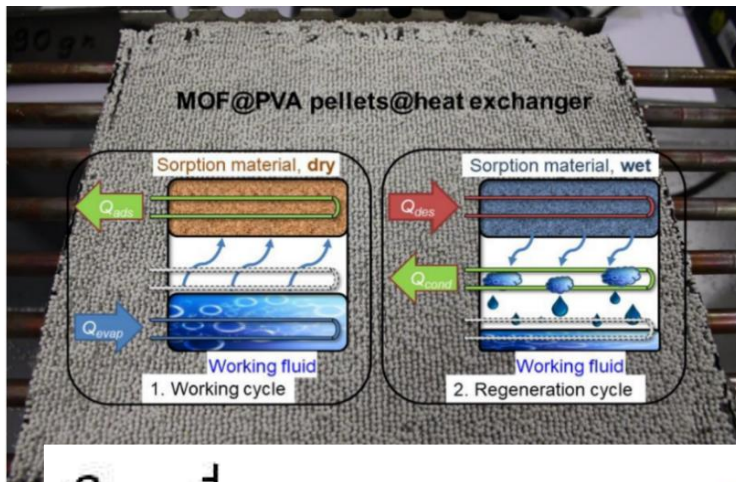
Pelletizing



+

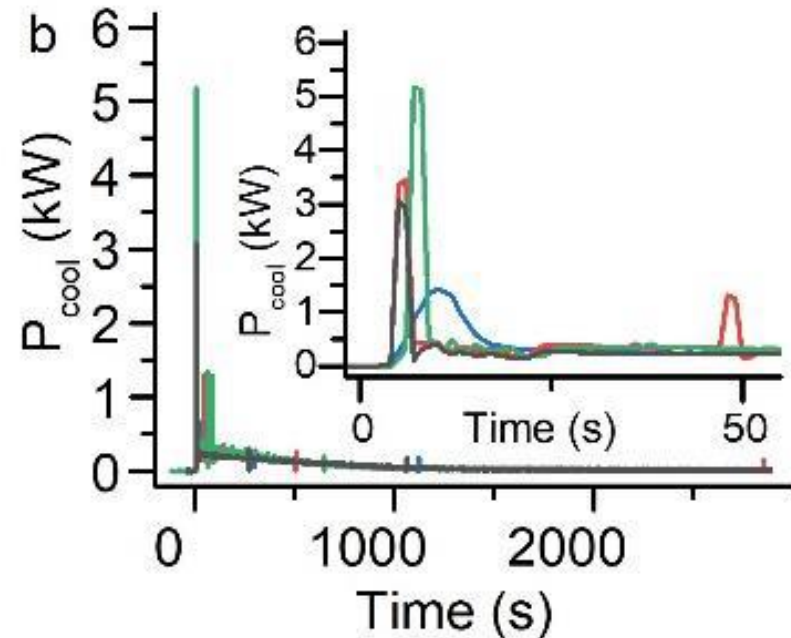
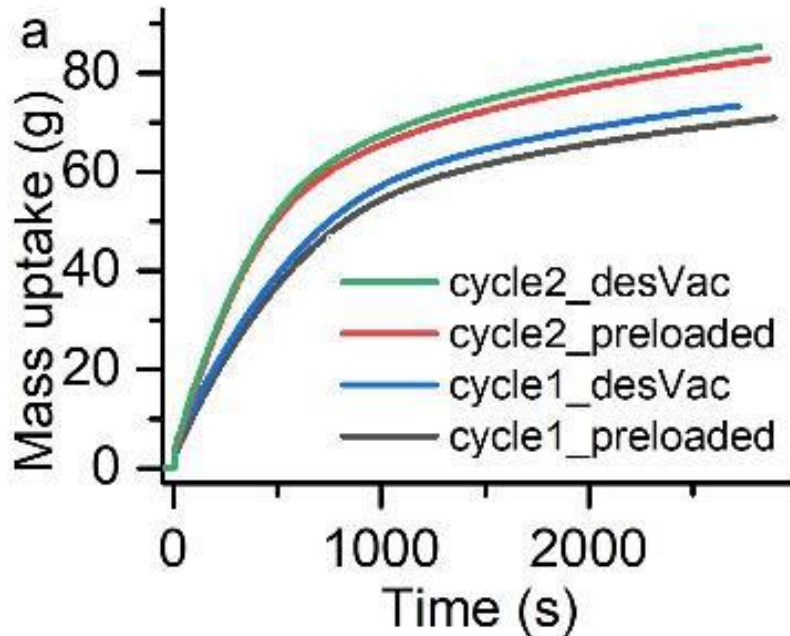


MOF formulation – formulating MOFs in grains/pellets



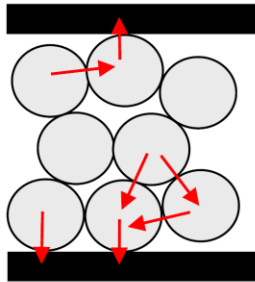
specific cooling powers (SCP)
from 349 up to 431 W/kg(adsorbent)

(better than silica gel grains in AHPs)

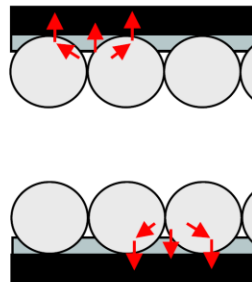


MOF shaping for handling and good heat transfer kinetics

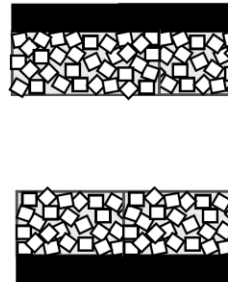
possible shapings for metal contact for heat transfer



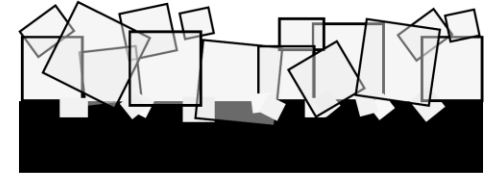
fixed bed



fixed monolayer



binder coating

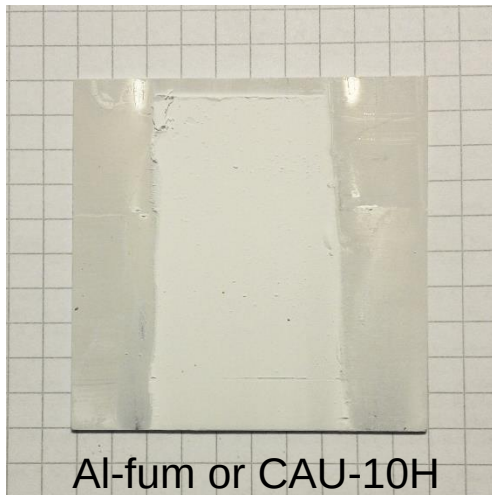


partial support transformation

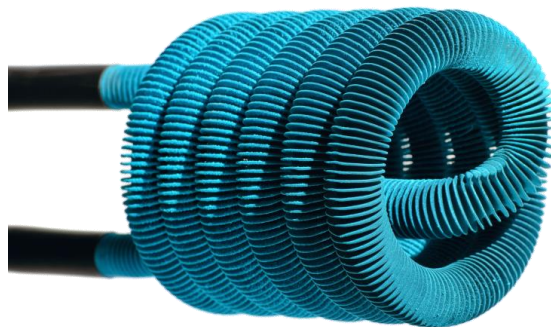
increasing heat transfer
increasing adsorption rate
increasing cooling power

MOF formulation – formulating MOFs for devices

- as surface coatings:

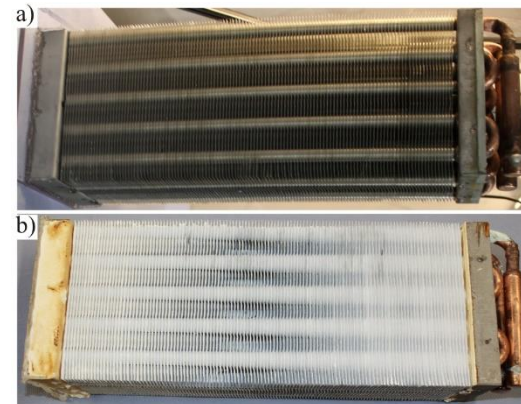


Al-fum or CAU-10H
@Al plate (5x5 cm)

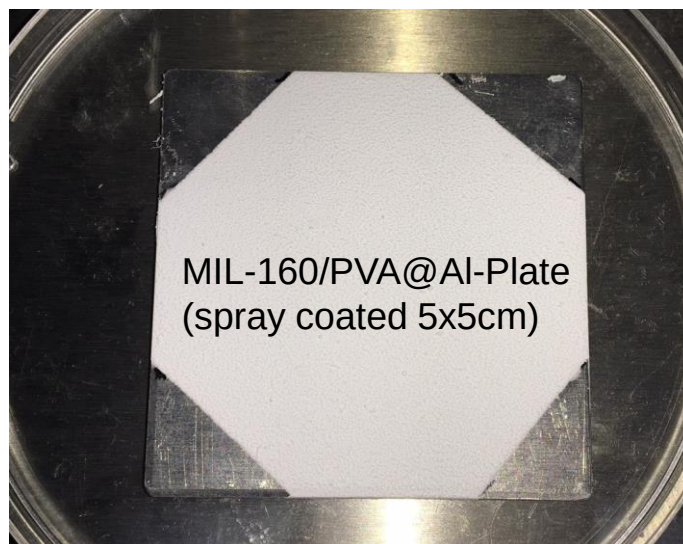


© Fraunhofer ISE

HKUST-1@heat exchanger



Al-fum@heat exchanger

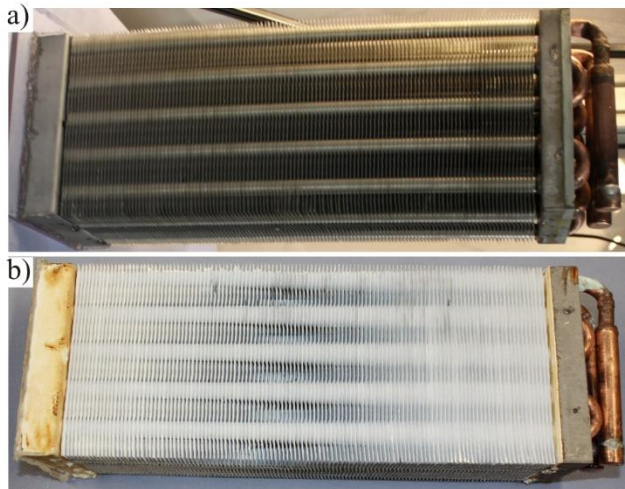


MIL-160/PVA@Al-Plate
(spray coated 5x5cm)

D. Fröhlich, E. Pantatosaki, P. D. Kolokathis, K. Markey, H. Reinsch, M. Baumgartner, M. A. van der Veen, D. E. De Vos, N. Stock, G. K. Papadopoulos, S. K. Henninger, C. Janiak, *J. Mater. Chem. A* **2016**, 4, 11859-11869.

H. Kummer, F. Jeremias, A. Warlo, G. Földner, D. Fröhlich, C. Janiak, R. Gläser, S. K. Henninger, *Ind. Eng. Chem. Res.* **2017**, 56, 8393-8398.

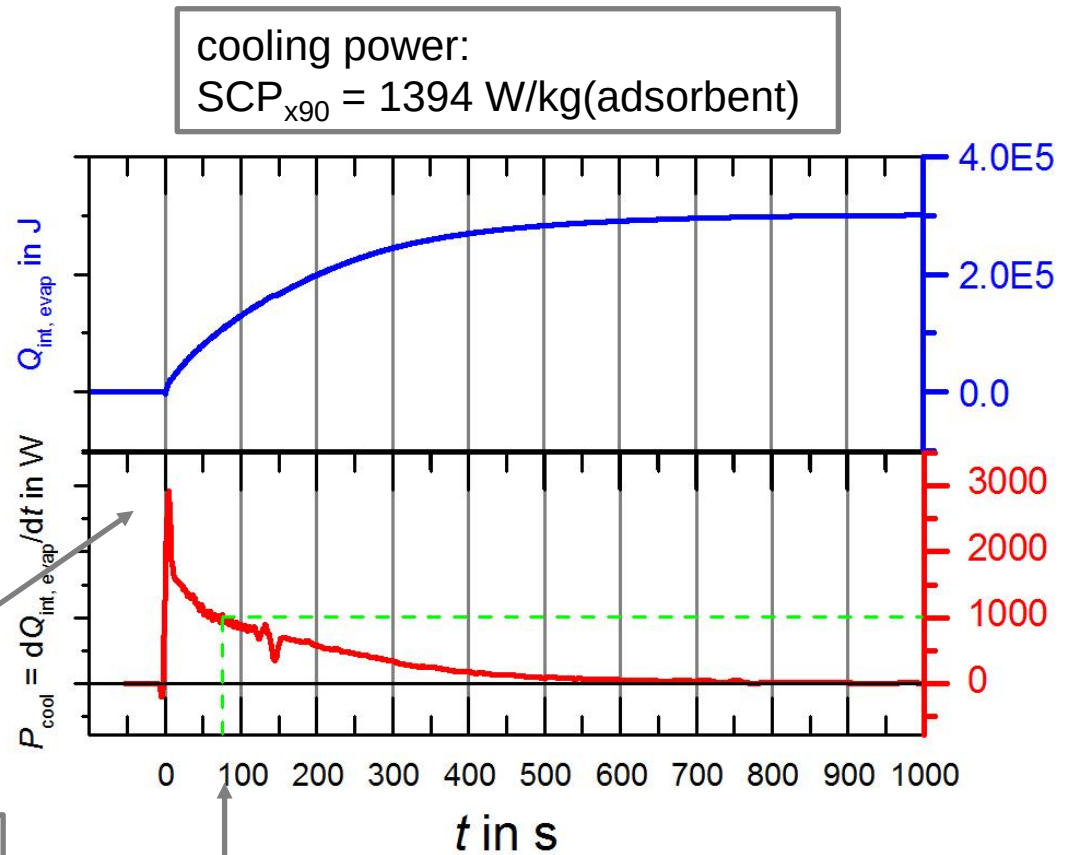
large scale MOF coating on metal for heat transfer



493 g Al-fum + binder in coated HX

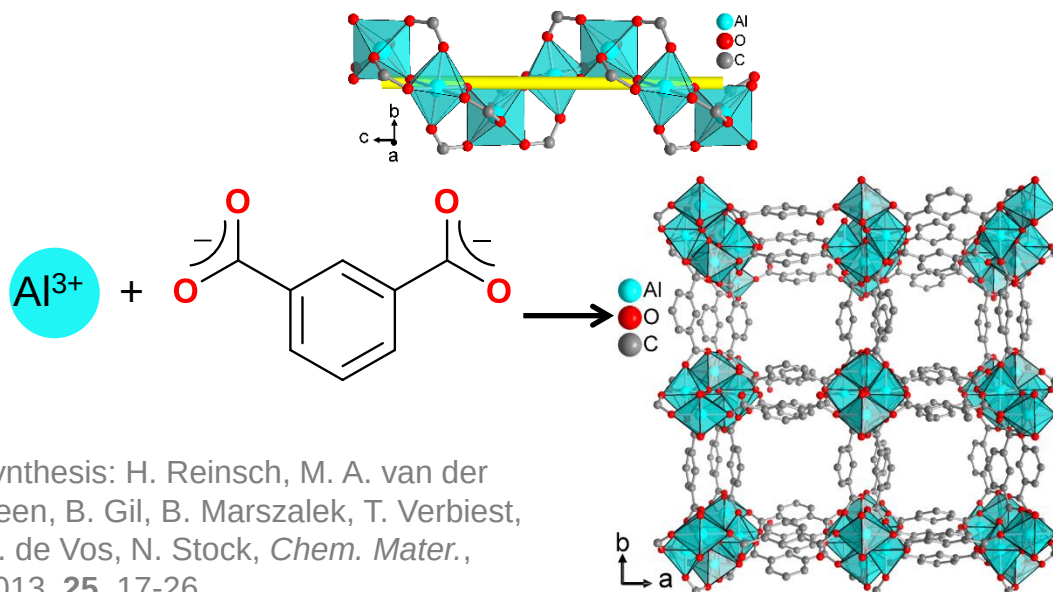
(binder = silicophen polymer)

$P_{\text{Cool}} = 2900 \text{ W}$ at the beginning of the working cycle

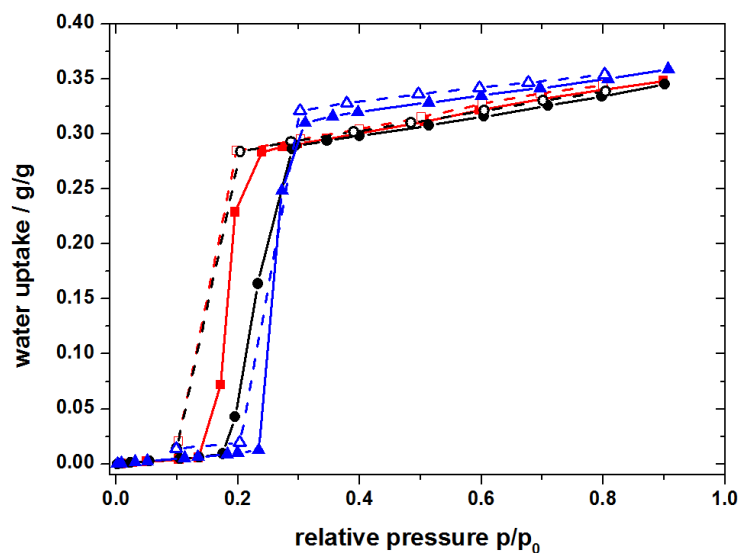
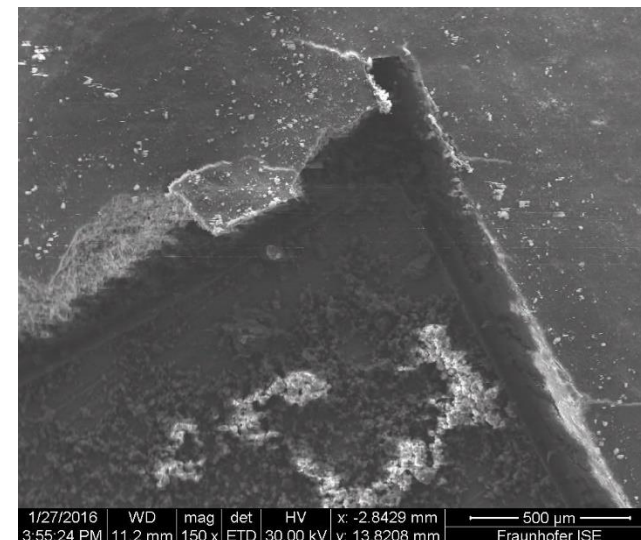


$P_{\text{Cool}} = 1000 \text{ W}$ for a half cycle time of $t = 74 \text{ s}$

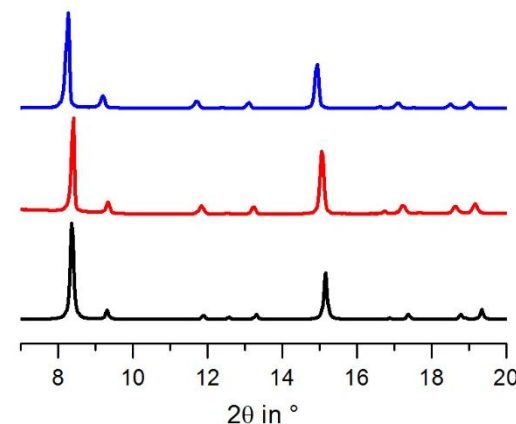
CAU-10-H MOF as coating on metal for heat transfer



CAU-10-H MOF on Al-metal plate:

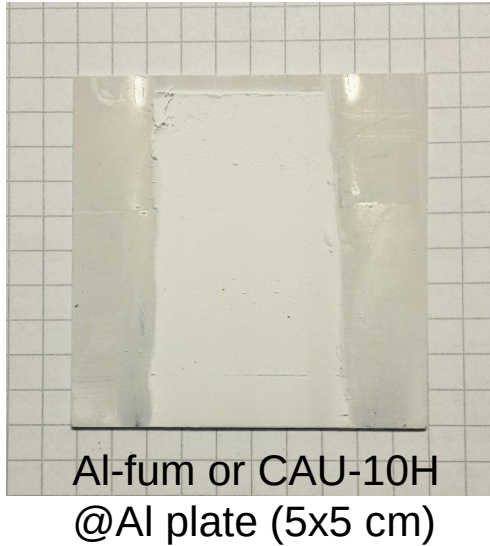


PXRD
- after 10000
H₂O vapor
sorption cycles
- before
- simulated
- no uptake loss



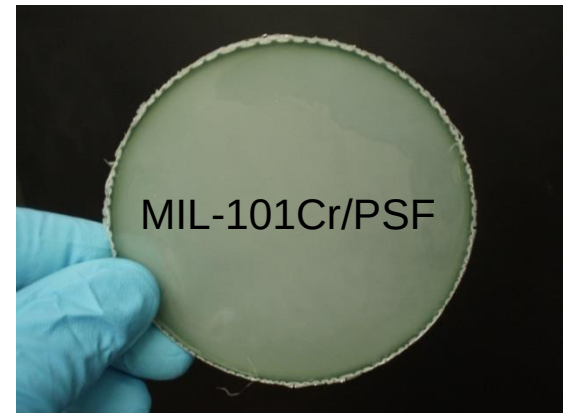
MOF formulation – formulating MOFs with polymer binders for devices

- as surface coatings:



Al-foam@heat exchanger

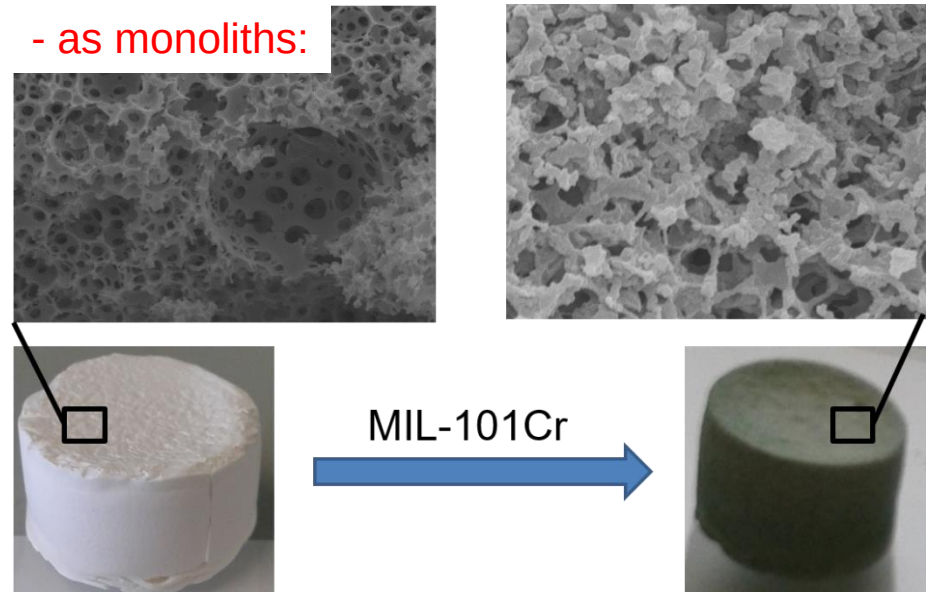
- as membranes:



- as grains:

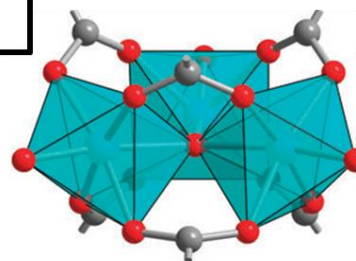
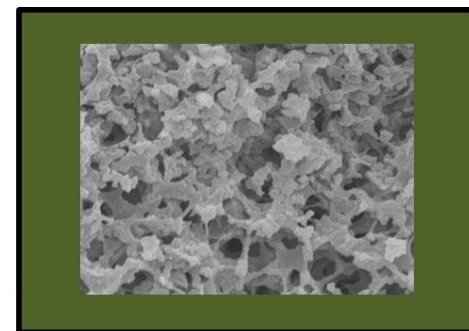
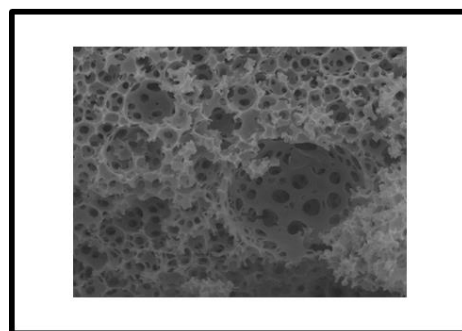
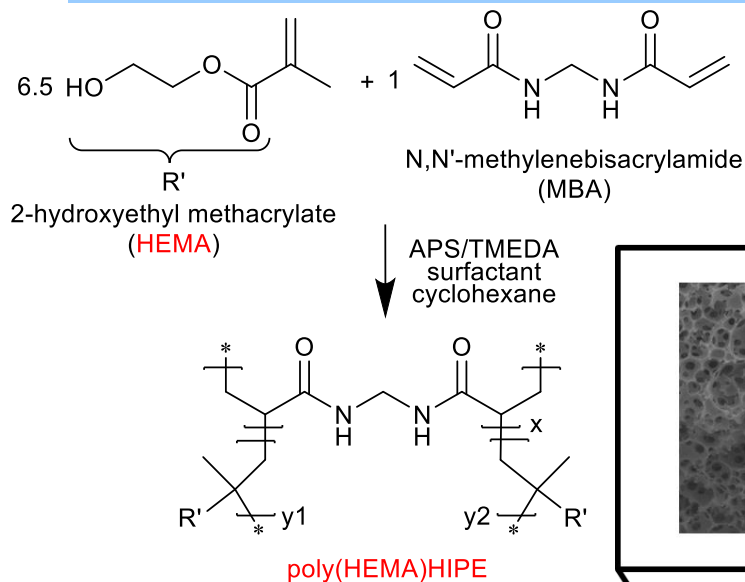


- as monoliths:

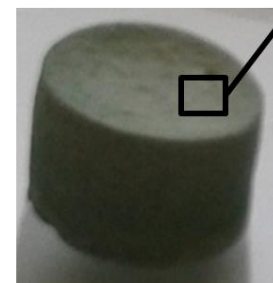
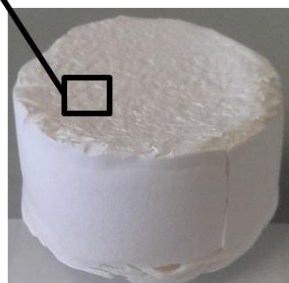


MOF formulation/monoliths with polymer-composites:

MIL-101(Cr)@poly(HEMA)HIPE

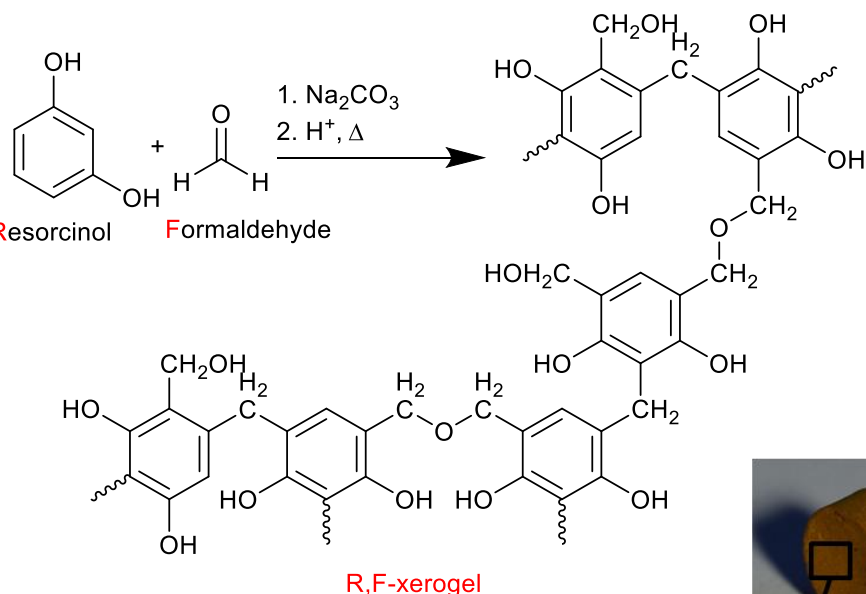


MIL-101(Cr)

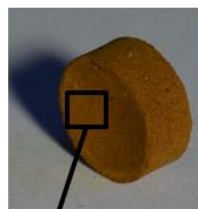


MOF formulation/monoliths with polymer-composites:

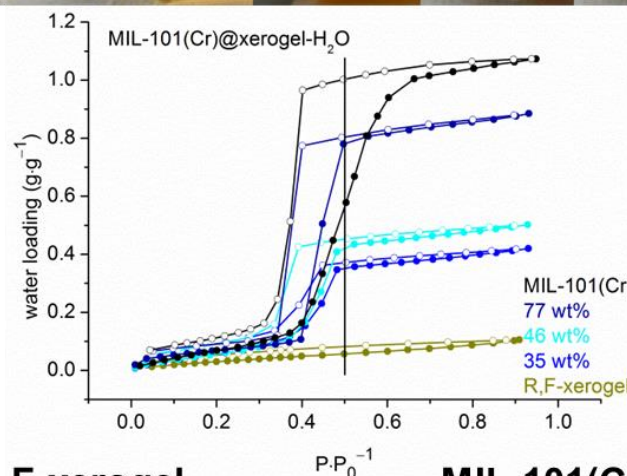
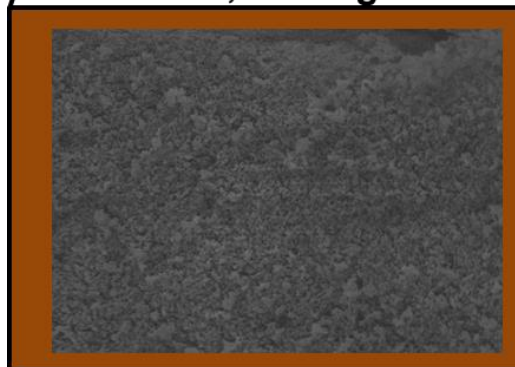
MIL-101(Cr)@R,F-xerogel



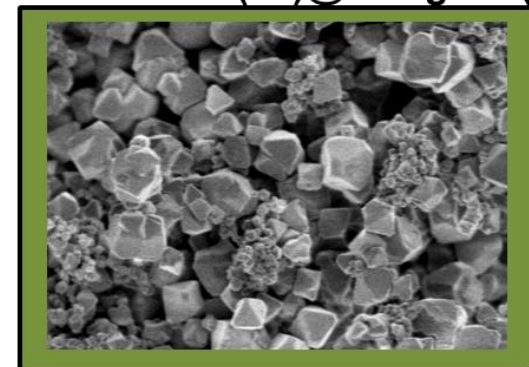
Heat induced pre-polymerization



Native R,F-xerogel



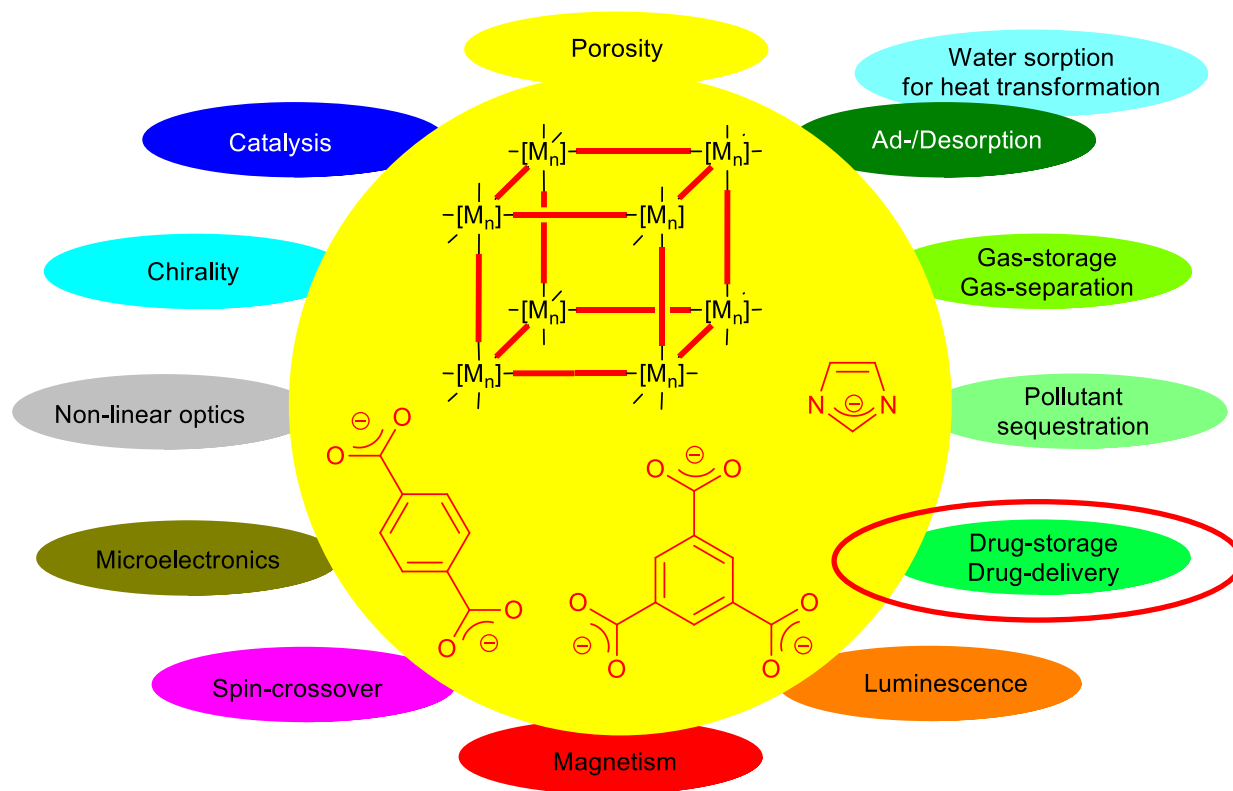
MIL-101(Cr)@xerogel



Micropor. Mesopor. Mater. **2015**,
215, 143-153.
DOI:
10.1016/j.micromeso.2015.05.017

Micropor. Mesopor. Mater. **2016**,
220, 258-269.
DOI:
10.1016/j.micromeso.2015.09.008

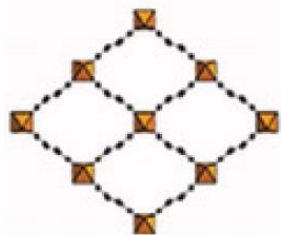
Metal-Organic Frameworks (MOFs): Application-oriented properties



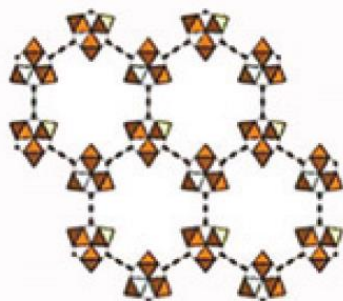
Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E.
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

Metal-organic networks: Drug-storage / drug-delivery

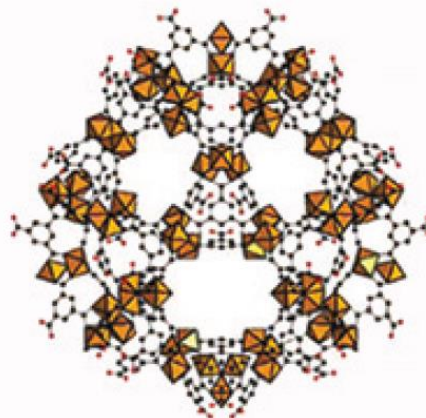
- Biodegradable porous iron carboxylates



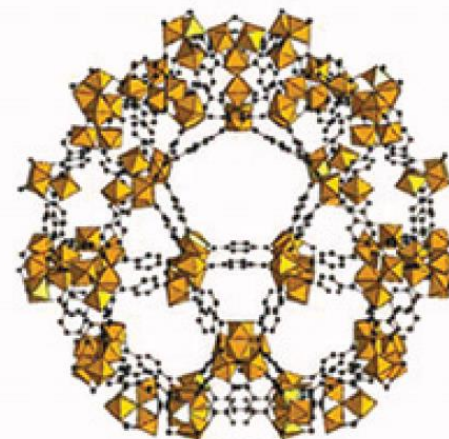
MIL-53
8 Å



MIL-88
6-11 Å

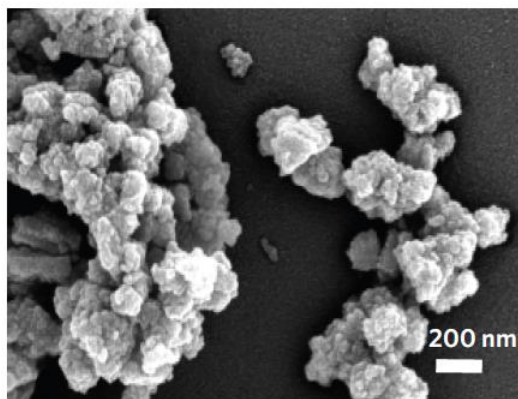


MIL-100
24-29 Å

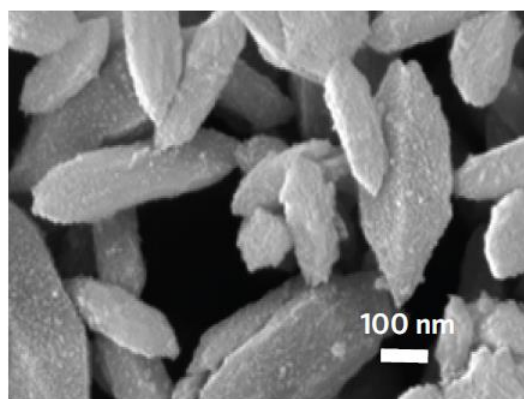


MIL-101
29-34 Å

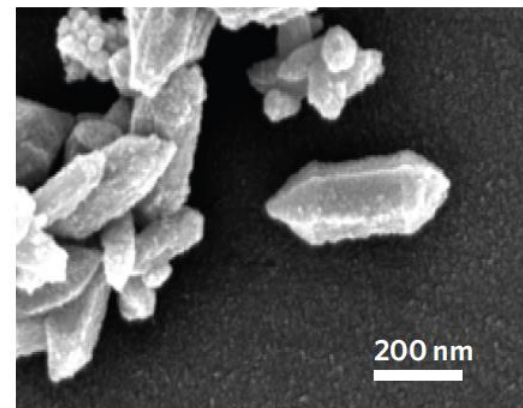
- non-toxic iron MOFs MIL-53, MIL-88A, MIL 88Bt, MIL-89, MIL-100 and MIL-101-NH₂;
- synthesis conditions adapted to obtain these materials as nanoparticles



MIL-100



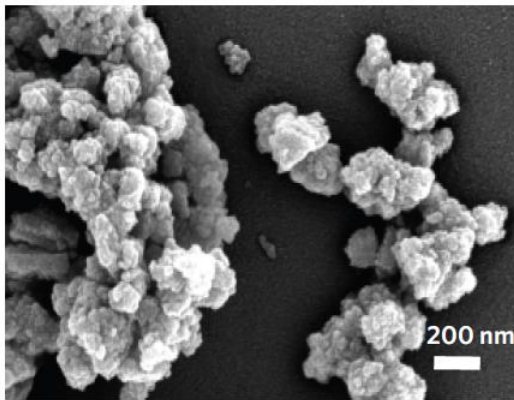
MIL-88A



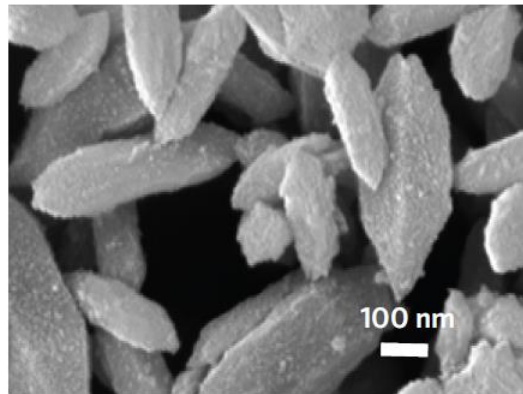
MIL-88A-PEG

Metal-organic networks: Drug-storage / drug-delivery

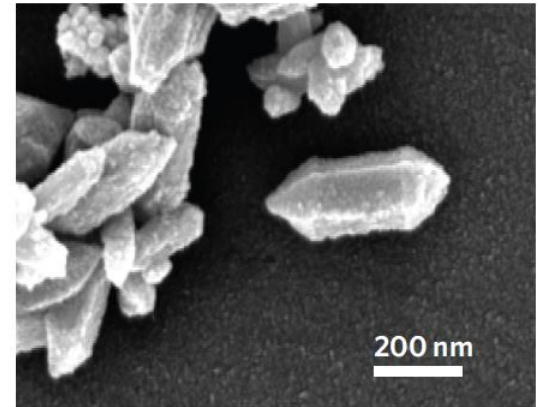
- non-toxic iron(III) carboxylate MOFs (MIL-53, MIL-88A, MIL 88Bt, MIL-89, MIL-100 and MIL-101-NH₂);
- synthesis conditions adapted to obtain these materials as nanoparticles



MIL-100



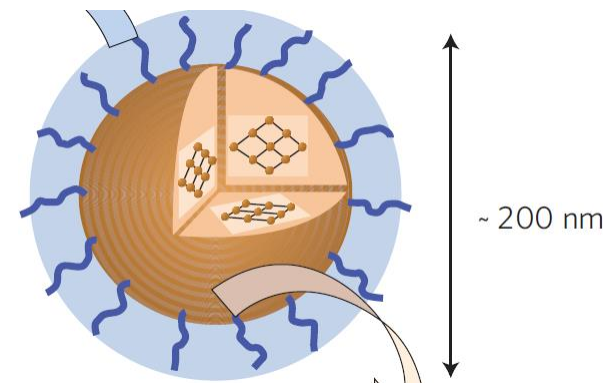
MIL-88A



MIL-88A-PEG

For biological applications, the nanoMOF surfaces were engineered by coating with several relevant polymers; this treatment prevented aggregation of the nanoparticles.

To **control crystal growth, poly(ethyleneglycol) PEG chains** with only one terminal reactive group (amino or carboxyl) were added during the course of the synthesis process. Thus, PEG led to the formation of a surface PEG `brush' sterically protecting the nanoparticles from aggregation.

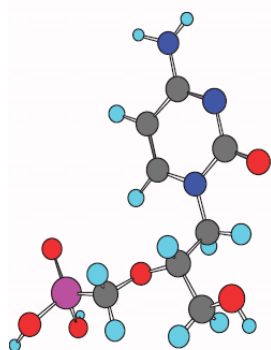


Metal-organic networks: Drug-storage / drug-delivery

efficiency as drug carriers tested with anticancer or antiviral drugs: busulfan, azidothymidine triphosphate, cidofovir and doxorubicin, which, except the latter, could not be successfully entrapped using existing nanocarriers (Table 1).

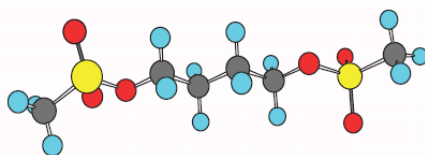
Some cosmetic molecules, such as caffeine (liporeductor), urea (hydrating agent), benzophenone 3 and benzophenone 4 (UVA and UVB filters) were also tested.

- Controlled release of challenging drugs



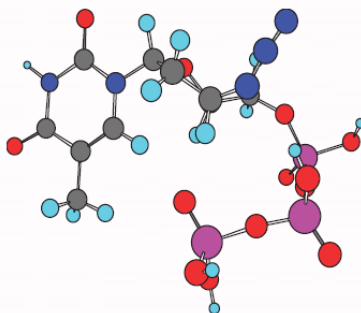
Cidofovir

CDV



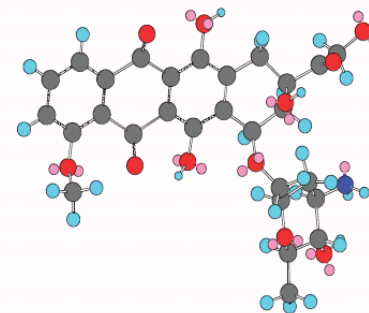
Busulfan

Bu



Azidothymidine
triphosphate

AZT-TP

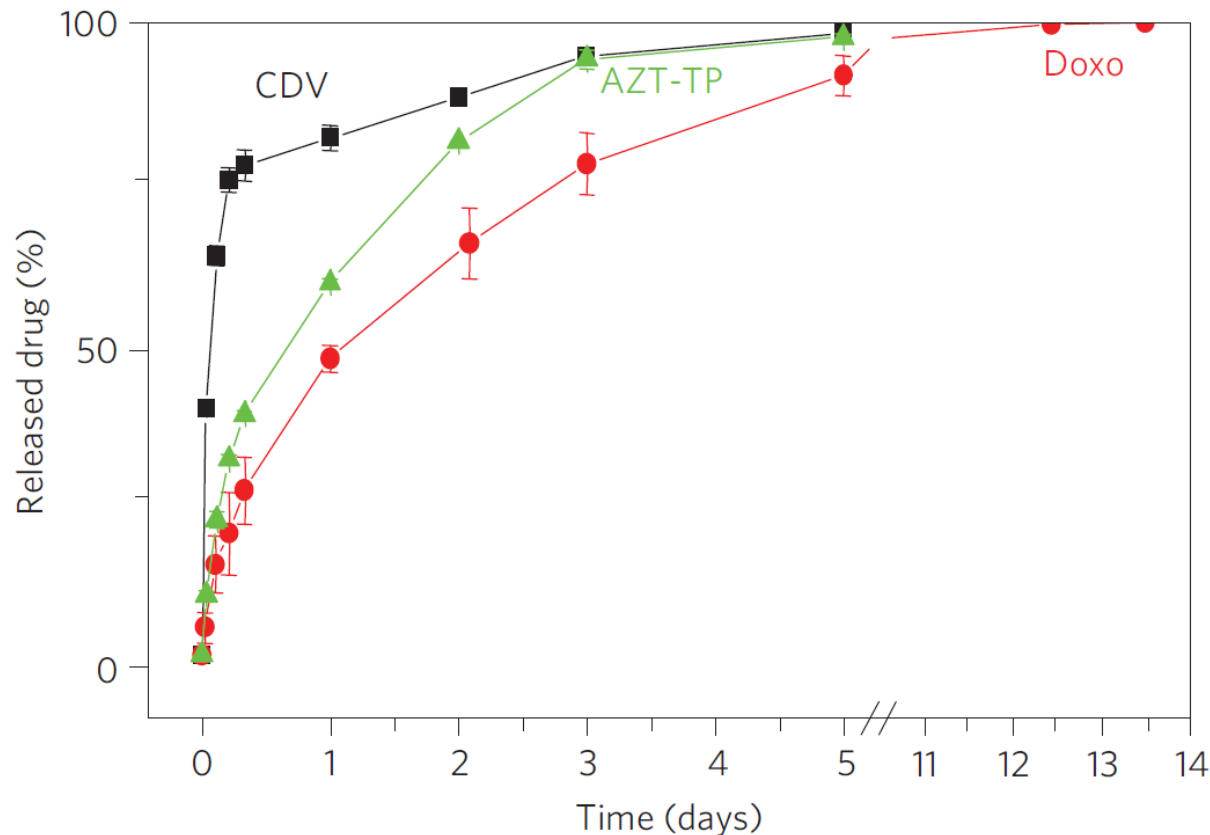


Doxorubicin

doxo

Metal-organic networks: Drug-storage / drug-delivery

A **progressive release** of the three active molecules (AZT-TP, CDV and doxo) is observed using MIL-100 nanoparticles , with **no 'burst effect'**.



CDV (black), doxo (red) and AZT-TP (green) delivery under simulated physiological conditions (37 °C) from MIL-100(Fe) nanoparticles. All experiments were carried out in quadruplicate.

Metal-organic networks: Drug-storage / drug-delivery

MOFgen – Nanoporous Materials for Wound Healing

Professor Russell Morris, University of St Andrews, Brian Mercer Award for Innovation 2012 and Industry Fellow 2011 – 14



Introduction

Chronic wound healing is a major issue in current healthcare. Wounds that do not heal are often infected with biofilms, a particularly difficult form of bacterial infection that are extremely resistant to antibiotics. We are developing a new wound healing technology based on the delivery of a powerful anti-biofilm agent, nitric oxide (NO), using a new type of nanoporous material called metal-organic frameworks (MOFs). MOFs are cheap to prepare and formulate (compared to many other advanced wound healing products) and can be used to precisely control the delivery of NO leading to highly effective anti-biofilm agents and a new generation of wound healing products.

Nitric Oxide – A Powerful Wound Healing Agent

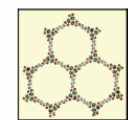
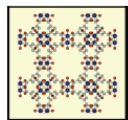
60% of chronic wounds contain biofilms
Antibiotics are often ineffective against biofilms

Nitric Oxide



An Anti-Biofilm Agent
*Killing bacteria
antibiotics cannot*

A Vasodilator
*Improving blood
supply*



A proven wound healing agent...

...but as a toxic gas it is difficult to administer safely

The solution: Use a new type of nanoporous solid, Metal-Organic Frameworks (MOFs) to store and deliver the gas.

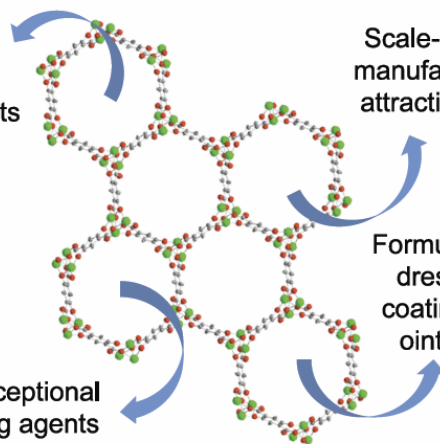
MOFs – A New generation of Nanoporous Delivery Agents

Powerful anti-bacterial and antifungal agents

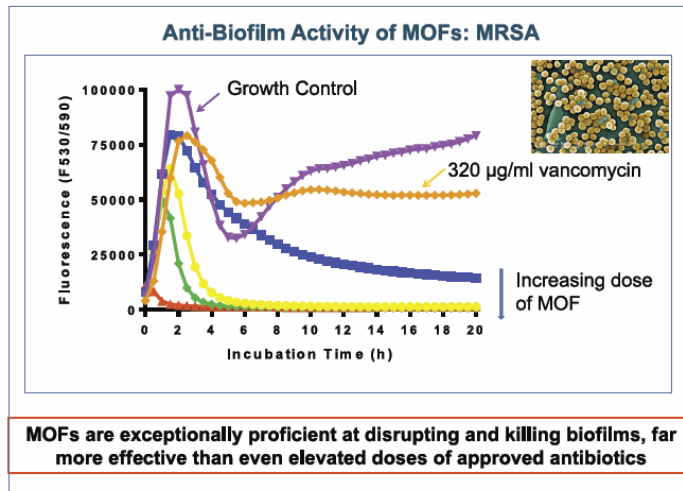
Scale-up and manufacture at attractive cost

Formulated as dressings, coatings and ointments

Delivering exceptional wound healing agents like nitric oxide



MOFgen Ltd is a University of St Andrews spin-out company that aims to develop MOF-based technology for several different applications. We develop the manufacture, formulation and applications of MOFs in healthcare and consumer applications.

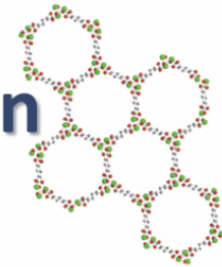


University of
St Andrews

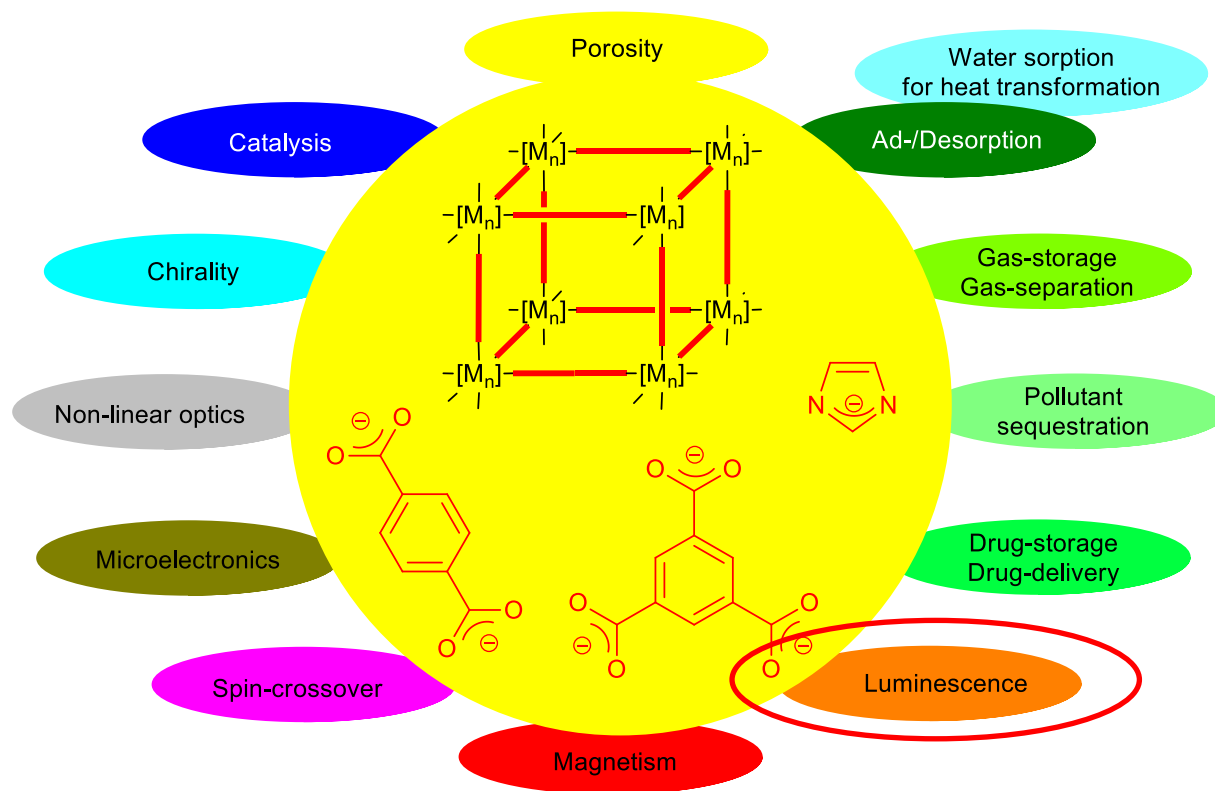
600
YEARS

MOFgen

*engineering
performance
materials*

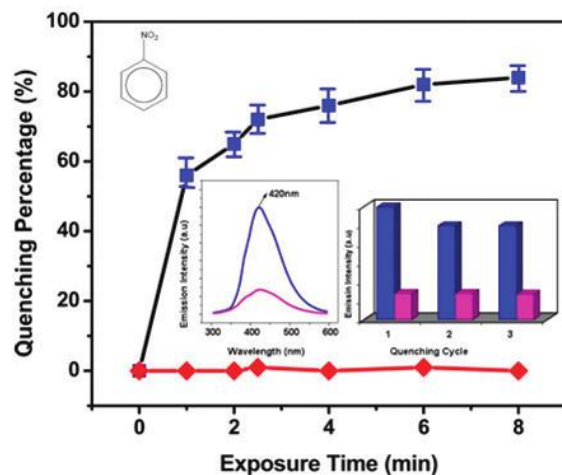


Metal-Organic Frameworks (MOFs): Application-oriented properties

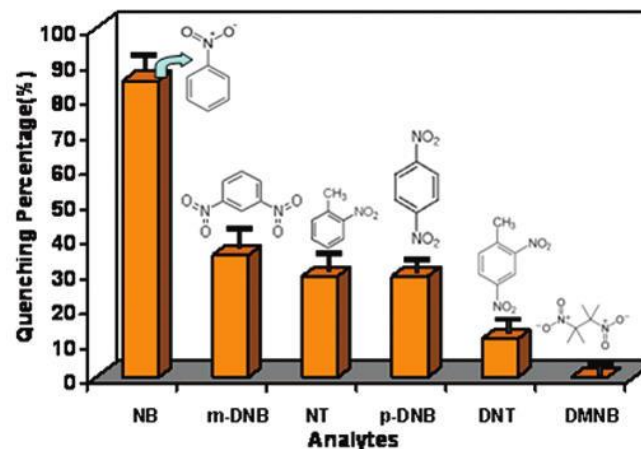


Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E.
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

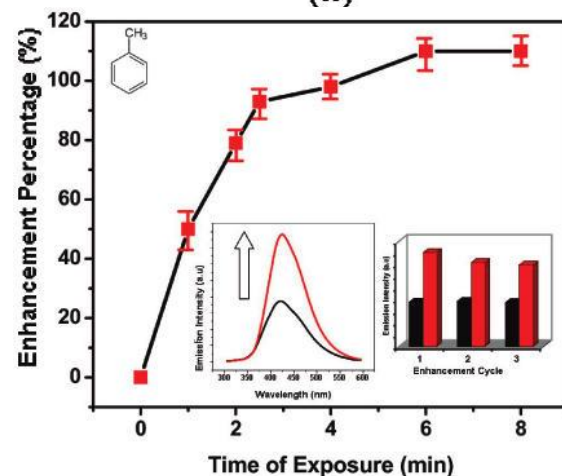
Fluorescence in MOFs - fluorescence quenching or enhancement for sensing



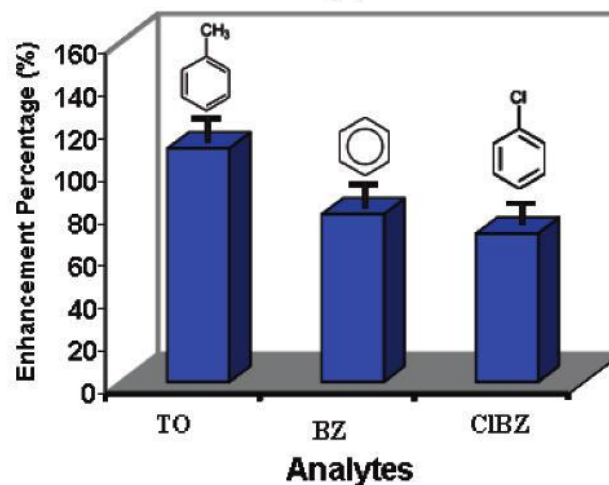
(a)



(b)



(a)

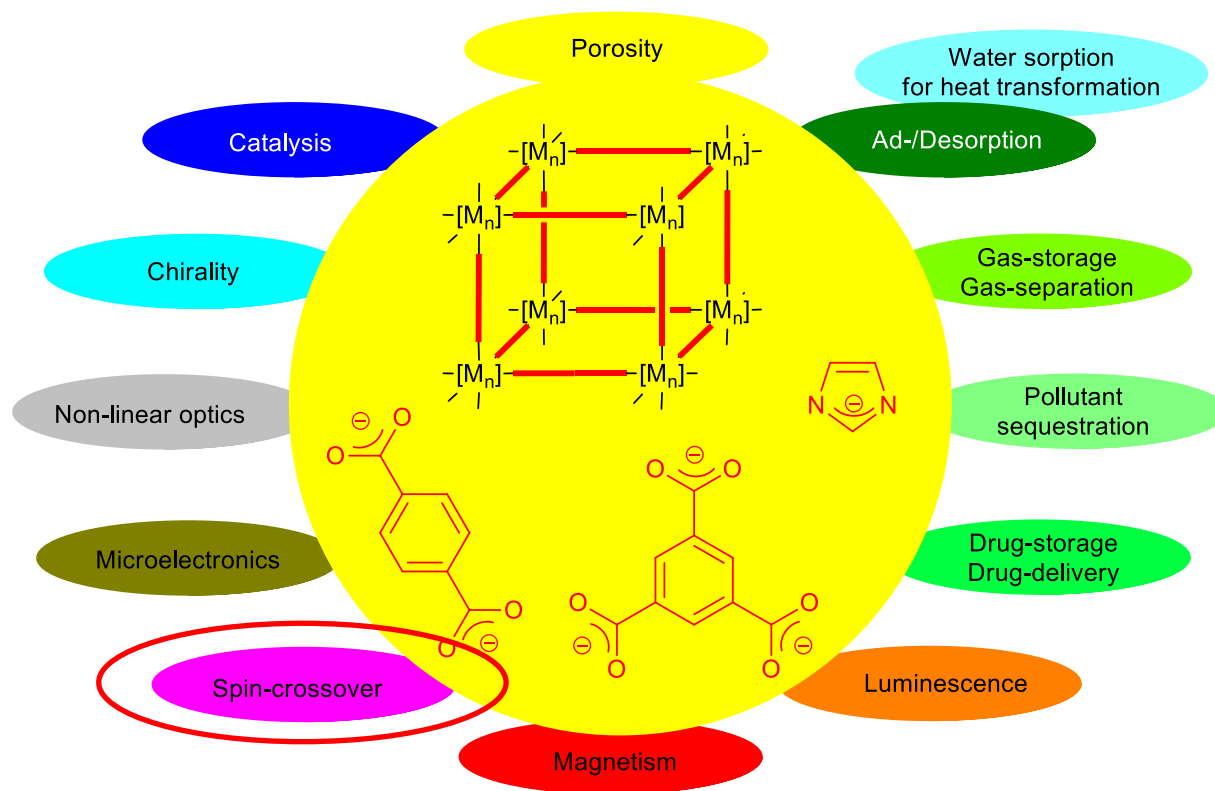


(b)

S. Pramanik, C. Zheng, X. Zhang, T. J. Emge, J. Li, *J. Am. Chem. Soc.* **2011**, 133, 4153.

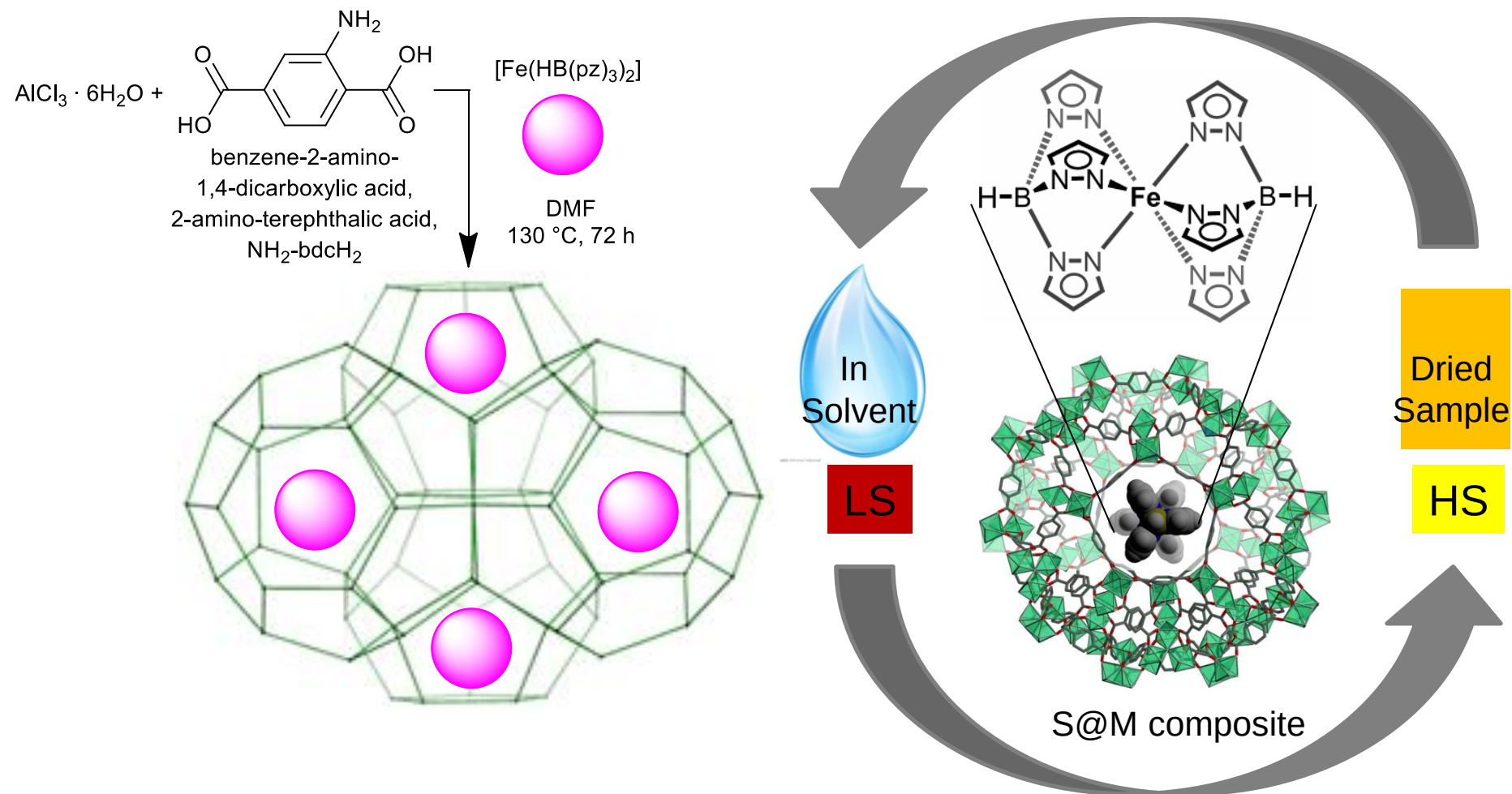
S. Pramanik, Z. Hu, X. Zhang, C. Zheng, S. Kelly, J. Li, *Chem. Eur. J.* **2013**, 19, 15964.

Metal-Organic Frameworks (MOFs): Application-oriented properties



Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E.
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

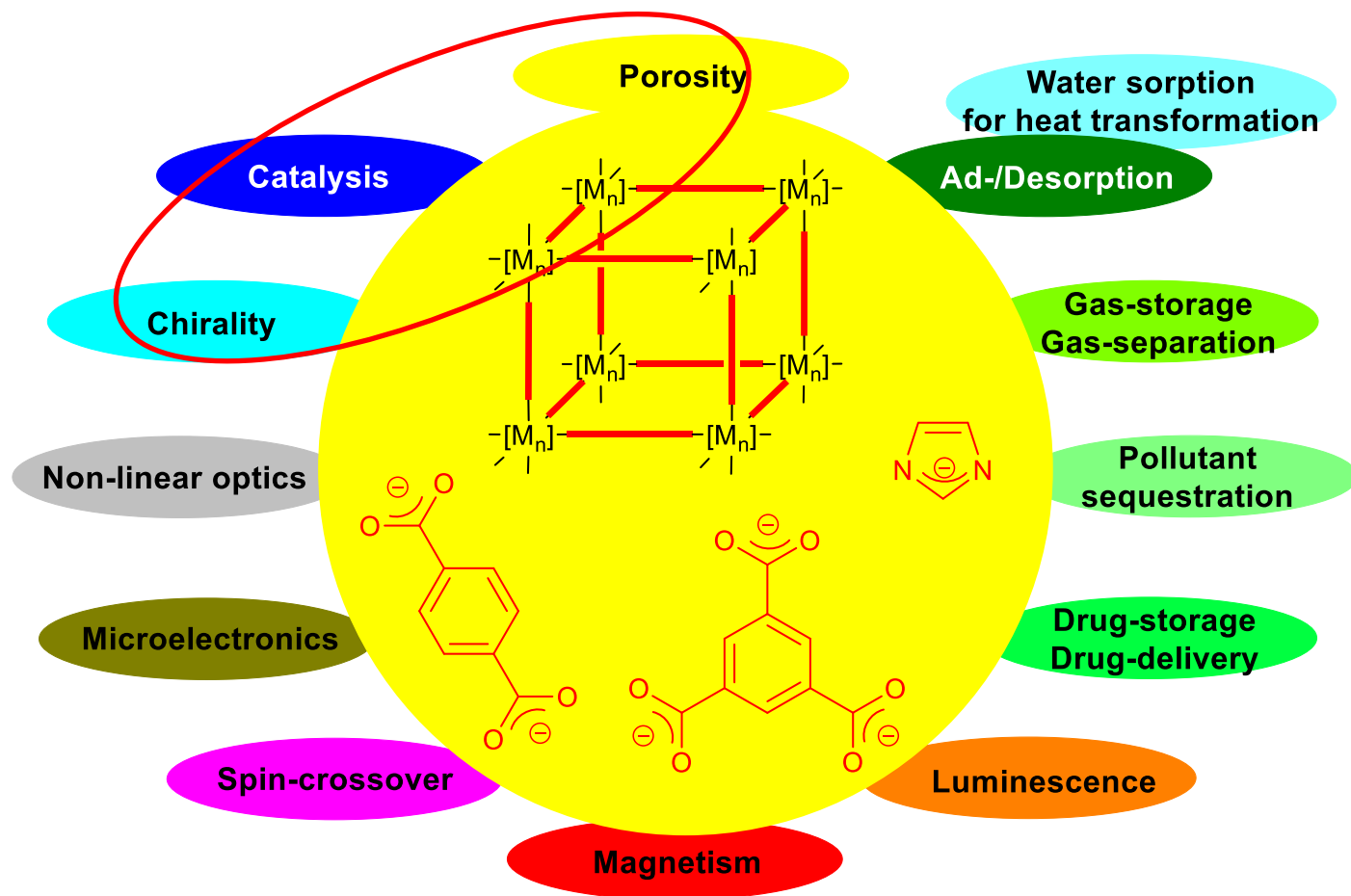
Solvent-triggered spin state switching of $[\text{Fe}(\text{HB}(\text{pz})_3)_2]$ in nano-confinement of MOF $\text{NH}_2\text{-MIL-101}(\text{Al})$



T. Zhao, I. Boldog, V. Spasojevic, A. Rotaru, Y. Garcia, C. Janiak,
J. Mater. Chem. C **2016**, 4, 6588-6601. DOI: 10.1039/c6tc01297c

Metal-Organic Frameworks (MOFs):

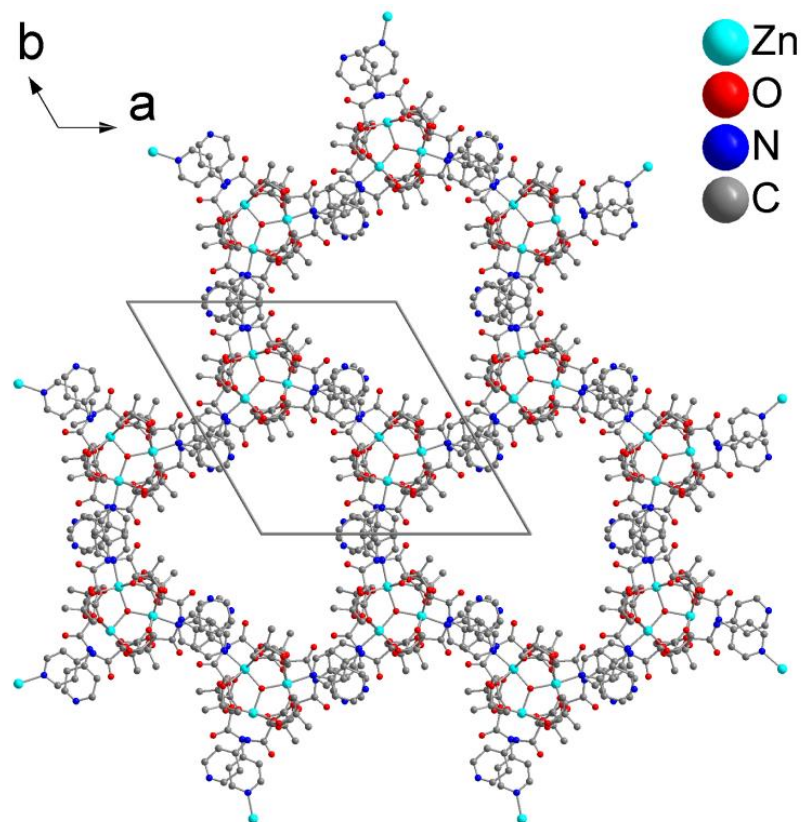
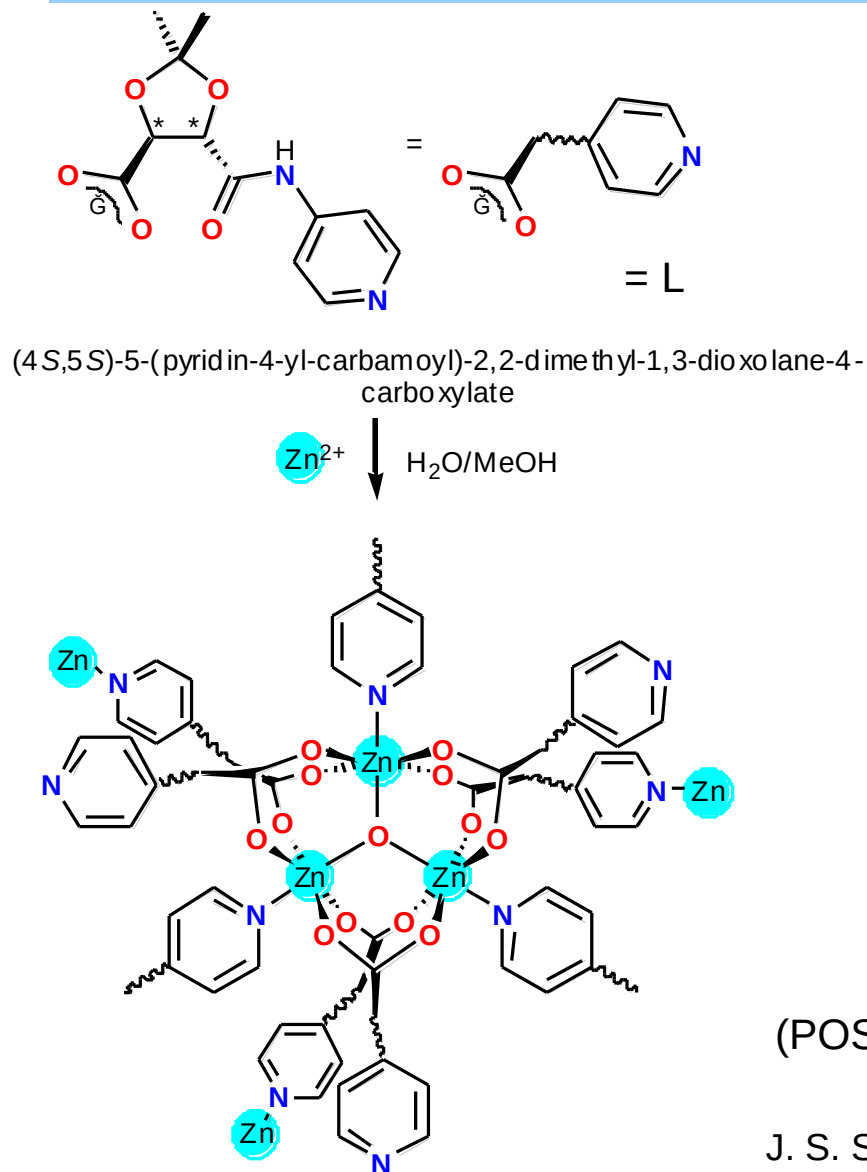
Application-oriented properties



Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E. (>400 citations)
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b. (>2500 citations)

MOFs:

Catalysis and chirality – network 2D- $\{[2H_3O][Zn_3(\mu_3-O)L_6]\}$

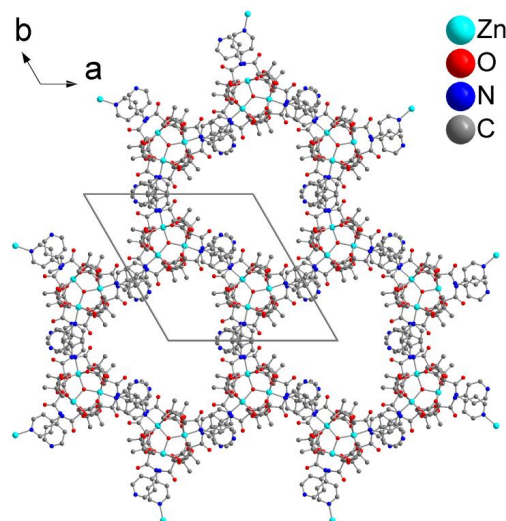
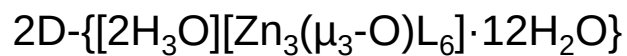
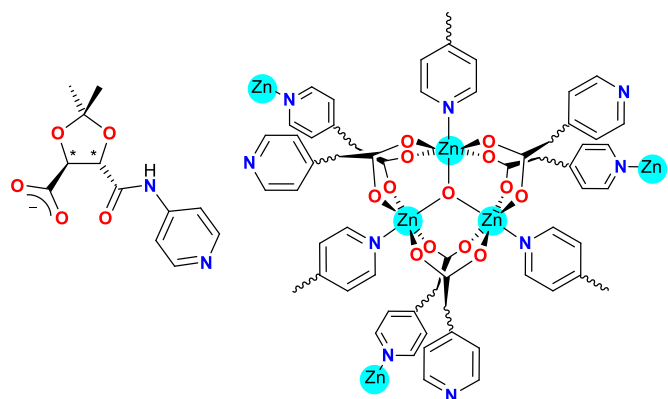


homochiral network 2D- $\{[2H_3O][Zn_3(\mu_3-O)L_6]\}$
 POST-1
 (POST = Pohang University of Science and Technology)

J. S. Seo, D. Whang, H. Lee, S. I. Jun, J. Oh, Y. J. Keon,
 K. Kim, *Nature* **2000**, 404, 982–986.

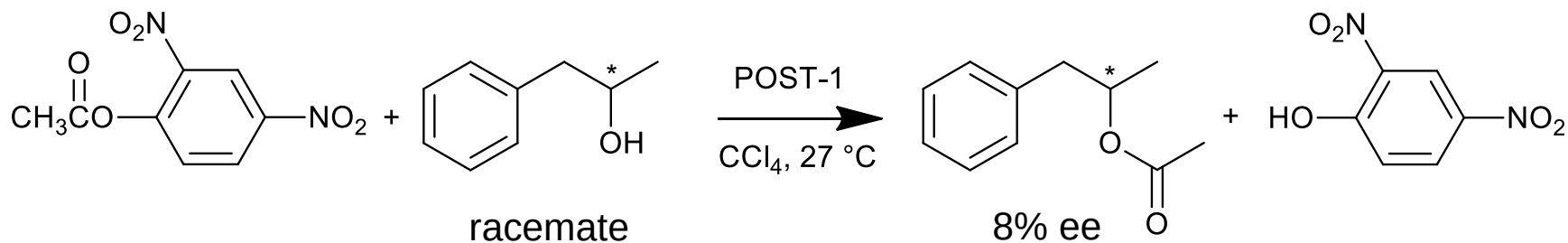
MOFs: Catalysis and chirality

First proof of principle for asymmetric catalysis with chiral MOF



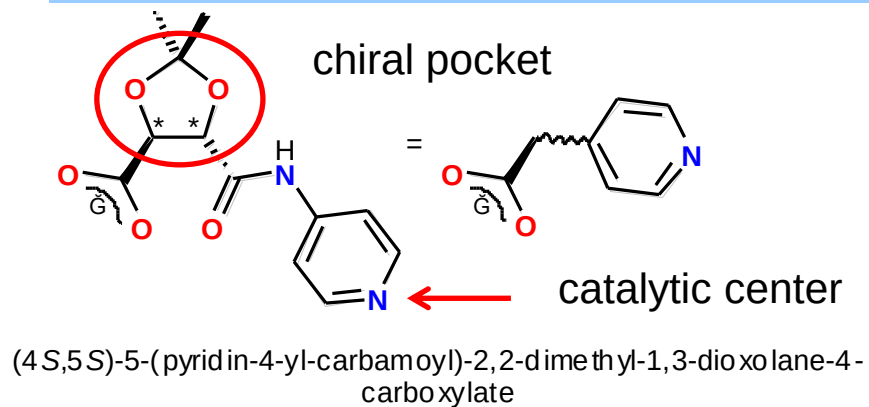
- homochiral
- chiral 1D channels
- 47% free volume

POST-1



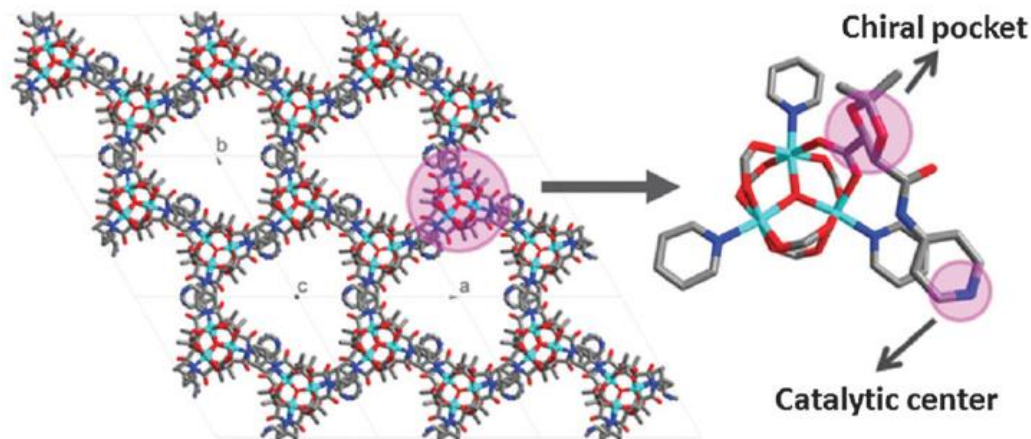
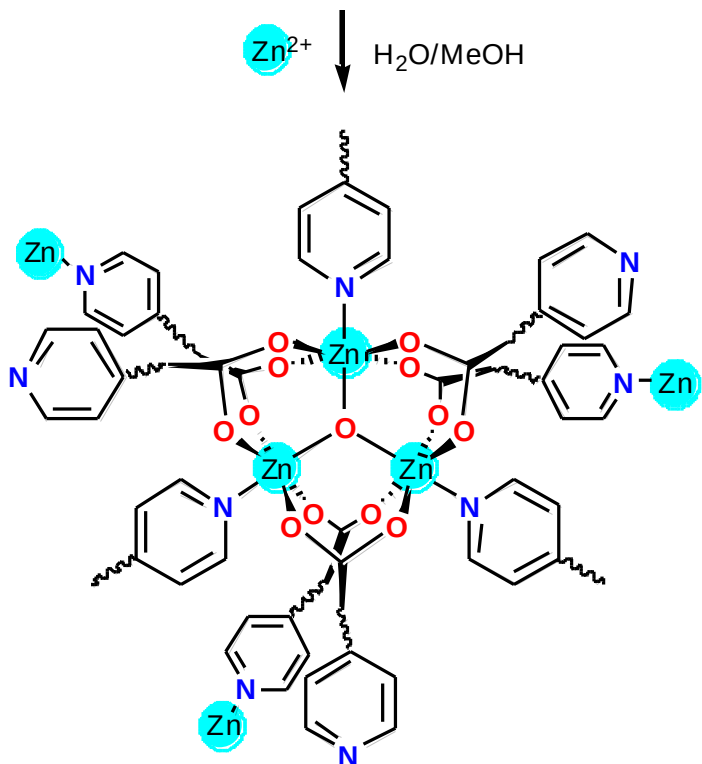
MOFs: Catalysis and chirality

First proof of principle for asymmetric catalysis with chiral MOF

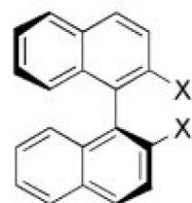


low enantioselectivity:
catalytically active free
pyridyl groups are too far from
the chiral sites of the pores.

ineffective to attempt to utilize
remote chirality for induction

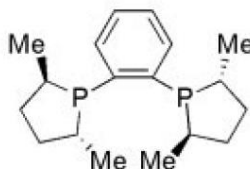


MOF-catalysis using enantiopure ligands "privileged ligands"



X=OH BINOL
X=PPh₂ BINAP

Diels-Alder
Mukaiyama aldol
aldehyde allylation
hydrogenation
alkene isomerization
Heck reaction



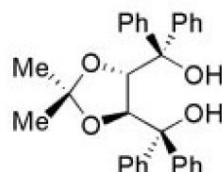
MeDuPhos

hydrogenation
hydrophosphination
hydroacylation
hydrosilylation
Bayer-Villager oxidation



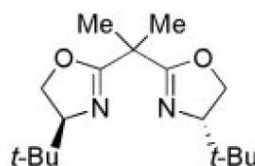
**Brintzinger's
ligand**

alkene reduction
imine reduction
alkene carbometallation
Ziegler-Natta
polymerization



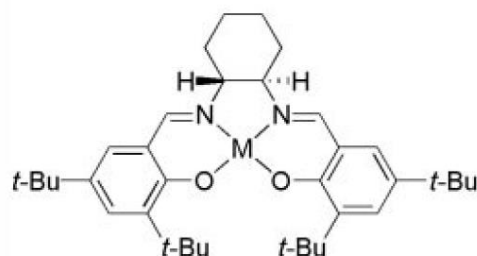
**TADDOLate
ligand**

Diels-Alder
aldehyde alkylation
ester alcoholysis
iodolactonization



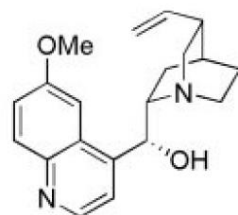
Bis(oxazoline)

Diels-Alder
Mukaiyama aldol
conjugate addition
cyclopropanation
aziridination



Salen complexes

epoxidation
epoxide ring-opening
Diels-Alder
imine cyanation
conjugate addition



**Cinchona alkaloid
derivatives**

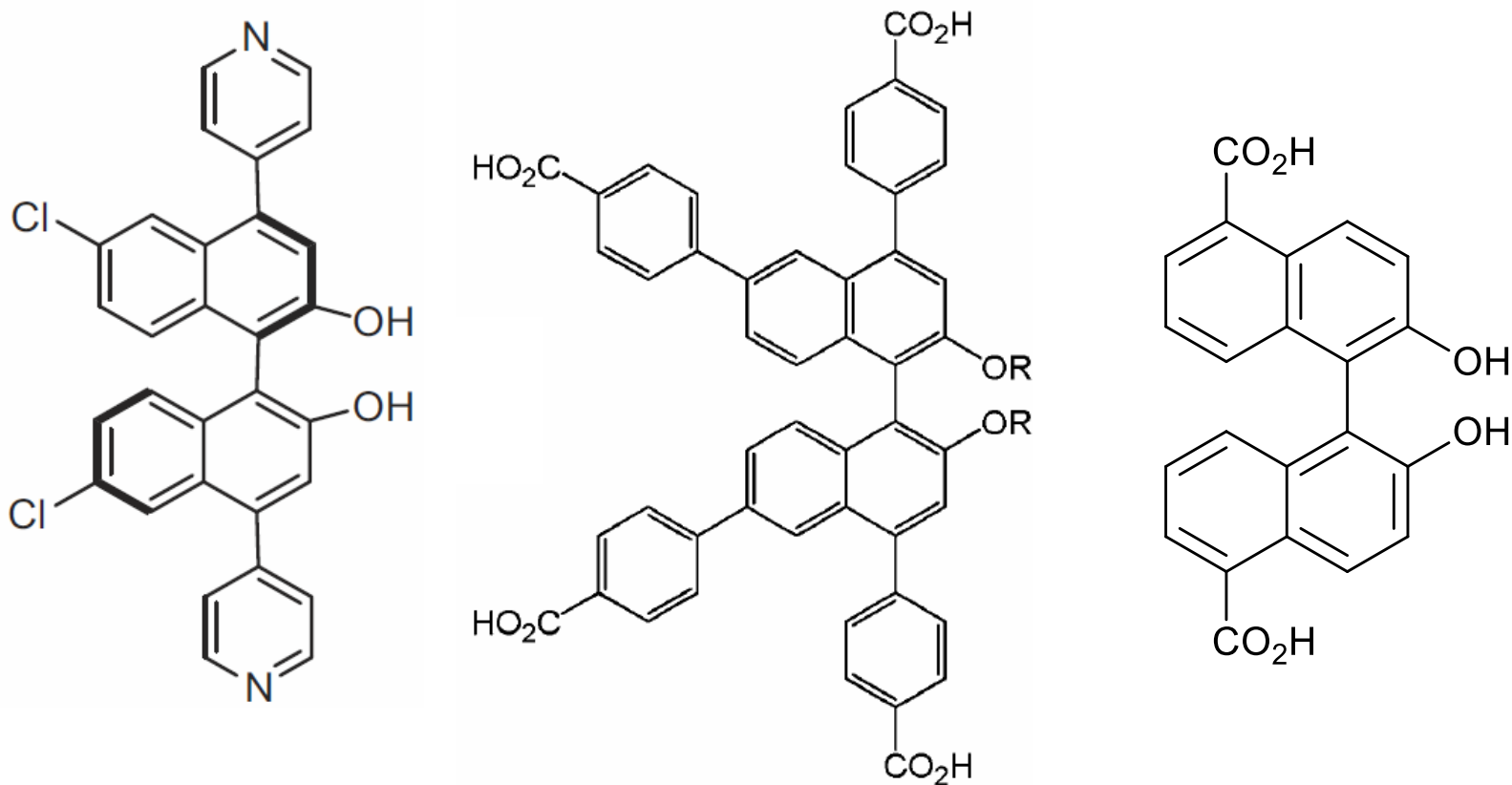
dihydroxylation
acylation
heterogeneous hydrogenation
phase transfer catalysis

T. P. Yoon, E. N. Jacobsen, *Science*
2003, 299, 1691-1693.

Fig. 1. Examples of privileged chiral ligands and catalysts.

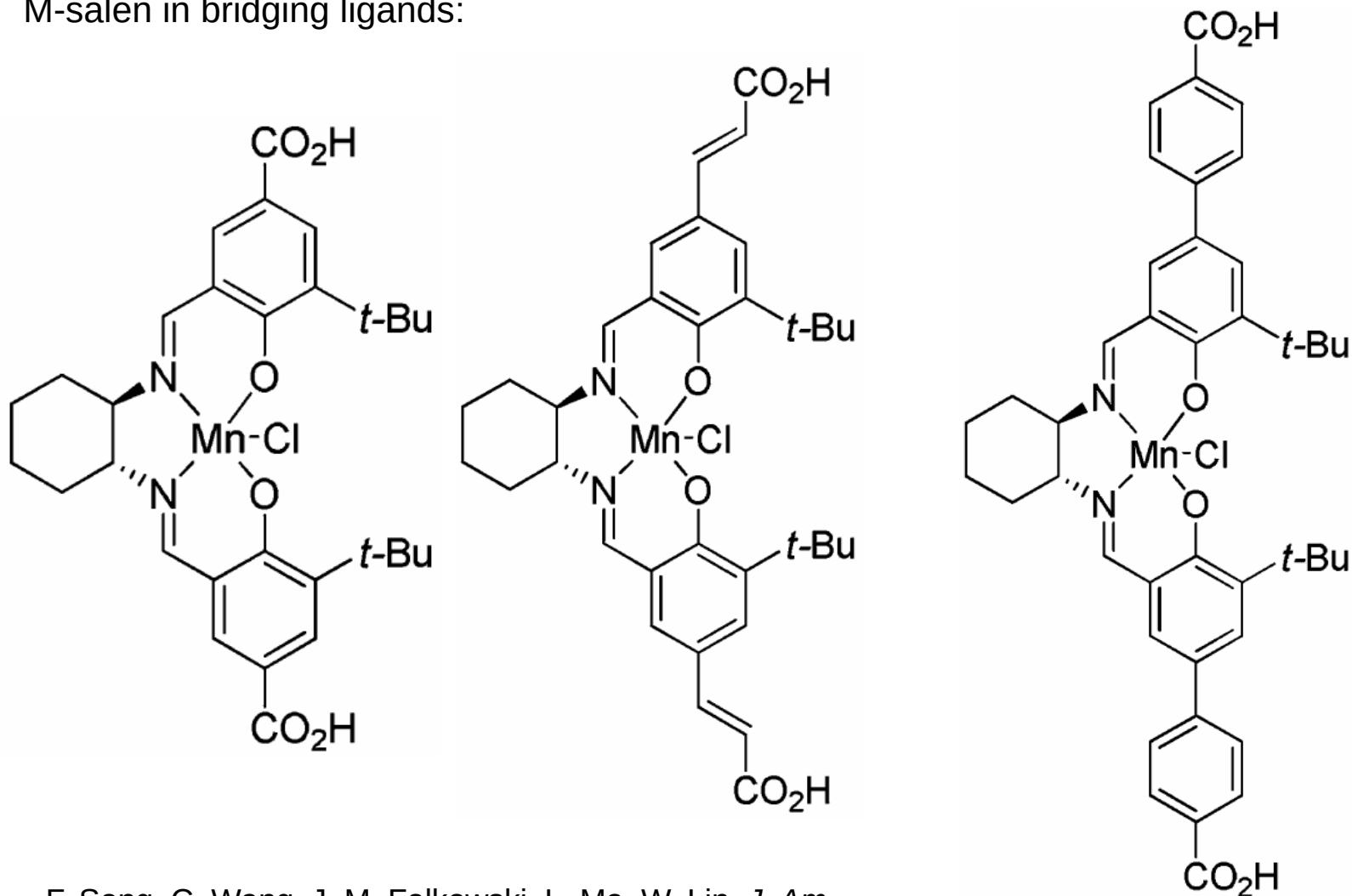
MOF-catalysis using enantiopure ligands "privileged ligands"

BINOL in bridging ligands:



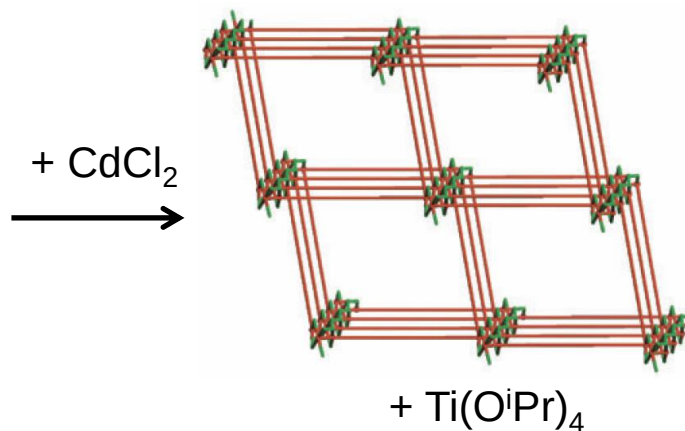
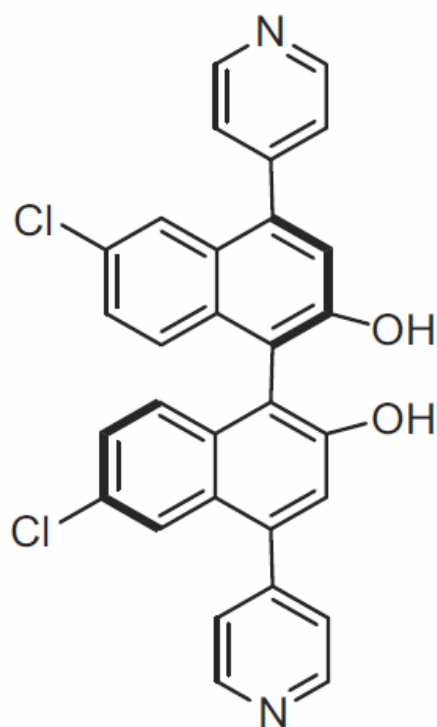
MOF-catalysis using enantiopure ligands "privileged ligands"

M-salen in bridging ligands:

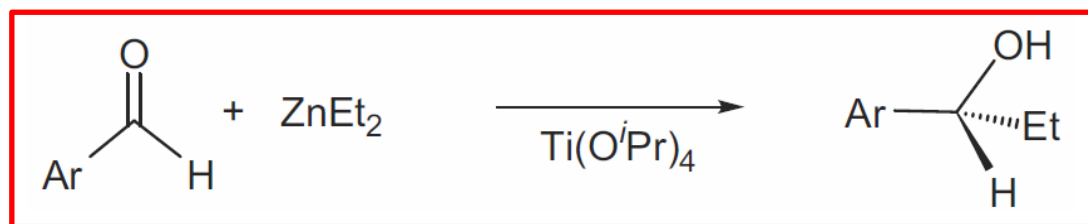


F. Song, C. Wang, J. M. Falkowski, L. Ma, W. Lin, *J. Am. Chem. Soc.* **2010**, 132, 15390–15398.

MOFs: privileged ligand catalysts - BINOL derivative

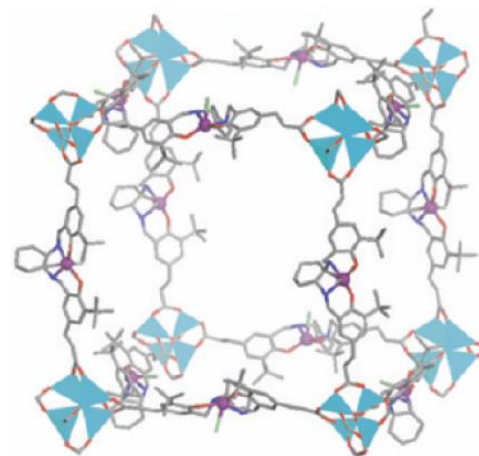
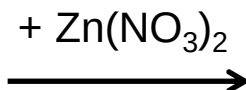
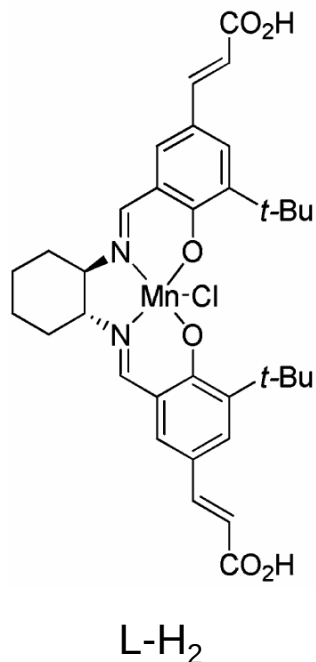


yield >99%
80-93% ee

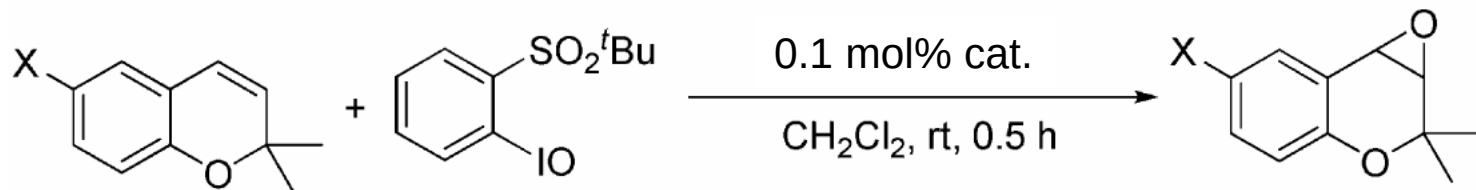


comparison: BINOL/ $\text{Ti}(\text{O}^i\text{Pr})_4$: yield >99%, 84-94% ee

MOFs: privileged ligand catalysts - salen derivative

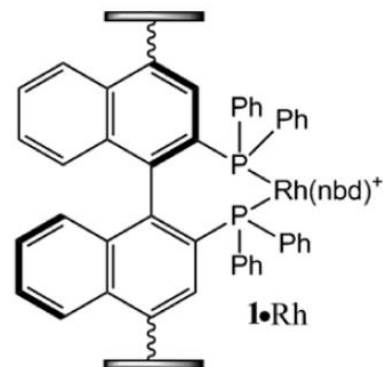
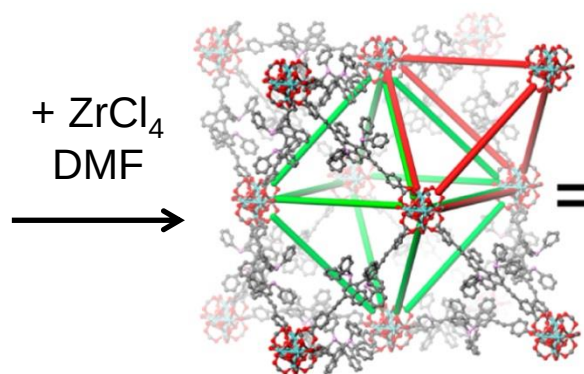
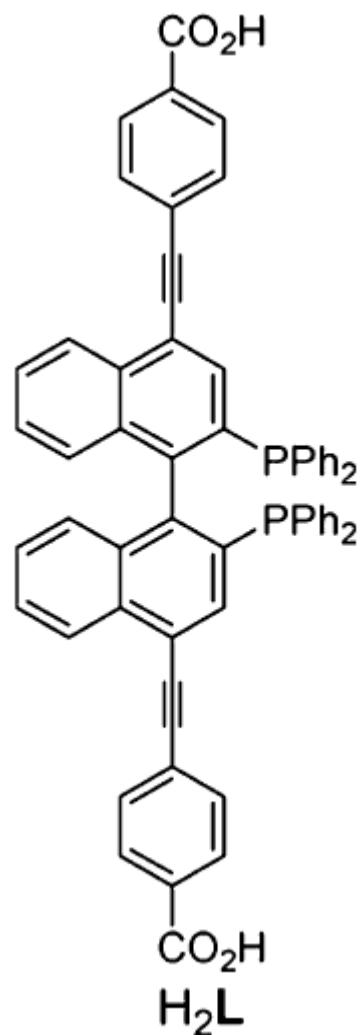


conv. 79-93%
75-85% ee

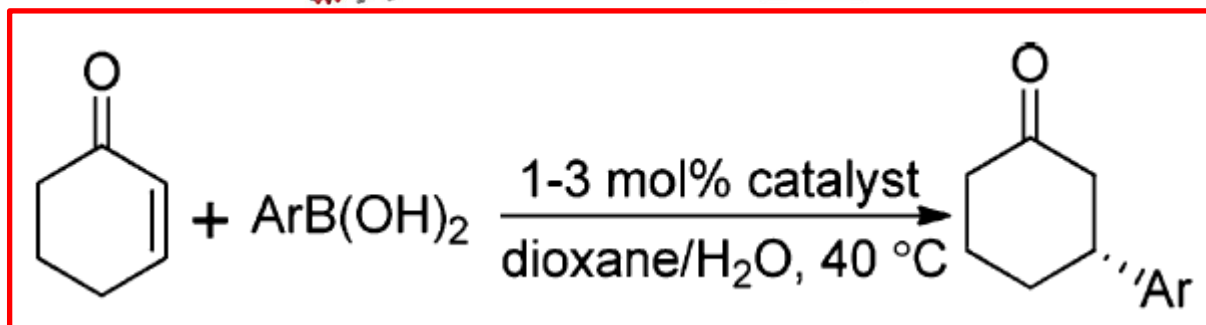


comparison: $L-Me_2$: yield 81-97%, 87-92% ee

MOFs: privileged ligand catalysts - BINAP derivative



yield 80-99%
>99% ee

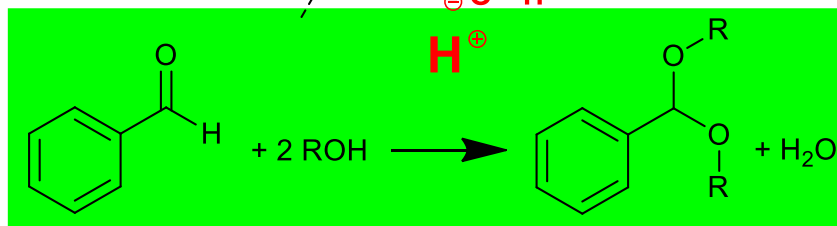
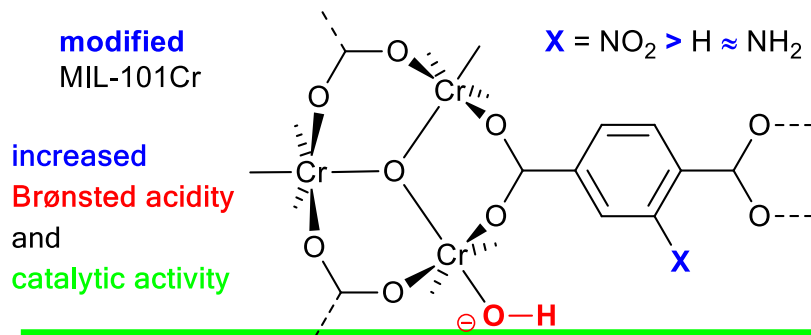
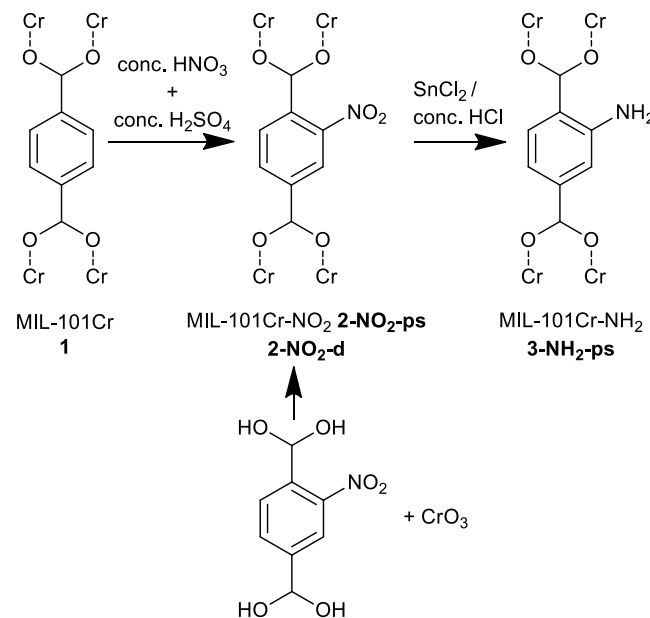
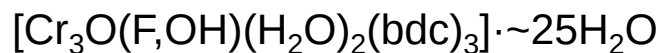
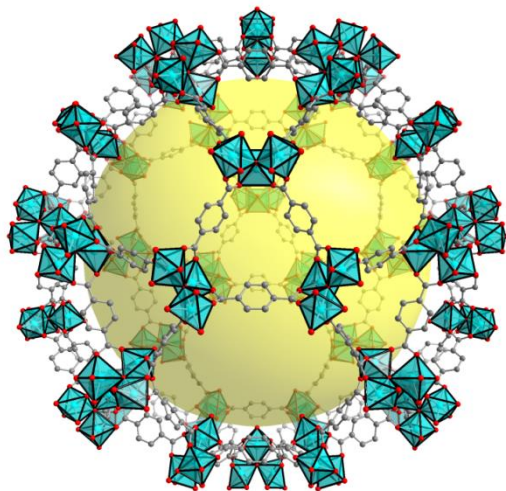


comparison: Rh-H₂L: yield <7%, ee not determined

comparison: Rh-BINAP: yield 29-46%, >99% ee

MOFs:

Catalysis with modified MIL-101

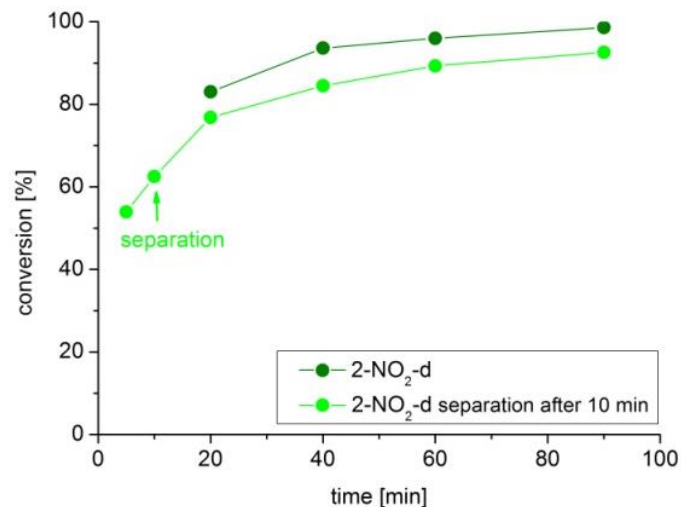
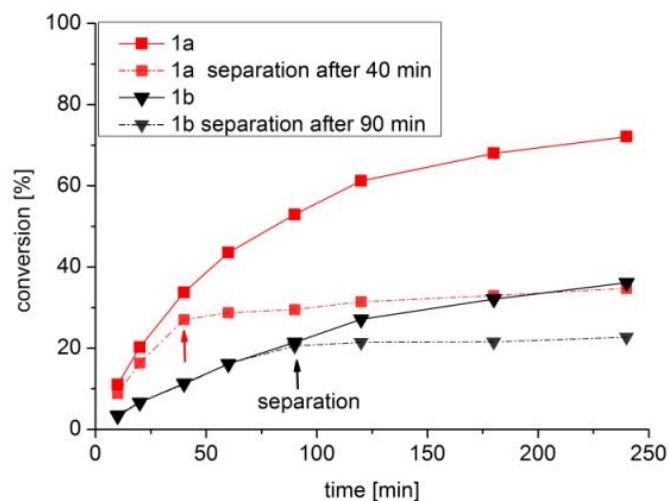


Inorg. Chem. **2014**, 53, 7319–7333.
DOI: 10.1021/ic5006456

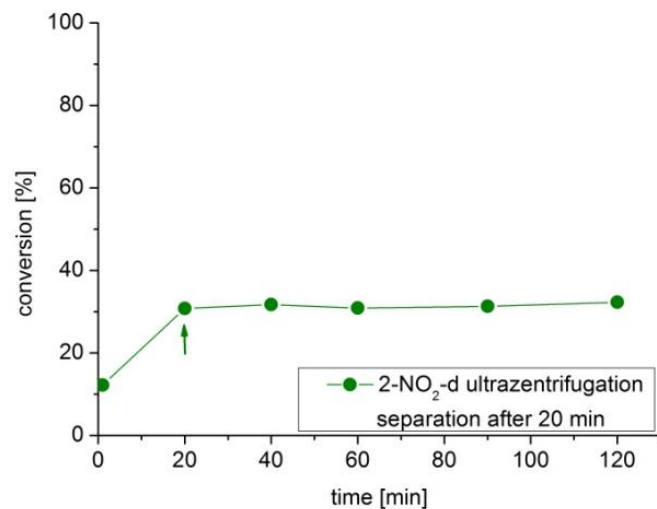
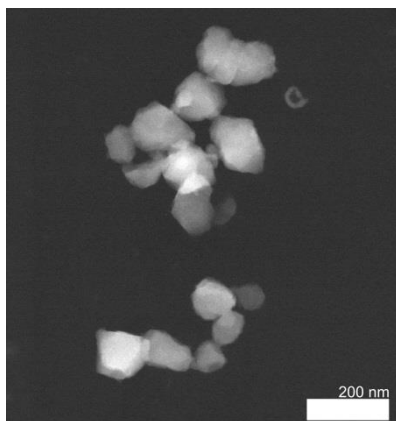
MOFs:

Catalysis with modified MIL-101

heterogeneous catalysis?

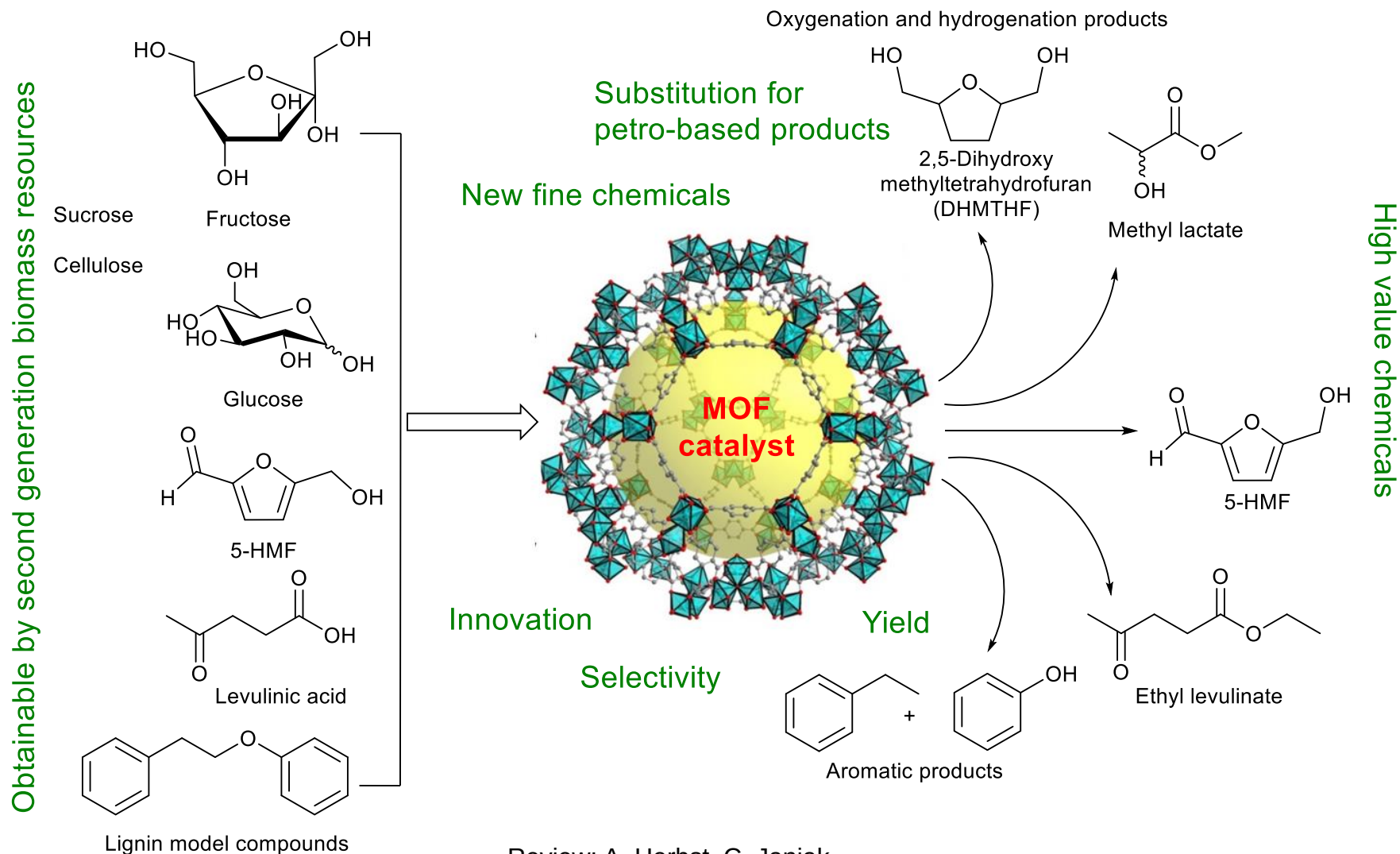


2-NO₂-d filtrate:
(nano-MOFs)



MOFs in biomass transformation:

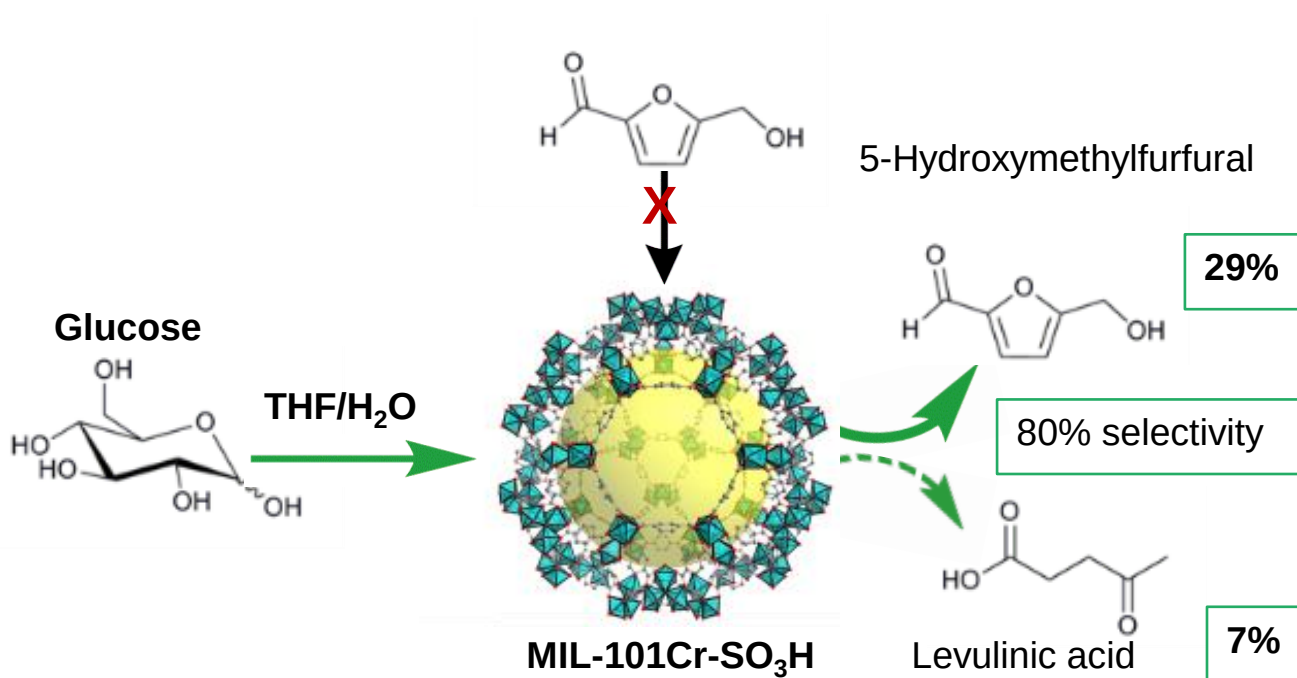
MOF catalysts in biomass upgrading



Review: A. Herbst, C. Janiak,
MOF catalysts in biomass upgrading towards value-added fine chemicals;
CrystEngComm **2017**, in press. DOI: 10.1039/C6CE01782G

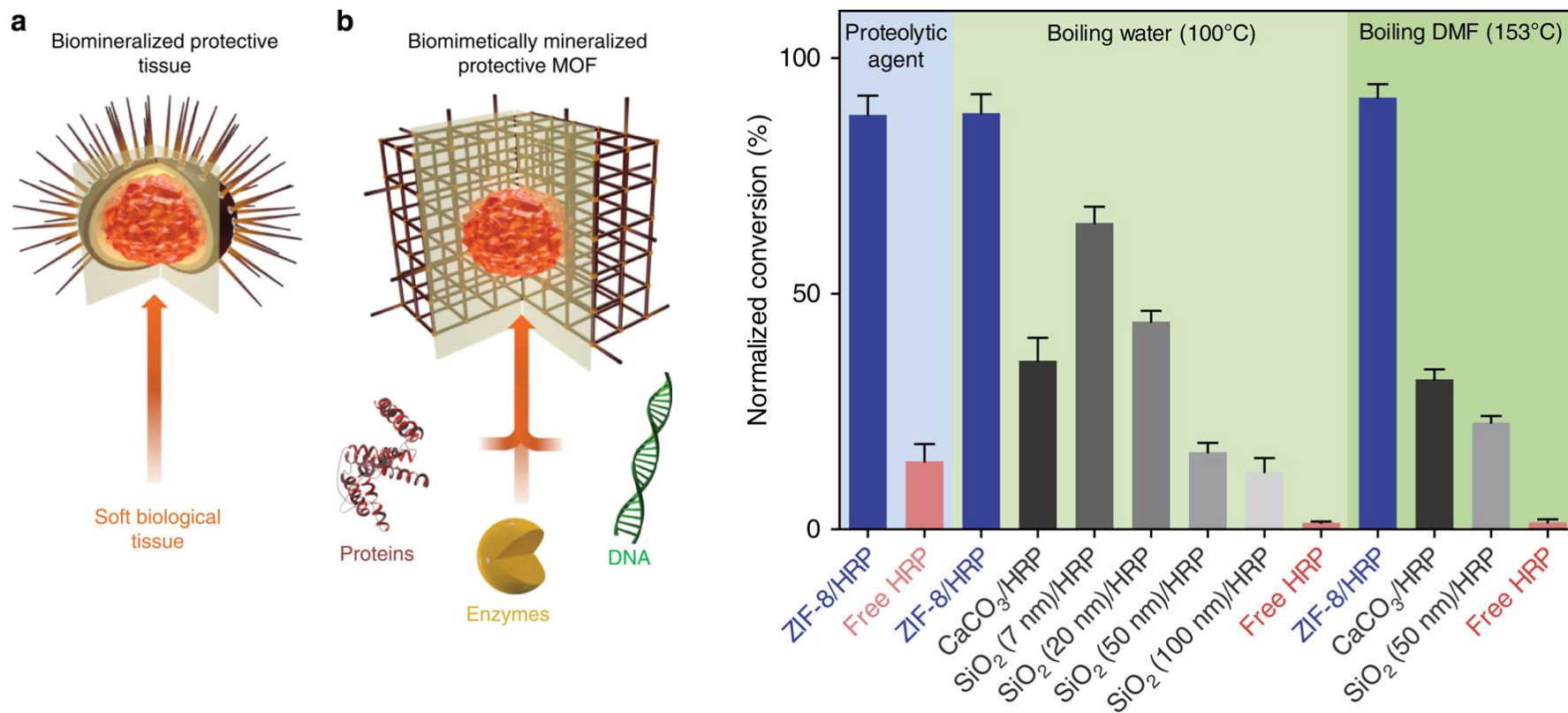
MOFs in biomass transformation:

Glucose to 5-HMF catalysis with modified MIL-101(Cr)



MOFs are making progress also in catalytic biomass transformation, here in the 5-HMF production with increased yield from glucose in low boiling solvents.

Biomimetic mineralization of MOFs as protective coatings for biomacromolecules (enzymes)



Lian et al. *Nature Communications* **2015**, 6, article number 7240

Last: The first commercial applications (?)

For the last 15 years, the elephant in the room has been “Where are the applications?”

MOFs find a use. By Nina Notman, ChemistryWorld 28 March 2017,
<https://www.chemistryworld.com/features/mofs-find-a-use/2500508.article>

Decco unveils freshness revolution

TruPick delivers benefits of 1-MCP in pioneering micro-absorbent technology

Decco Worldwide has announced the EPA registration in the US of TruPick, its next generation 1-MCP technology to maintain postharvest freshness in fruits and vegetables by controlling the effects of naturally occurring ethylene during storage and transportation.

The company describes the product as the first gel formulation in water-soluble pouches delivering the benefits of 1-methylcyclopropene (1-MCP) in a novel, micro-absorbent technology.

“TruPick’s performance has been rigorously evaluated in many of the globally important apple markets and confirmed by private and university researchers as highly effective in maintaining firmness and suppressing ethylene production in both RA (refrigerated air) and CA (controlled atmosphere) storage,” Decco said in a press release.

TruPick is a proprietary and patented release system, discovered and developed by Dr. Nazir A. Mir of MirTech, and distributed by Decco. The companies have set up a joint partnership, Essentiv, to develop, register and manufacture pre- and post-harvest solutions for the produce industry.

“We believe that TruPick and other new products in our pipeline and the Decco Worldwide technical, sales and service organisation will not only deliver the same well known benefits packers expect from 1-MCP but also unlock value to expand its use to more packers and to more fruit and vegetables as we pursue registrations in all key markets around the world”, said Dr Lynn Oakes, global project manager for Decco Worldwide and general manager for Essentiv.



Changing channels: produce distribution in 2...

Trump triggers NAFTA renegotiation

Vanguard expands Pride Packing team

Stage set for Fresh Produce Forum China

Macri flies flag for Argentine fruit in China

Chile-China MOU will boost mutual trade

Chile hopeful of Indonesia deal

Mark Pay joins Rockit Global

New 1000 look hits US

... <http://www.fruitnet.com/americafruit/article/169350/decco-unveils-freshness-revolution>[23.05.2017 15:50:34]

Last: The first commercial applications (?)

MOF Technologies has become the world's first metal organic framework (MOF) technology provider to successfully commercialize its research. Teaming up with leading **fruit and vegetable storage provider, Decco Worldwide Post-Harvest Holdings**, MOF Technologies identified and delivered the **ideal metal organic framework** required **for** a **novel food application known as TruPick™ postharvest freshness management** – Decco's next generation in storage provision for selected fruits and vegetables for postharvest freshness management.

Specifically, the **metal organic framework** provided by MOF Technologies is central **to** the effective **control of naturally occurring, but unwanted, ethylene** – often found when storing and transporting fruits and vegetables.

Another MOF product launched last year was TruPick, developed by spin-out company [MOF Technologies](#) in collaboration with Decco, a global company with expertise in fruit and vegetable storage. The pores in TruPick's MOFs hold the gas 1-methylcyclopropene (1-MCP). '1-MCP retards the ripening of fruit,' explains Stuart James, chair of inorganic chemistry at Queen's University Belfast in the UK and co-founder of MOF Technologies. 'Ethylene is a plant hormone, and when it is released by fruit it triggers ripening in other fruit. What 1-MCP does is bind to the ethylene binding site in the enzymes in the fruit and block the effect of ethylene.'

1-MCP gas has been used for over a decade to lengthen the shelf life of fruit and vegetables in warehouse storage containers, but this is the first time it has been supplied in a way that eliminates the need for gas cylinders. TruPick is a disposable sachet, which looks very similar to the silica gel desiccant packs put in shoe boxes, that when put in water releases the 1-MCP gas into the air. According to MOF Technologies, within 24 hours the released 1-MCP will effectively block the effects of ethylene on all the fruit in a storage room. TruPick is now approved for use, and being sold for use on hard fruits in the US and Turkey, and is going through the registration processes in other fruit-growing parts of the world including the EU.

MOFs find a use. By Nina Notman, ChemistryWorld 28 March 2017,
<https://www.chemistryworld.com/features/mofs-find-a-use/2500508.article>

ION-X. These gas cylinders contain proprietary MOFs holding very toxic gases such as arsine, phosphine and boron trifluoride (typically stored in cylinders under high pressure. In the ION-X cylinders, the gas molecules are stored below atmospheric pressure meaning a vacuum needs to be applied to get them out. 'If you store these gases at high pressure and you have a leak, things will flow from the cylinder outward.



Omar Farha was not able to say which electronic manufacturers are now using ION-X, nor give a ballpark figure of the kind of volumes his company are currently supplying. NuMat has however made public a collaboration with global gas supplier the Linde Group to develop MOF-based next generation separation and storage technologies – but as yet none of these have hit the market.

MOFs find a use. By Nina Notman, ChemistryWorld 28 March 2017,
<https://www.chemistryworld.com/features/mofs-find-a-use/2500508.article>

In 2000, long before other academics started spinning MOF companies out from their labs, Yaghi began working alongside chemical giant [BASF](#) to explore the commercial potential of MOFs. To date nothing has hit the shelves yet, but a number of different products are in the pipeline.

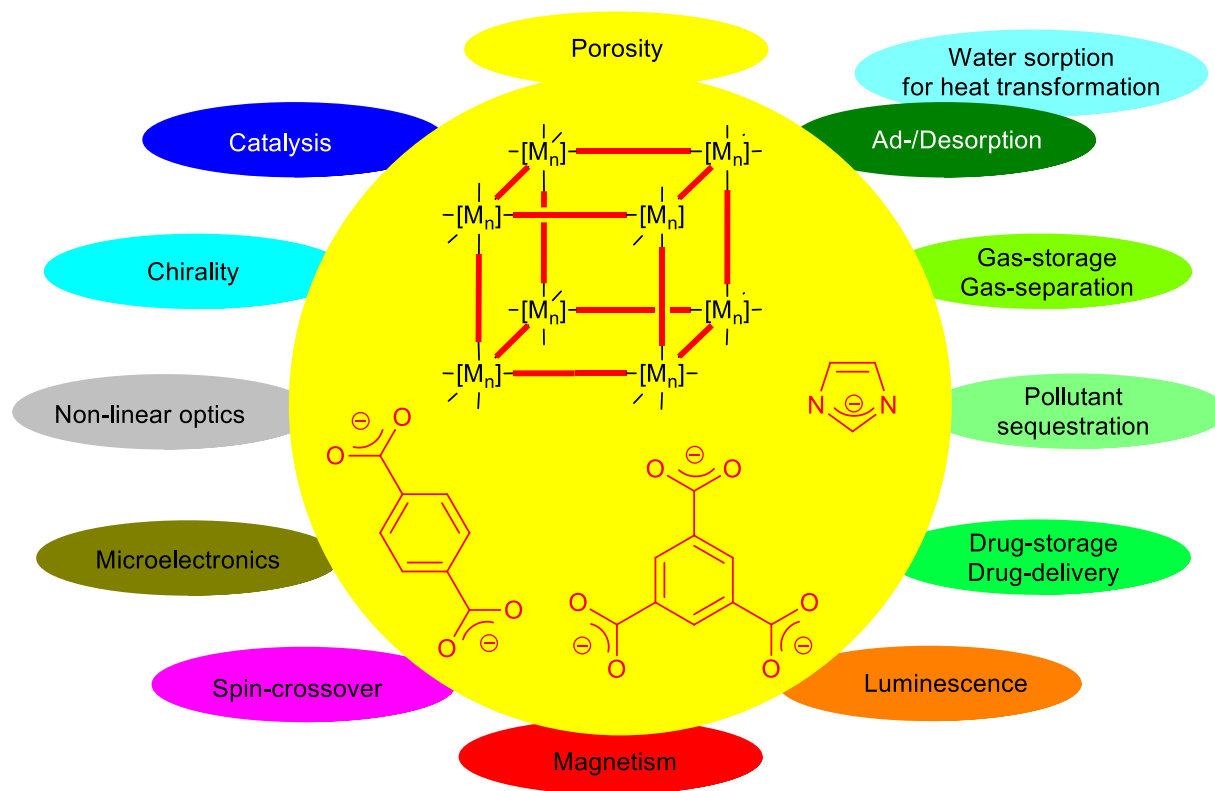
The currently mothballed project that has progressed the furthest forward to date is using MOFs in tanks for methane-powered vehicles. 'We can store double or triple the amount of natural gas in a tank containing a MOF compared to a tank that does not, under the same temperature and pressure,' Yaghi says. 'A fuel tank [containing a MOF] has been tested with the methane uptake and release cycled to many, many cycles, to be equivalent to the lifetime of the automobile without seeing any degradation in the MOF behaviour or properties.'

The project was very successful technically, agrees Ulrich Mueller, who leads the MOF research group at BASF in Ludwigshafen, Germany. 'We had a car fleet here at Ludwigshafen and we had a truck in the US running from east coast to west coast,' he says.

"We had a methane-fuelled MOF car fleet, but when the oil price dropped sharply, the economic incentive was no longer there."

End: Metal-Organic Frameworks (MOFs):

Application-oriented properties



Reviews: *New J. Chem.* **2010**, 34, 2366 DOI: 10.1039/C0NJ00275E.
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

Acknowledgements

- the Team counts



Dr. Stefan Henninger,
Fraunhofer ISE, Freiburg



Dr. Felix Jeremias



SPP 1708

Subarna Dey
Dominik Fröhlich
Sebastian Glomb
Irina Gruber
Emrah Hastürk
Ülkü Kökcam-Demir
Niels Tannert

Johannes Ehrenmann
Dr. Hesham Mena
Dr. Vasile Lozan
Dr. Martin Wickenheisser
Dr. Anupam Khutia
Dr. Asamanjoy Bhunia
Dr. Annika Herbst

international
collaboration projects
- China
- India
- Spain
- Moldova
- Bangladesh
- Egypt

Deutsche
Forschungsgemeinschaft
DFG



Federal Ministry
of Economics
and Technology

BUSINESS.
GROWTH.
PROSPERITY.

*Science brings Europe
and India closer*



Alexander von Humboldt
Stiftung/Foundation

DAAD

Deutscher Akademischer Austausch Dienst
German Academic Exchange Service



Federal Ministry
of Education
and Research



China Scholarship Council
www.csc.edu.cn